

CompTIA Server+

Automation of Server Administration using Scripts

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 - Exercise 3 - User Creation Using Scripts
 - Review
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Introduction

Server+

Powershell Script

Linux Script

Batch Script File

Variables

Conditions

Welcome to the **Automation of Server Administration using Scripts** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Script files can be used on Windows or Linux operating systems to automate repetitive tasks minimizing administrative effort. These tasks can include scheduling an automated backup, restarting devices at specific times or remapping shared network drives at start-up. Different operating systems use different file types to create scripts.

In this module, different scripting techniques for the different operating systems will be explored.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Different Script Types

- Exercise 2 - Basic Script Constructs
- Exercise 3 - User Creation using Scripts

After completing this module, you should be able to:

- Create a Linux Script
- Create a Basic Powershell Script
- Create a Basic Batch Script File
- Create a Script File with Different Variables and Conditions
- Create a Script that Creates Users in Active Directory

Exam Objectives

The following exam objectives are covered in this module:

2.6 Summarize scripting basics for server administration

- Script types
- Environment variables
- Basic script constructs
- Common server administration scripting tasks

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.6 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Creating Basic Scripts

Different operating systems use different file types that can be used for scripting. Microsoft operating system can use Powershell scripts with a file extension of .ps or Command Prompt file types with a .bat file extension. In comparison, script files for Linux operating systems use the bash shell to create a file with a .sh extension.

In this exercise, different scripting files will be identified, and basic scripts will be created.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a Linux Script
- Create a Basic Powershell Script
- Create a Basic Batch Script File

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.6 - Stand-alone Linux Workstation)

Task 1 - Create a Linux Script

Script files can be used to automate repetitive administration tasks to minimize administrative effort. The most commonly used file for creating a script using a Linux operating system in the bash shell is the .sh file type.

In this task, a simple Linux script will be created.

Step 1

Connect to **PLABALMA**.

Click the **Activities** icon in the top lefthand corner.

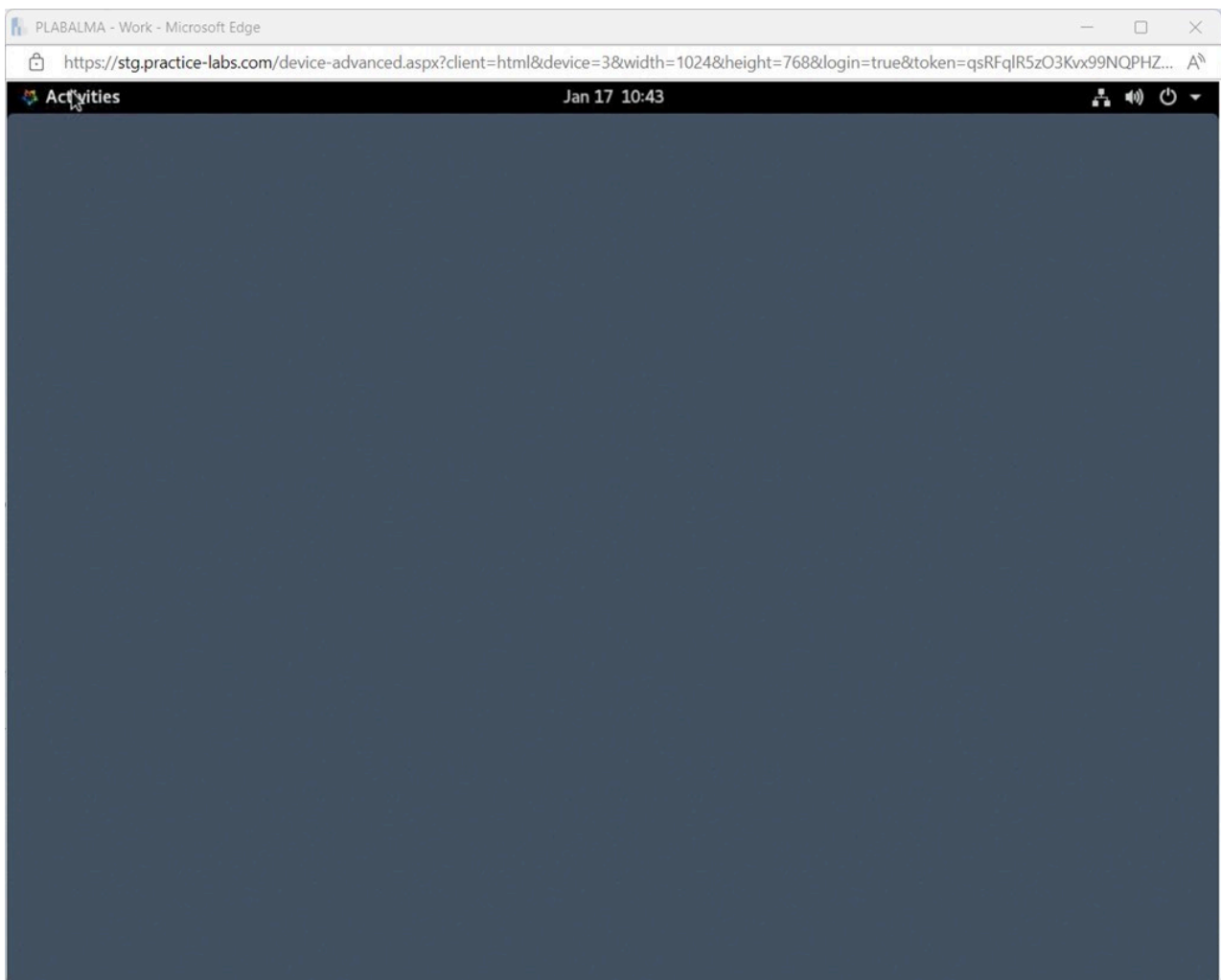


Figure 1.1 Screenshot of PLABALMA: Displaying selecting the Activities icon in the top lefthand corner.

Step 2

Select **Terminal** in the fly-out menu.

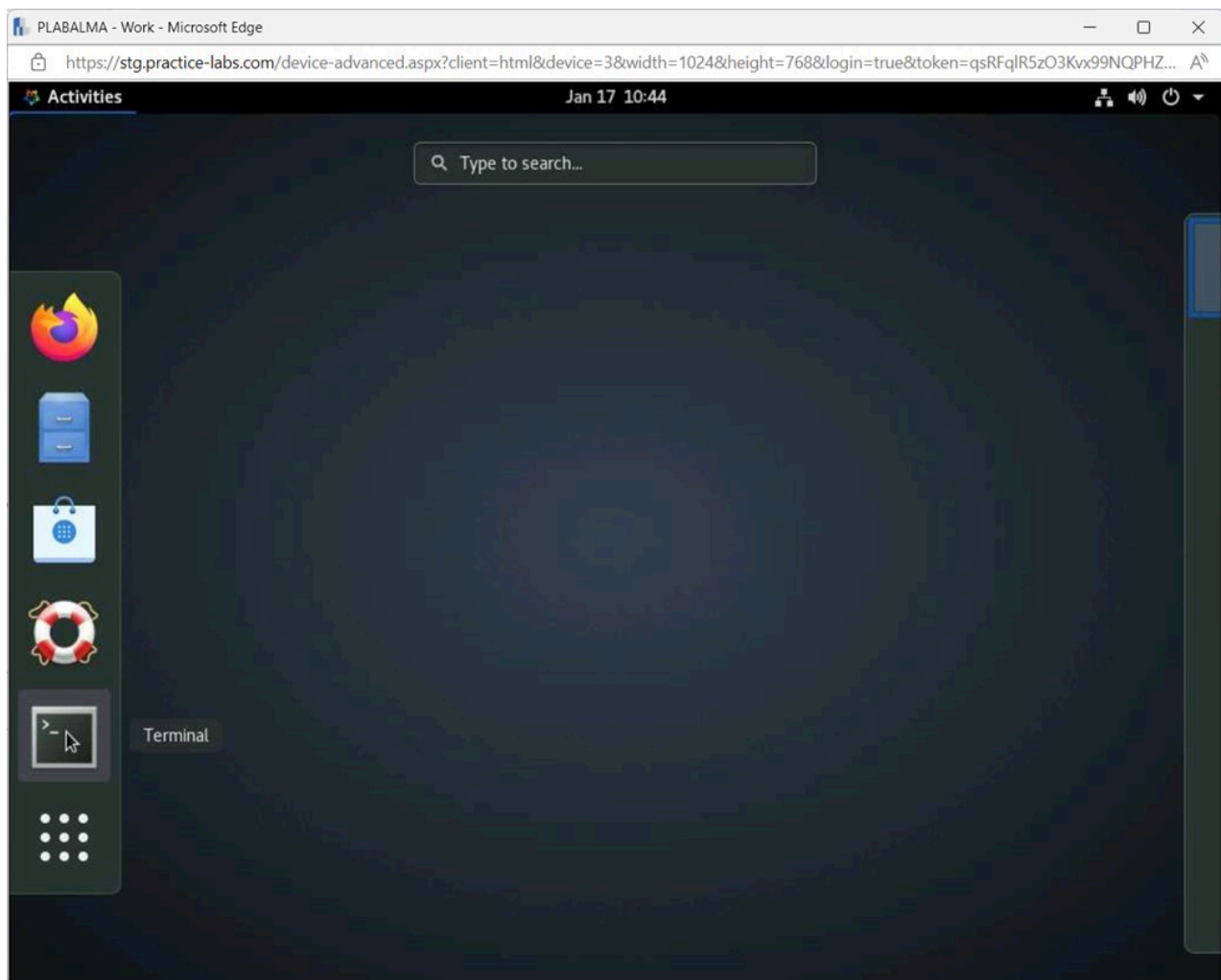


Figure 1.2 Screenshot of PLABALMA: Displaying selecting Terminal from the fly-out menu.

Step 3

In the **Terminal** window, type the following and press **Enter**:

```
touch Documents/plabdoc{1..10}.txt
```

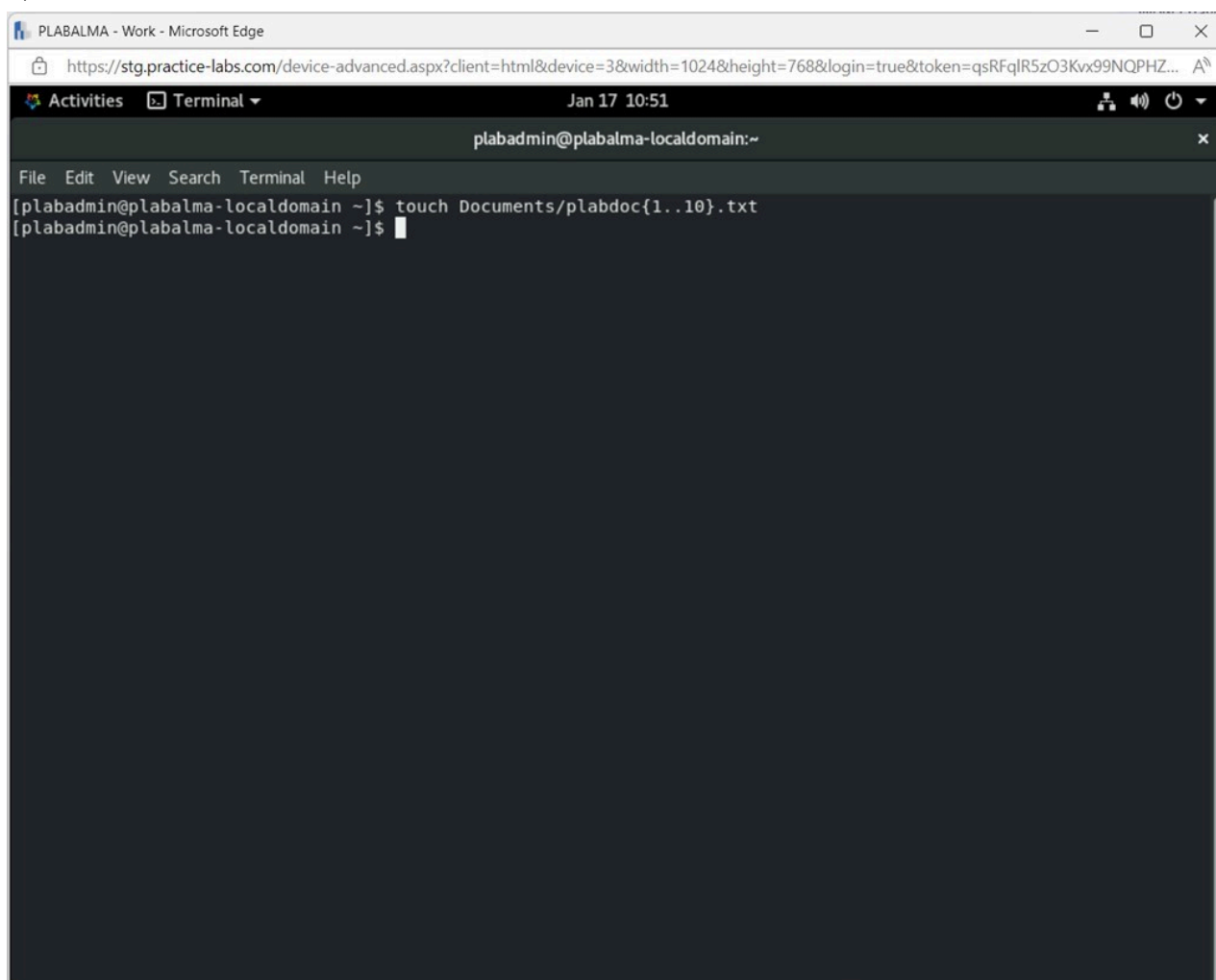


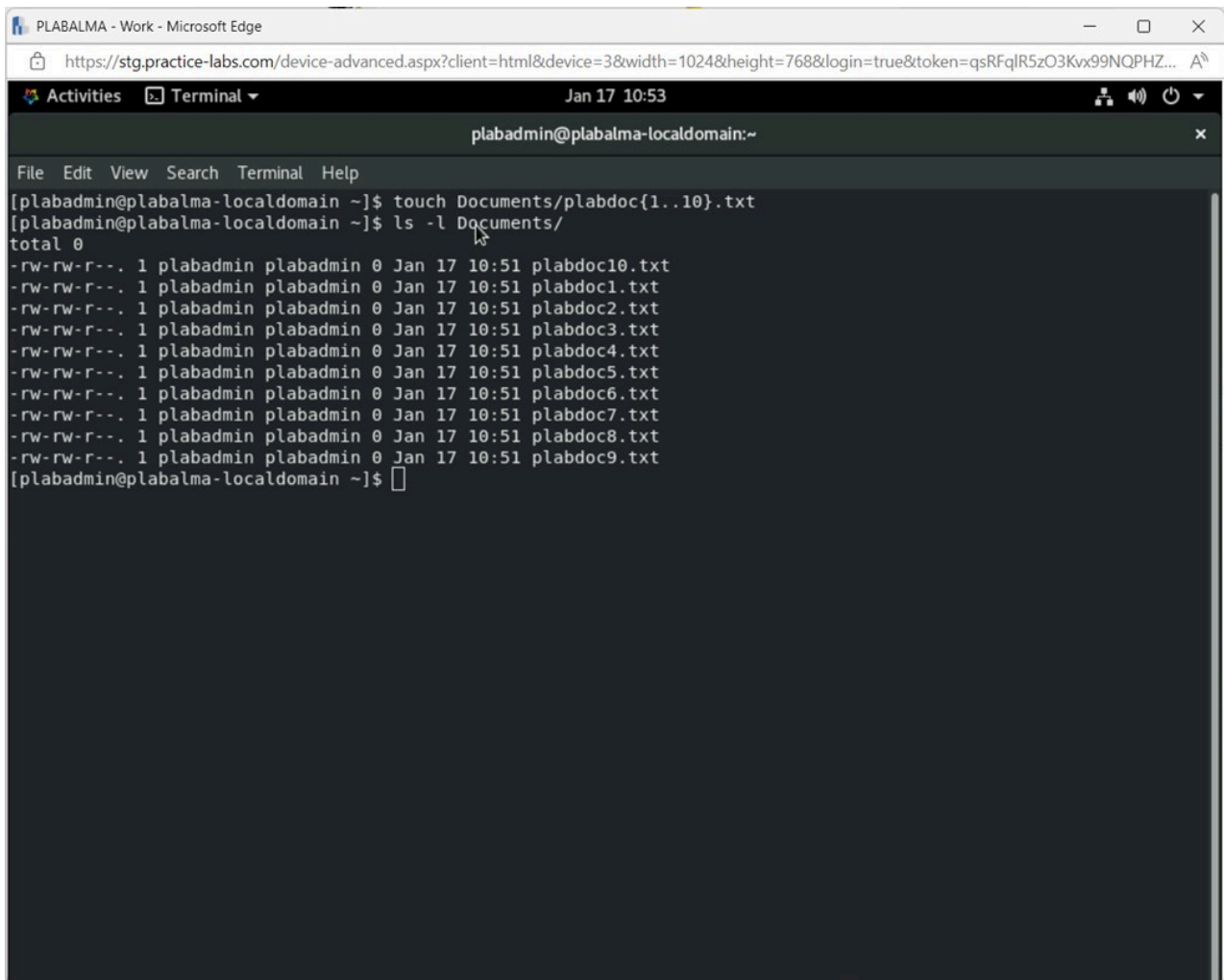
Figure 1.3 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window.

***Note:** Executing the **touch** command will create files that will be backed up later using a script.*

Step 4

In the **Terminal** window, type the following and press **Enter**:

```
ls -l Documents/
```



The screenshot shows a web browser window titled 'PLABALMA - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=38&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...'. The browser's 'Activities' and 'Terminal' tabs are visible. The terminal window, titled 'plabadmin@plabalma-localdomain:~', shows the following commands and output:

```
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc10.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$
```

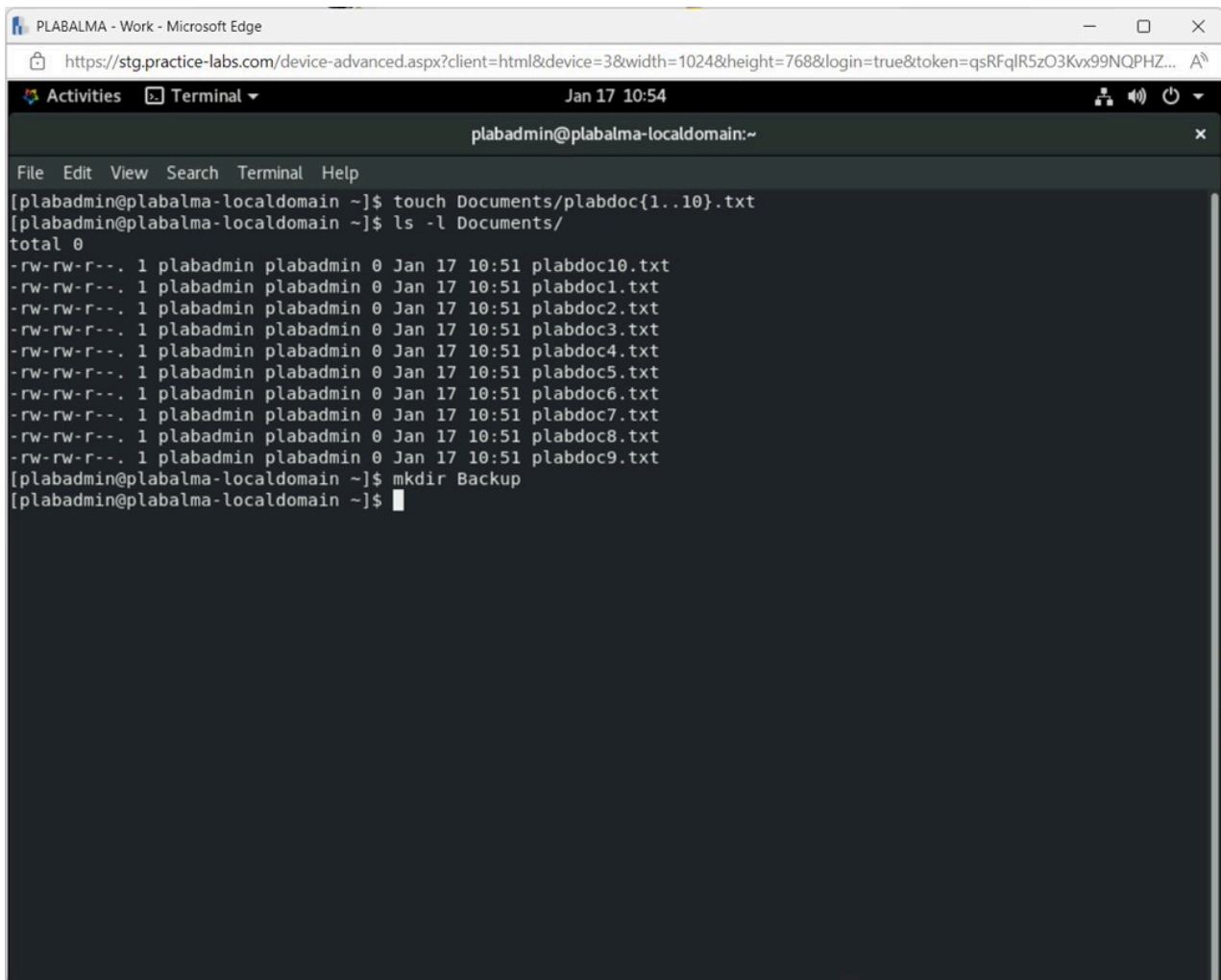
Figure 1.4 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window.

Note: Executing the command will create ten text files in the Documents folder.

Step 5

In the **Terminal** window, type the following and press **Enter**:

```
mkdir Backup
```



The screenshot shows a web browser window titled 'PLABALMA - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...'. Below the browser is a terminal window titled 'plabadmin@plabalma-localdomain:~'. The terminal shows the following commands and output:

```
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc10.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$
```

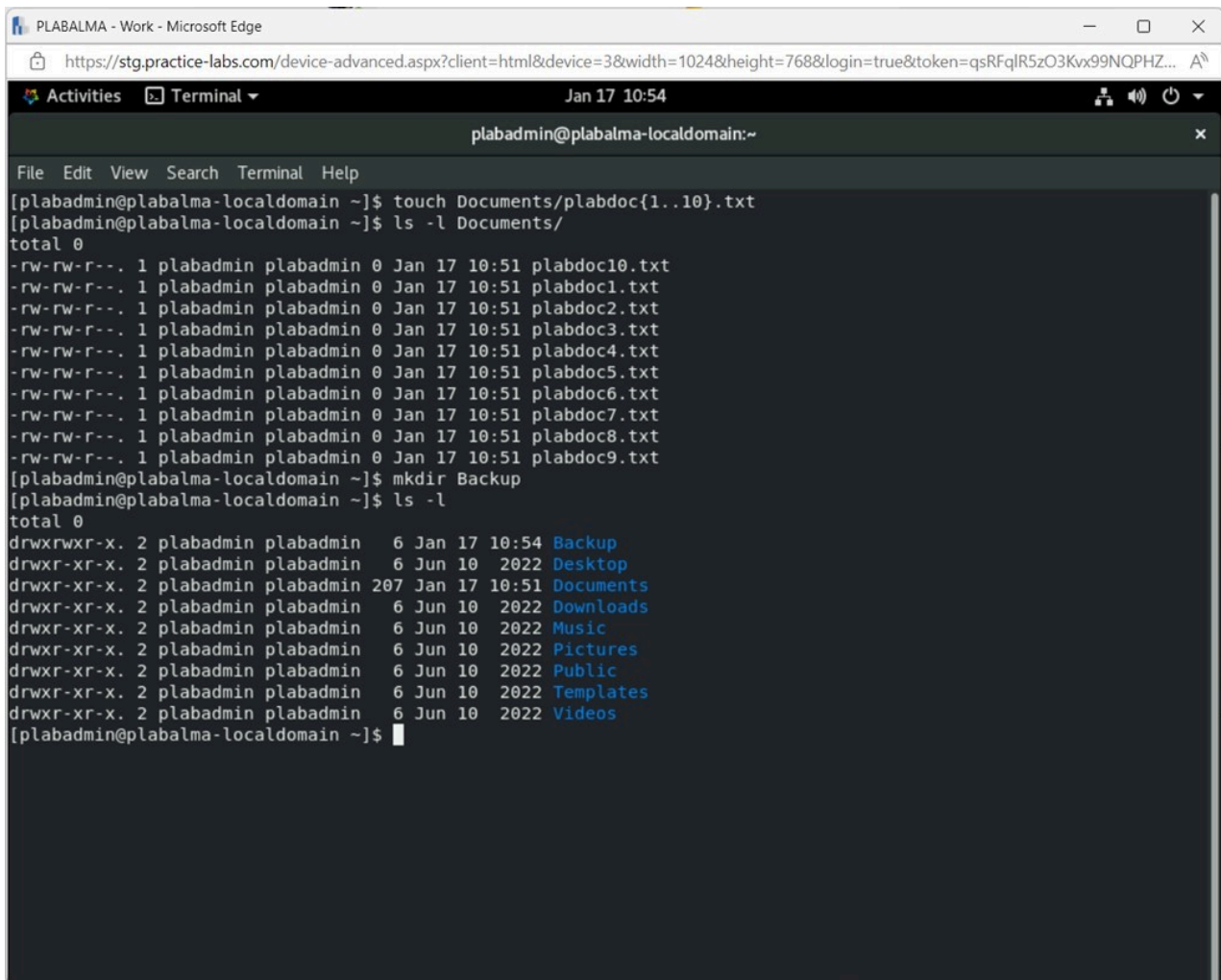
Figure 1.5 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window to create a backup directory.

Note: The above command will create a Backup directory.

Step 6

In the **Terminal** window, type the following and press **Enter**:

```
ls -l
```

The screenshot shows a web browser window titled 'PLABALMA - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqlR5zO3Kvx99NQPHZ...'. Below the browser window is a terminal window titled 'Jan 17 10:54' and 'plabadmin@plabalma-localdomain:~'. The terminal shows the following commands and output:

```
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc10.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$ ls -l
total 0
drwxrwxr-x. 2 plabadmin plabadmin  6 Jan 17 10:54 Backup
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Desktop
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Downloads
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Music
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Pictures
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Public
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Templates
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Videos
[plabadmin@plabalma-localdomain ~]$
```

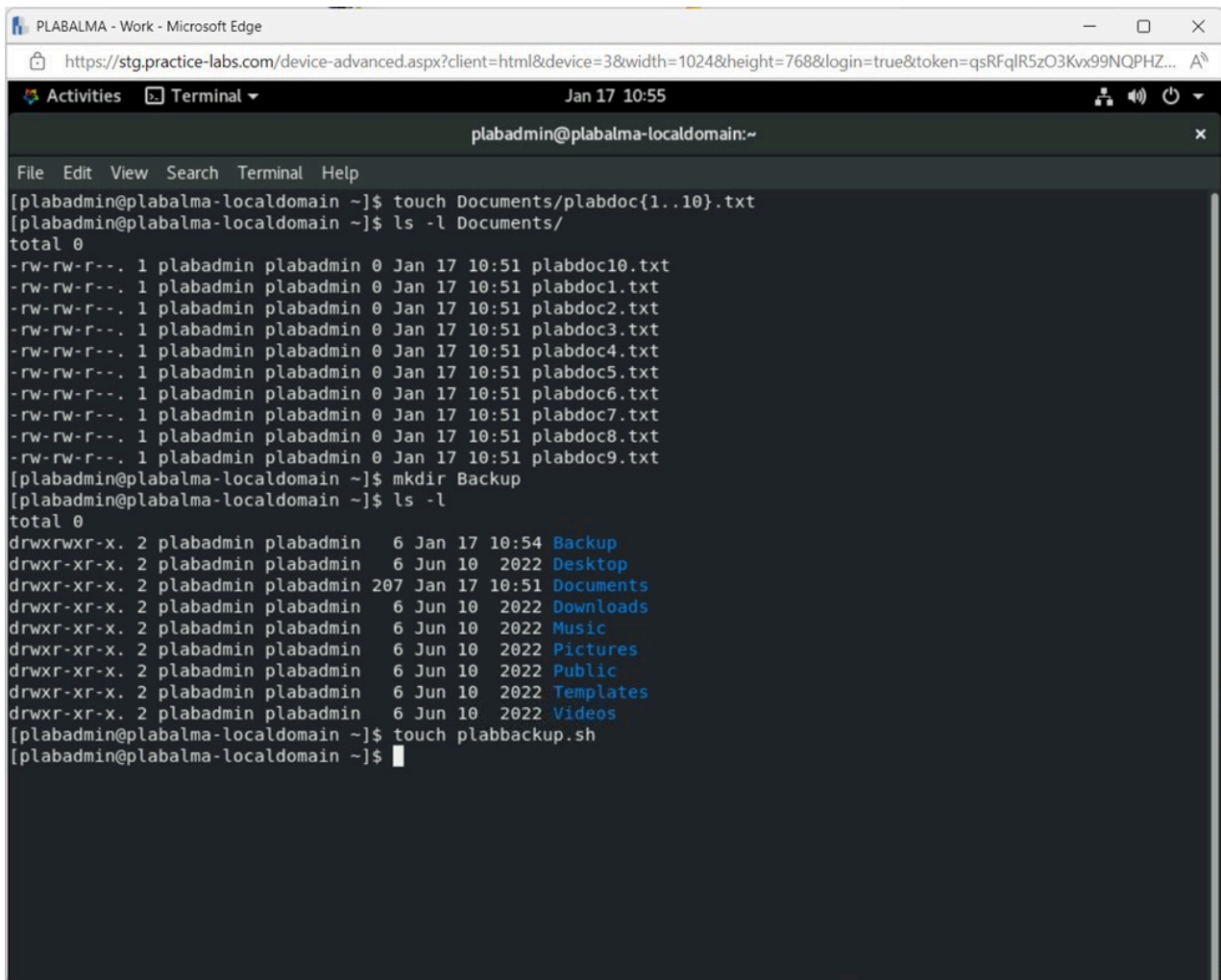
Figure 1.6 Screenshot of PLABALMA: Displaying executing the required command to list the user's home folder content.

Note: Executing the command will list the user's home folder content.

Step 7

In the **Terminal** window, type the following and press **Enter**:

```
touch plabbackup.sh
```



```
PLABALMA - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...
Activities Terminal Jan 17 10:55
plabadmin@plabalma-localdomain:~
File Edit View Search Terminal Help
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc10.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$ ls -l
total 0
drwxrwxr-x. 2 plabadmin plabadmin 6 Jan 17 10:54 Backup
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Desktop
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Downloads
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Music
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Pictures
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Public
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Templates
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Videos
[plabadmin@plabalma-localdomain ~]$ touch plabbackup.sh
[plabadmin@plabalma-localdomain ~]$
```

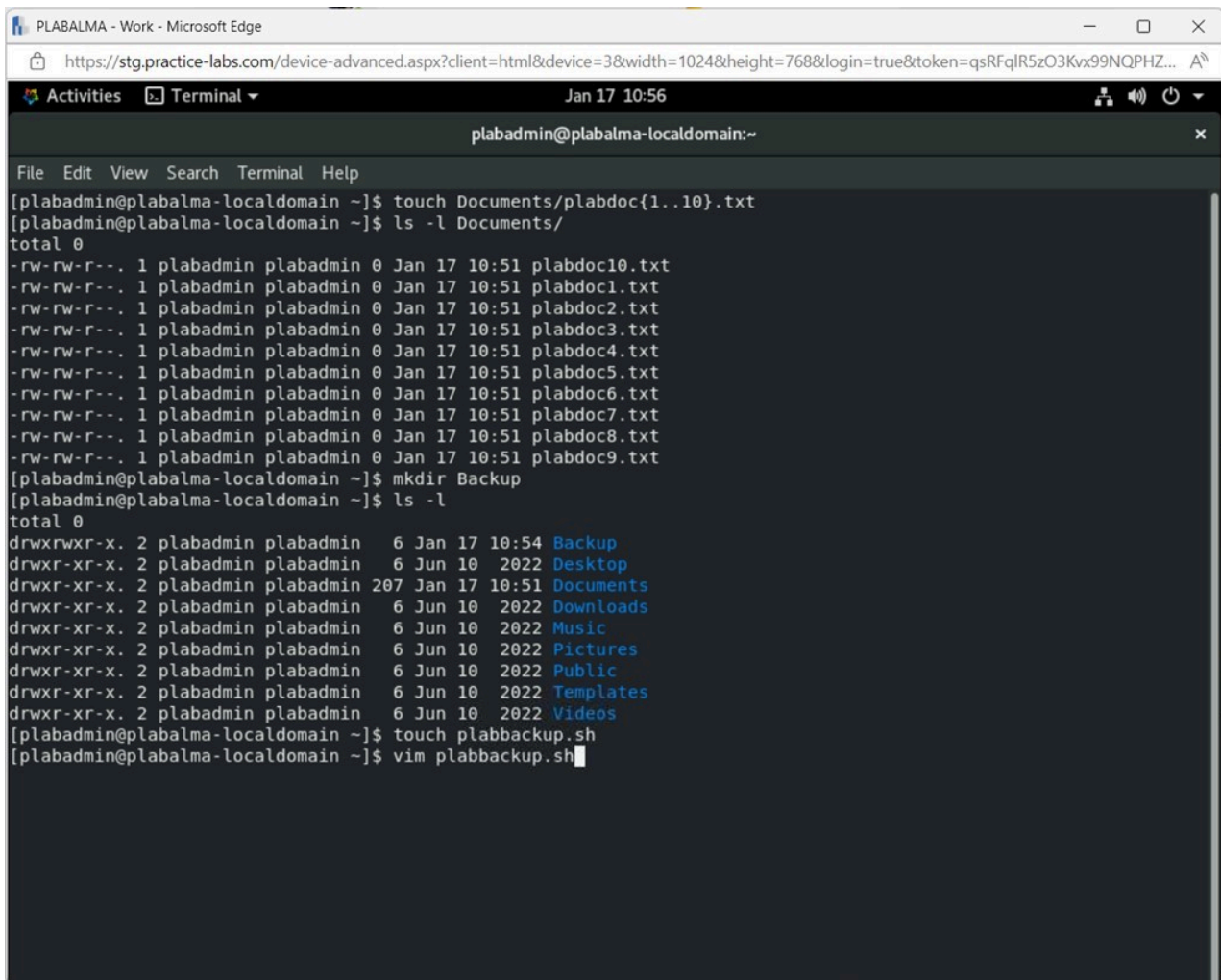
Figure 1.7 Screenshot of PLABALMA: Displaying executing the required command to create a file for the script.

Note: Executing the **touch** command will create a file that will be used to create a backup script.

Step 8

In the **Terminal** window, type the following and press **Enter**:

```
vim plabbackup.sh
```



The screenshot shows a web browser window titled "PLABALMA - Work - Microsoft Edge" with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqlR5zO3Kvx99NQPHZ...>. The browser's address bar shows the URL. Below the browser window, a terminal window is open, displaying the following commands and output:

```
plabadmin@plabalma-localdomain:~  
File Edit View Search Terminal Help  
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt  
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/  
total 0  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc1.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc2.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc3.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc4.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc5.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc6.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc7.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc8.txt  
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 10:51 plabdoc9.txt  
[plabadmin@plabalma-localdomain ~]$ mkdir Backup  
[plabadmin@plabalma-localdomain ~]$ ls -l  
total 0  
drwxrwxr-x. 2 plabadmin plabadmin 6 Jan 17 10:54 Backup  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Desktop  
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Downloads  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Music  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Pictures  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Public  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Templates  
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Videos  
[plabadmin@plabalma-localdomain ~]$ touch plabbackup.sh  
[plabadmin@plabalma-localdomain ~]$ vim plabbackup.sh
```

Figure 1.8 Screenshot of PLABALMA: Displaying executing the vim command in the Terminal window.

Step 9

In the **Terminal** window, press the “i” key.

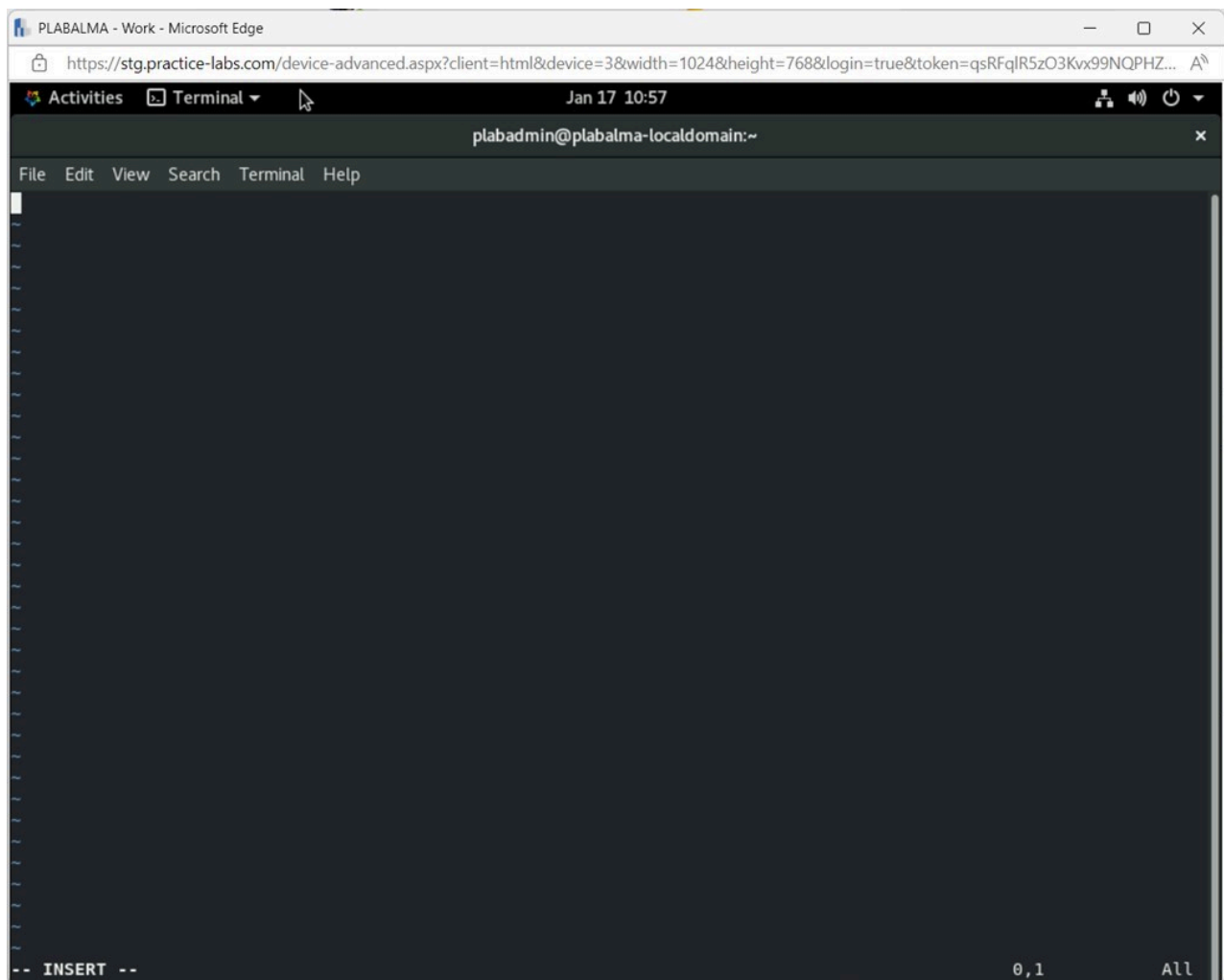


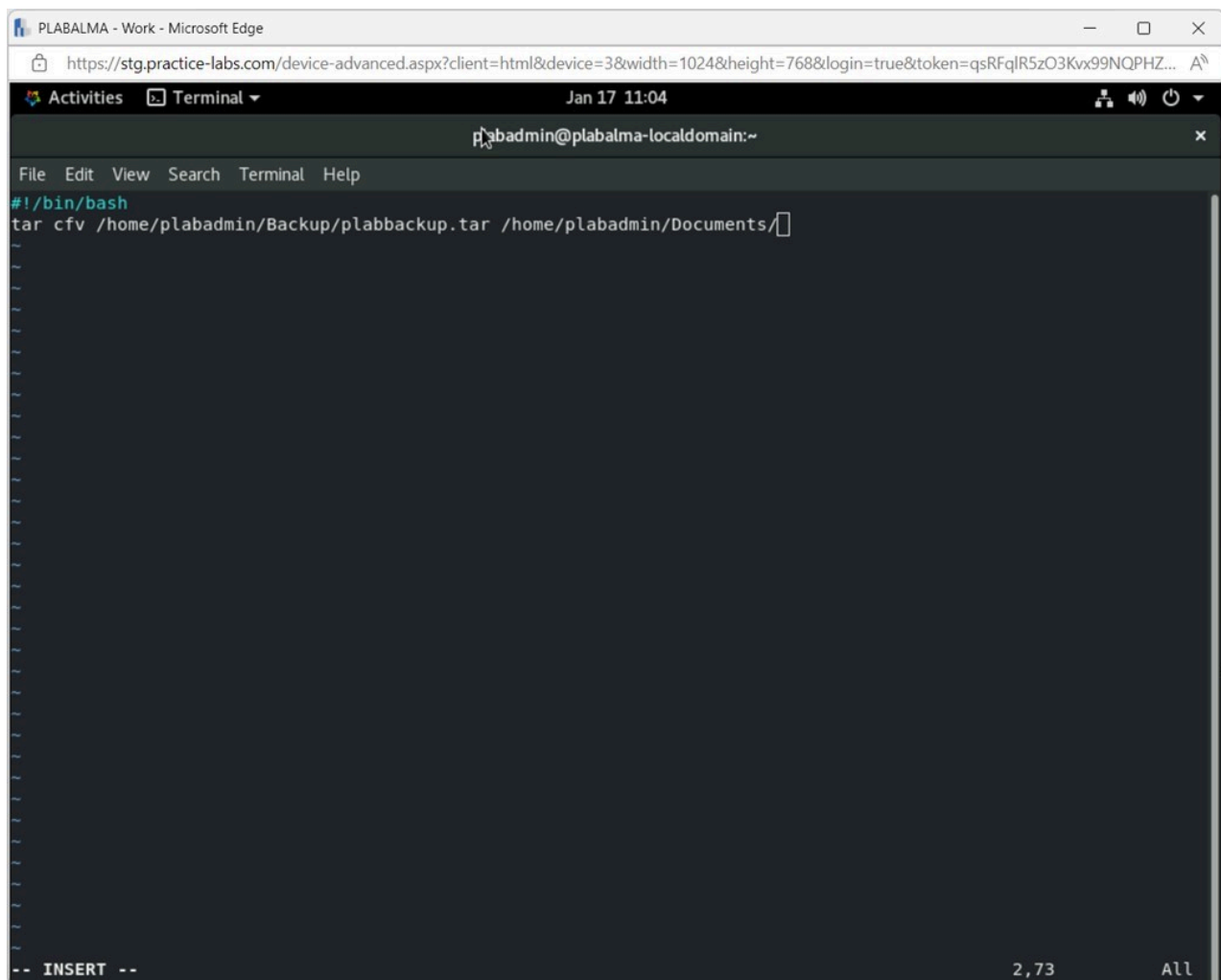
Figure 1.9 Screenshot of PLABALMA: Displaying pressing the “i” key in the Terminal window.

Note: Pressing the “i” key enables editing the file using the vim application.

Step 10

In the **Terminal** window, type the following and press **Enter**:

```
#!/bin/bash
tar cfv /home/plabadmin/Backup/plabbackup.tar
/home/plabadmin/Documents/
```



The screenshot shows a web browser window titled 'PLABALMA - Work - Microsoft Edge' with the URL <https://stg.practice-labs.com/device-advanced.aspx?client=html&device=38&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...>. Below the browser is a terminal window titled 'Jan 17 11:04' and 'plabadmin@plabalma-localdomain:~'. The terminal has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The prompt is '#!/bin/bash' and the command 'tar cfv /home/plabadmin/Backup/plabbackup.tar /home/plabadmin/Documents/' is being entered. The terminal status bar at the bottom shows '-- INSERT --', '2,73', and 'All'.

Figure 1.10 Screenshot of PLABALMA: Displaying creating a script file using the vim application in the Terminal window.

Note: The vim application is used to create a basic script file that will back up the documents in the Documents folder to the backup directory.

Step 11

In the **vim** application, press the **Esc** key.

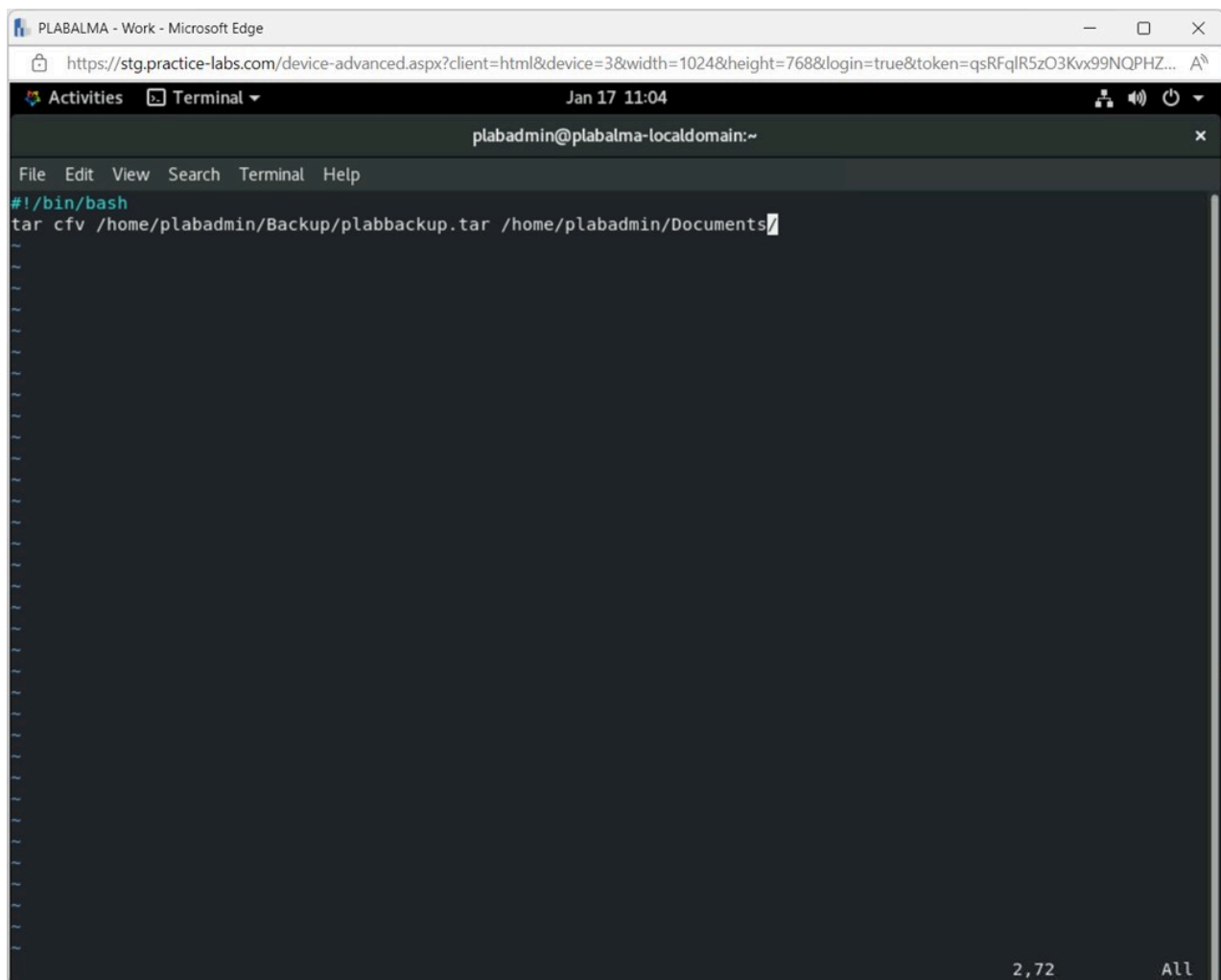


Figure 1.11 Screenshot of PLABALMA: Displaying saving the script file.

Step 12

In the **Vim** application, type the following and press **Enter**:

```
:wq
```

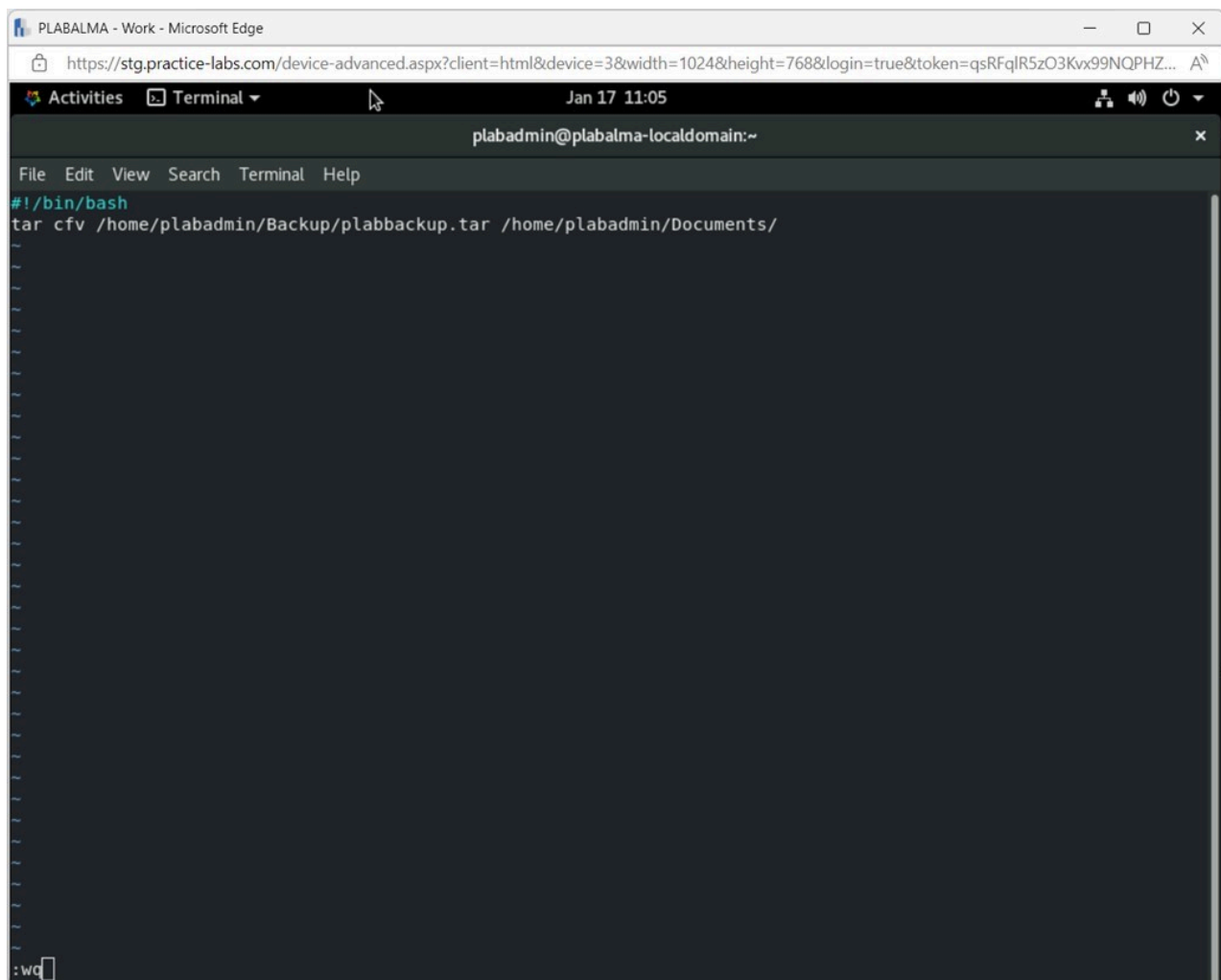
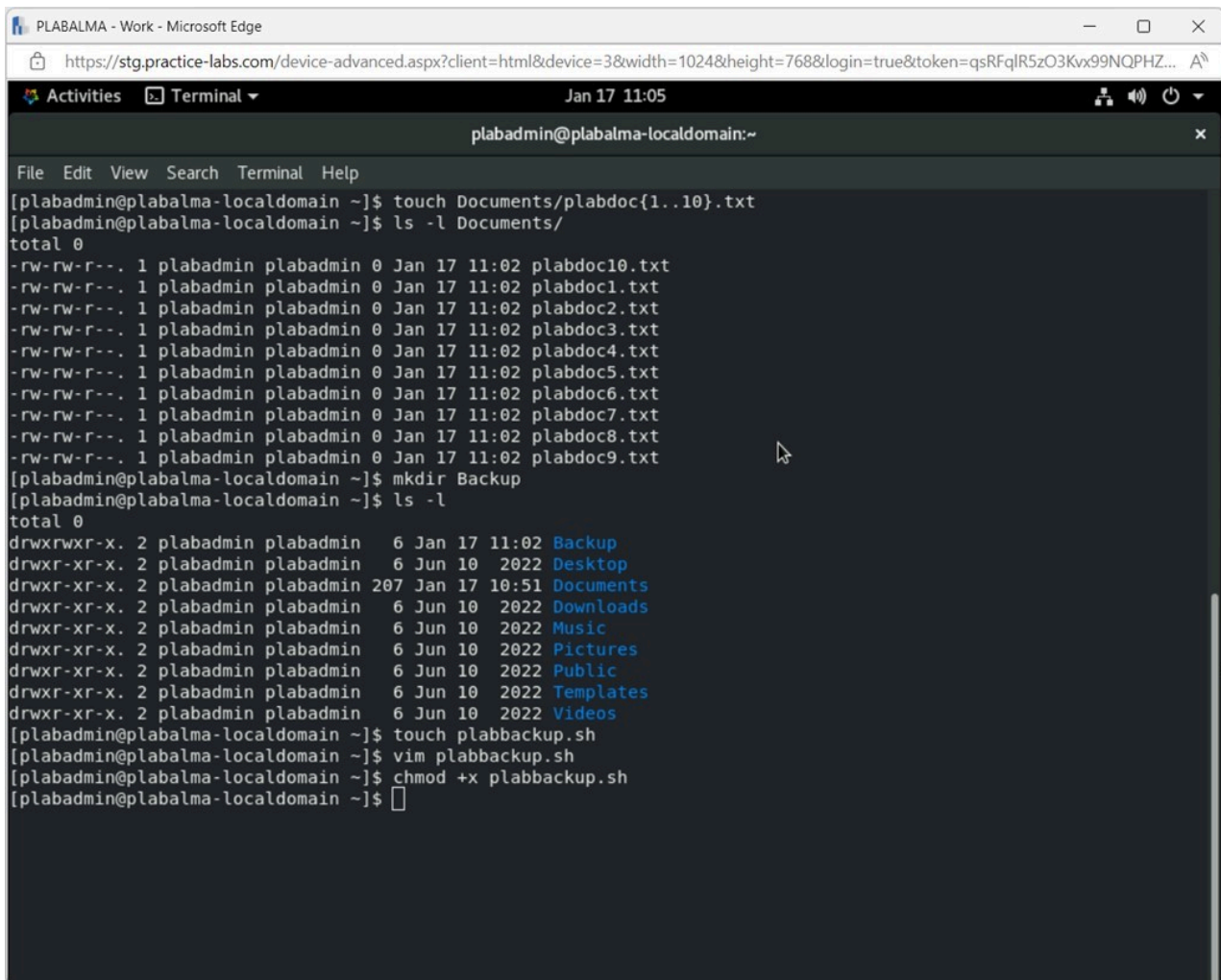


Figure 1.12 Screenshot of PLABALMA: Displaying exiting out of the vim application.

Step 13

In the **Terminal** window, type the following and press **Enter**:

```
chmod +x plabbackup.sh
```

The screenshot shows a web browser window titled 'PLABALMA - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...'. Below the browser is a terminal window titled 'Jan 17 11:05' and 'plabadmin@plabalma-localdomain:~'. The terminal shows the following commands and output:

```
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$ ls -l
total 0
drwxrwxr-x. 2 plabadmin plabadmin  6 Jan 17 11:02 Backup
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Desktop
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Downloads
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Music
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Pictures
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Public
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Templates
drwxr-xr-x. 2 plabadmin plabadmin  6 Jun 10 2022 Videos
[plabadmin@plabalma-localdomain ~]$ touch plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ vim plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ chmod +x plabbackup.sh
[plabadmin@plabalma-localdomain ~]$
```

Figure 1.13 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window.

Note: The **chmod** command is to be used with the **+x** parameter to make the script file executable.

Step 14

In the **Terminal** window, type the following and press **Enter**:

```
ls -l plabbackup.sh
```



```

PLABALMA - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...
Activities Terminal Jan 17 11:06
plabadmin@plabalma-localdomain:~
File Edit View Search Terminal Help
[plabadmin@plabalma-localdomain ~]$ touch Documents/plabdoc{1..10}.txt
[plabadmin@plabalma-localdomain ~]$ ls -l Documents/
total 0
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc10.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc1.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc2.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc3.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$ ls -l
total 0
drwxrwxr-x. 2 plabadmin plabadmin 6 Jan 17 11:02 Backup
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Desktop
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Downloads
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Music
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Pictures
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Public
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Templates
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Videos
[plabadmin@plabalma-localdomain ~]$ touch plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ vim plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ chmod +x plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ ls -l plabbackup.sh
-rwxrwxr-x. 1 plabadmin plabadmin 85 Jan 17 11:05 plabbackup.sh
[plabadmin@plabalma-localdomain ~]$

```

Figure 1.14 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window to display the properties of the created script file.

Note: The command displays the properties of the created script file.

Step 15

In the **Terminal** window, type the following and press **Enter**:

```
./plabbackup.sh
```

```

PLABALMA - Work - Microsoft Edge
https://stg.practice-labs.com/device-advanced.aspx?client=html&device=3&width=1024&height=768&login=true&token=qsRFqIR5zO3Kvx99NQPHZ...
Activities Terminal Jan 17 11:08
plabadmin@plabalma-localdomain:~
File Edit View Search Terminal Help
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc4.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc5.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc6.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc7.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc8.txt
-rw-rw-r--. 1 plabadmin plabadmin 0 Jan 17 11:02 plabdoc9.txt
[plabadmin@plabalma-localdomain ~]$ mkdir Backup
[plabadmin@plabalma-localdomain ~]$ ls -l
total 0
drwxrwxr-x. 2 plabadmin plabadmin 6 Jan 17 11:02 Backup
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Desktop
drwxr-xr-x. 2 plabadmin plabadmin 207 Jan 17 10:51 Documents
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Downloads
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Music
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Pictures
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Public
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Templates
drwxr-xr-x. 2 plabadmin plabadmin 6 Jun 10 2022 Videos
[plabadmin@plabalma-localdomain ~]$ touch plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ vim plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ chmod +x plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ ls -l plabbackup.sh
-rwxrwxr-x. 1 plabadmin plabadmin 85 Jan 17 11:05 plabbackup.sh
[plabadmin@plabalma-localdomain ~]$ ./plabbackup.sh
tar: Removing leading '/' from member names
/home/plabadmin/Documents/
/home/plabadmin/Documents/plabdoc1.txt
/home/plabadmin/Documents/plabdoc2.txt
/home/plabadmin/Documents/plabdoc3.txt
/home/plabadmin/Documents/plabdoc4.txt
/home/plabadmin/Documents/plabdoc5.txt
/home/plabadmin/Documents/plabdoc6.txt
/home/plabadmin/Documents/plabdoc7.txt
/home/plabadmin/Documents/plabdoc8.txt
/home/plabadmin/Documents/plabdoc9.txt
/home/plabadmin/Documents/plabdoc10.txt
[plabadmin@plabalma-localdomain ~]$

```

Figure 1.15 Screenshot of PLABALMA: Displaying executing the required command in the bash Konsole window.

Note: The script that was created will create a backup of the user's Documents folder to the Backup folder. The script can be automated to run at a specific time.

Close the **Terminal** window.

Task 2 - Create a Basic Powershell Script

Windows Powershell can be used to create scripts to automate specific administrative tasks, for example, mapping a shared network drive. Creating scripts can reduce the administrative effort needed to complete repetitive tasks.

In this task, a basic Powershell script will be created.

Step 1

Connect to **PLABDC01**.

Click the **File Explorer** icon on the **Taskbar**.

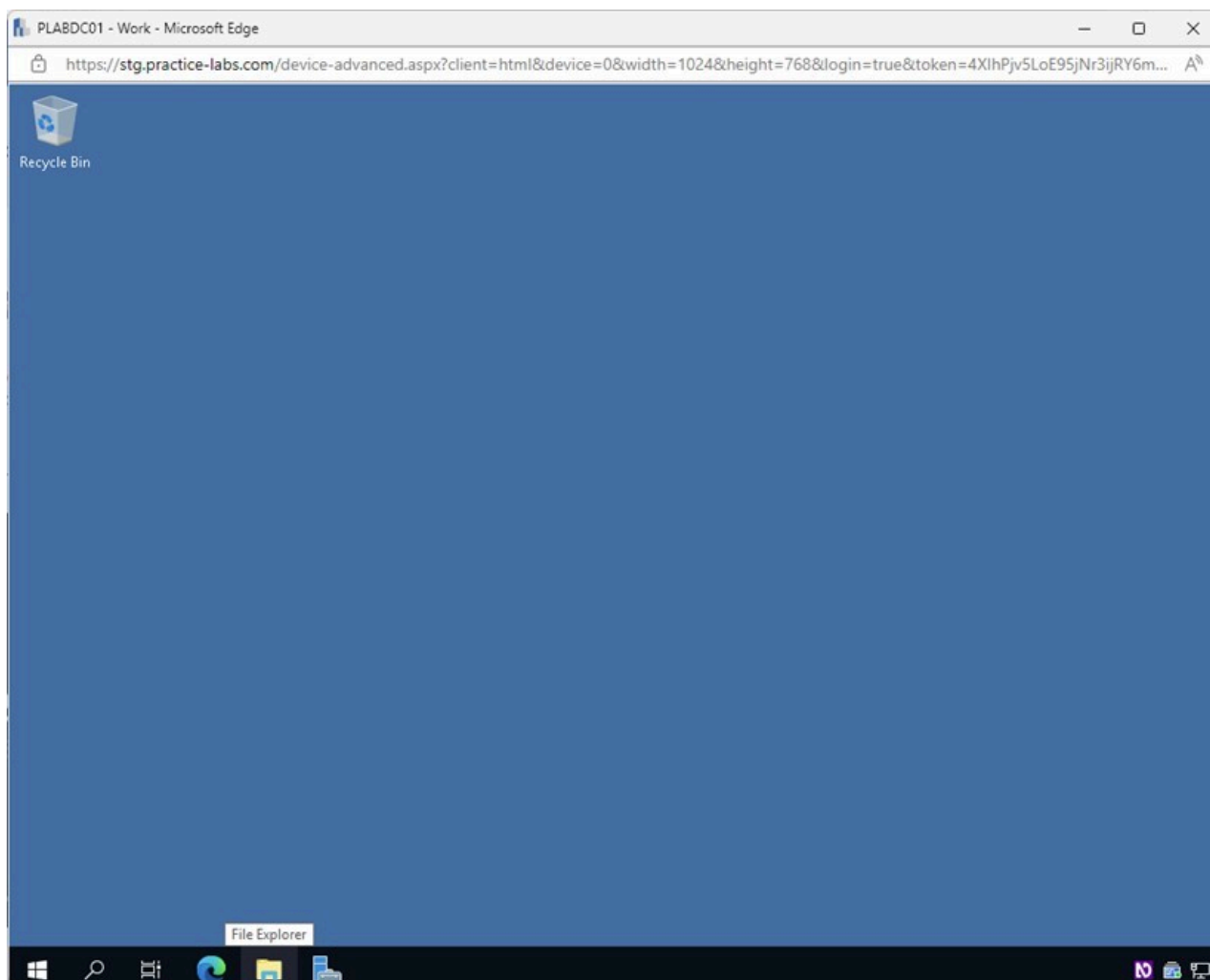


Figure 1.16 Screenshot of PLABDC01: Displaying opening File Explorer from the Taskbar.

Step 2

In the **File Explorer** window, expand **This PC** and select **Local Disk (C:)**.

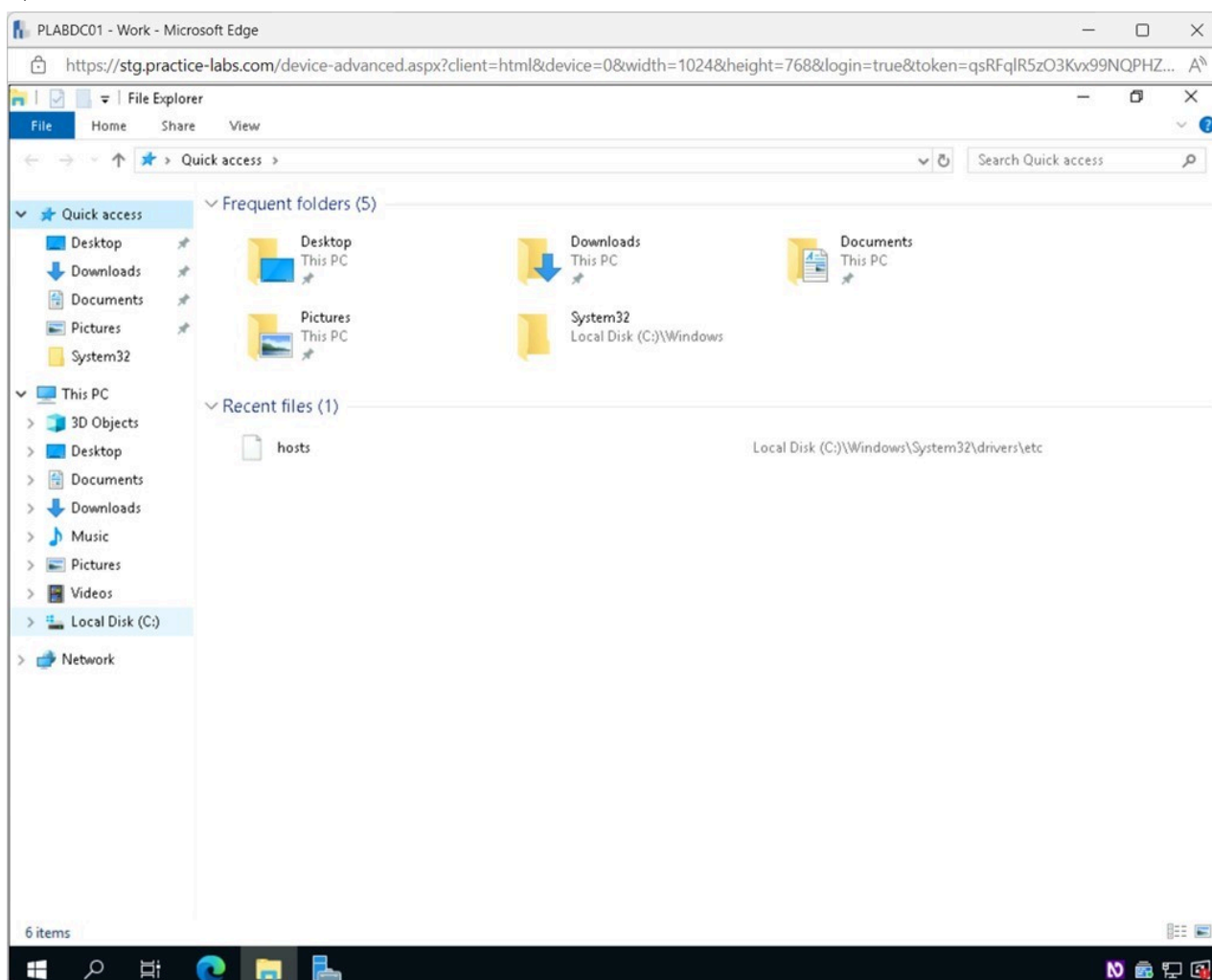


Figure 1.17 Screenshot of PLABDC01: Displaying selecting Local Disk (C:) in the File Explorer window.

Step 3

Right-click in the right pane and select **New > Folder**.

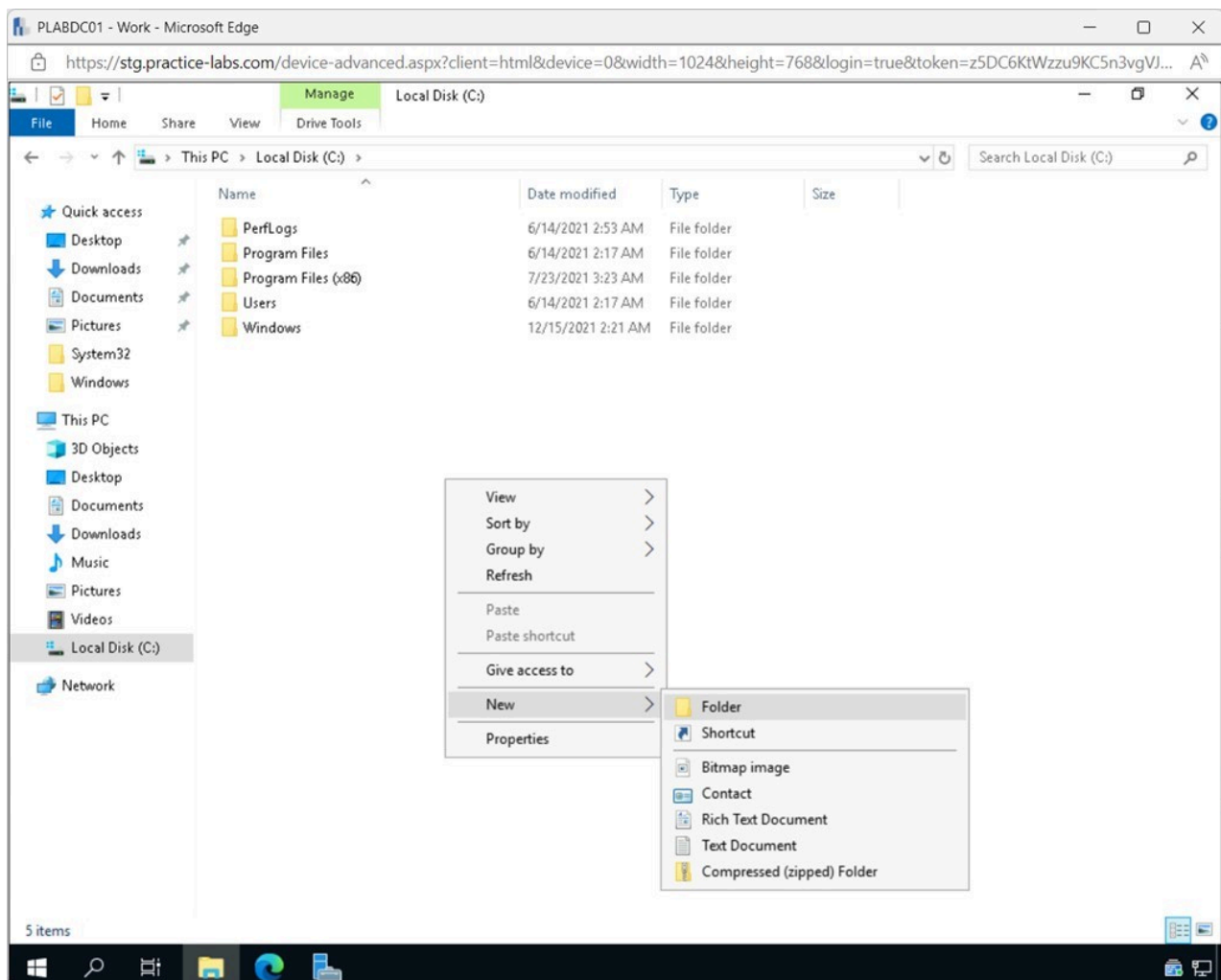


Figure 1.18 Screenshot of PLABDC01: Displaying right-clicking in the right pane and selecting New > Folder.

Step 4

Rename the **New folder** to the following, and press **Enter**:

Marketing

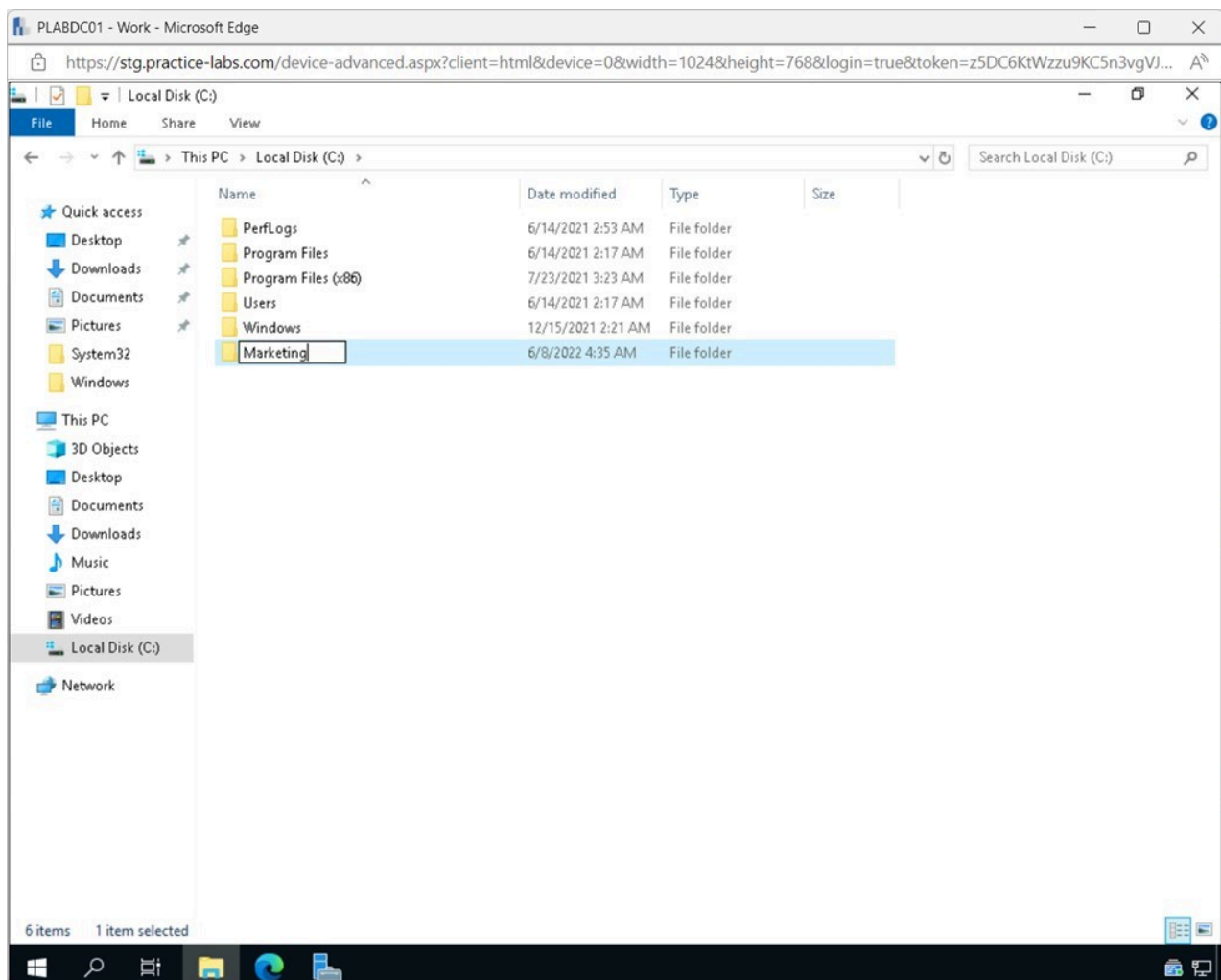
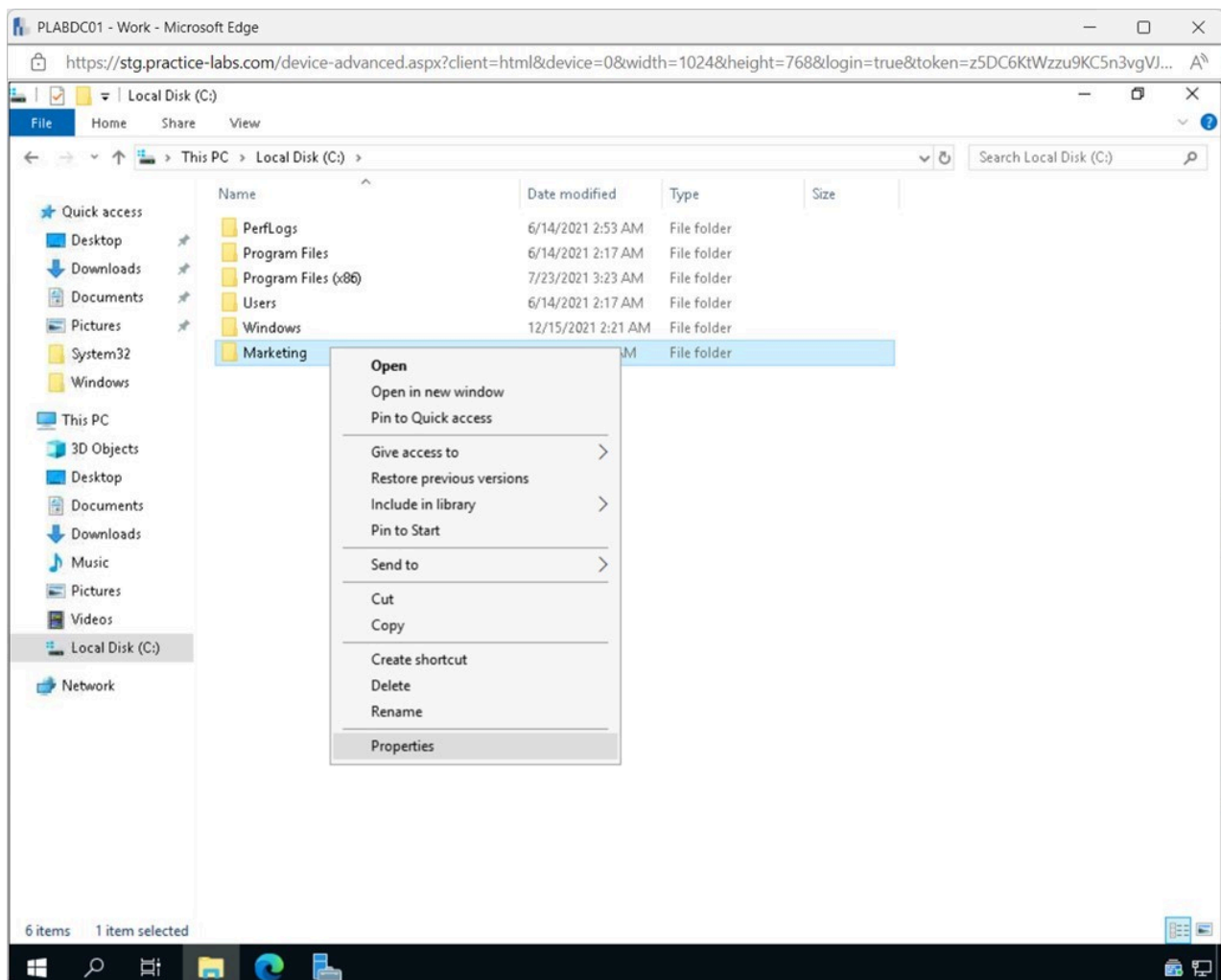


Figure 1.19 Screenshot of PLABDC01: Displaying creating a new folder called Marketing in File Explorer.

Step 5

Right-click on the **Marketing** folder and select **Properties**.



Step 1.20 Screenshot of PLABDC01: Displaying right-clicking on the Marketing folder and selecting Properties.

Step 6

On the **Marketing Properties** window, select the **Sharing** tab.

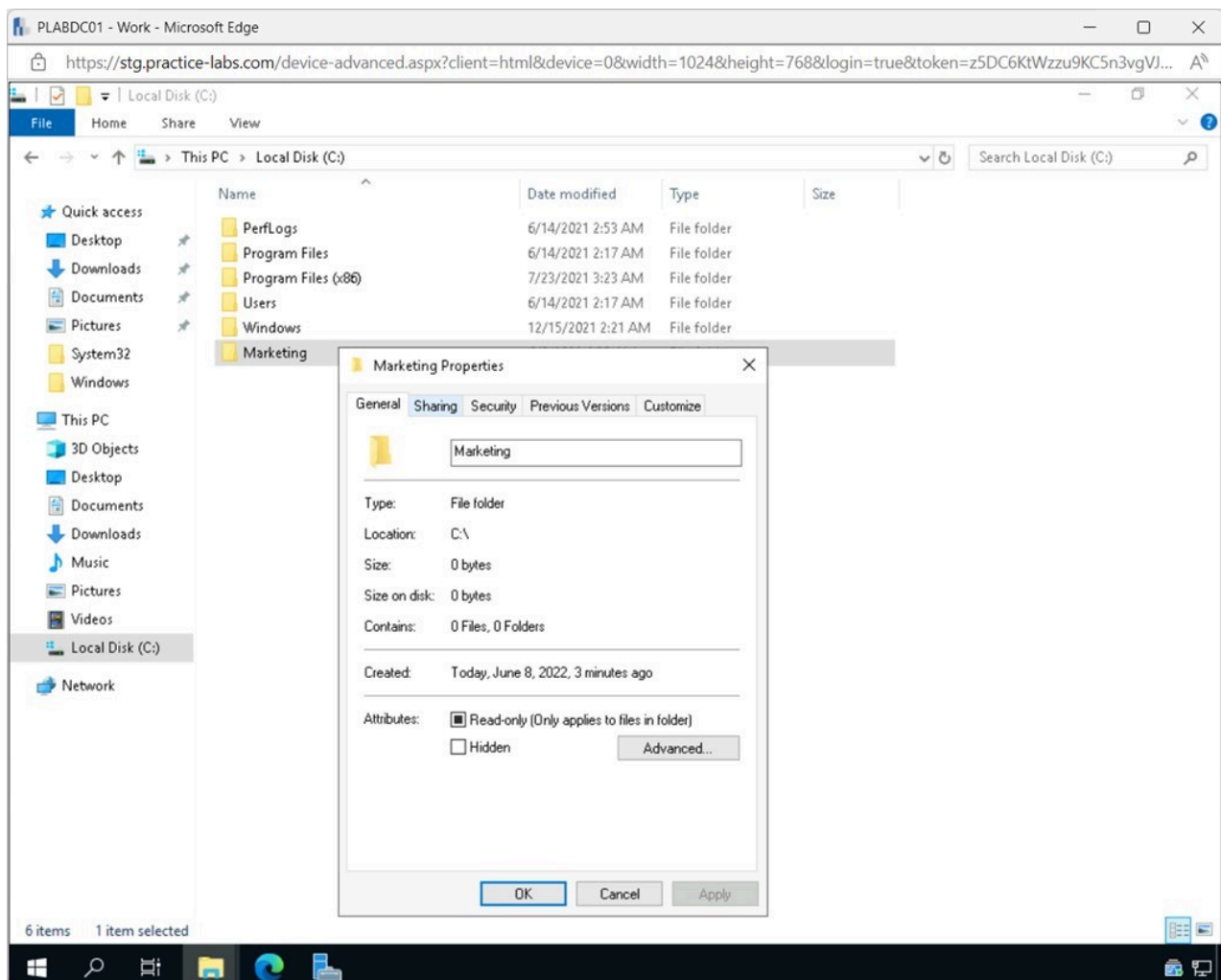


Figure 1.21 Screenshot of PLABDC01: Displaying selecting the Sharing tab on the Marketing Properties window.

Step 7

On the **Sharing** tab, select **Advanced Sharing**.

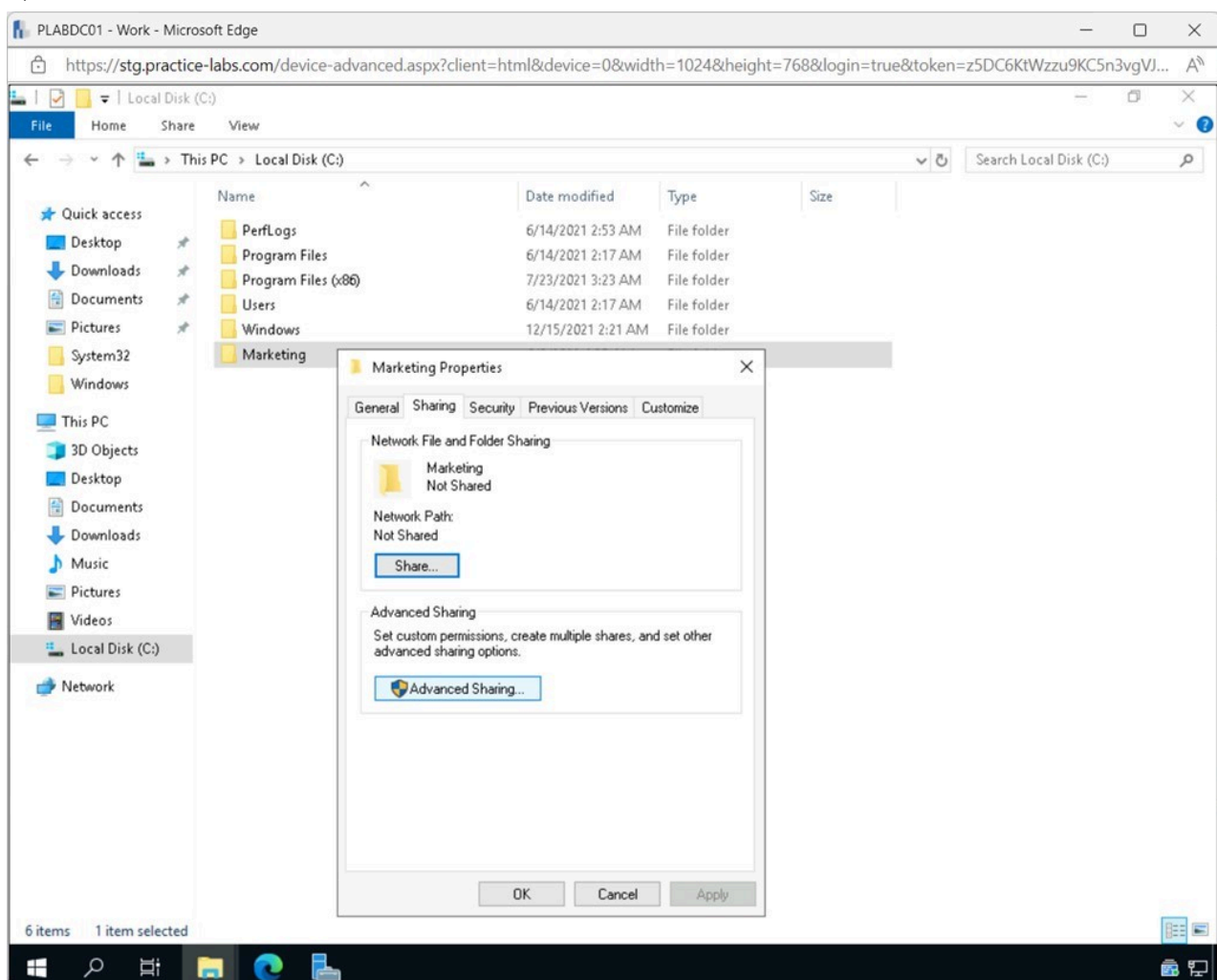


Figure 1.22 Screenshot of PLABDC01: Displaying selecting Advanced Sharing on the Sharing tab.

Step 8

In the **Advanced Sharing** window, tick the **Share this folder** checkbox and click **OK**.

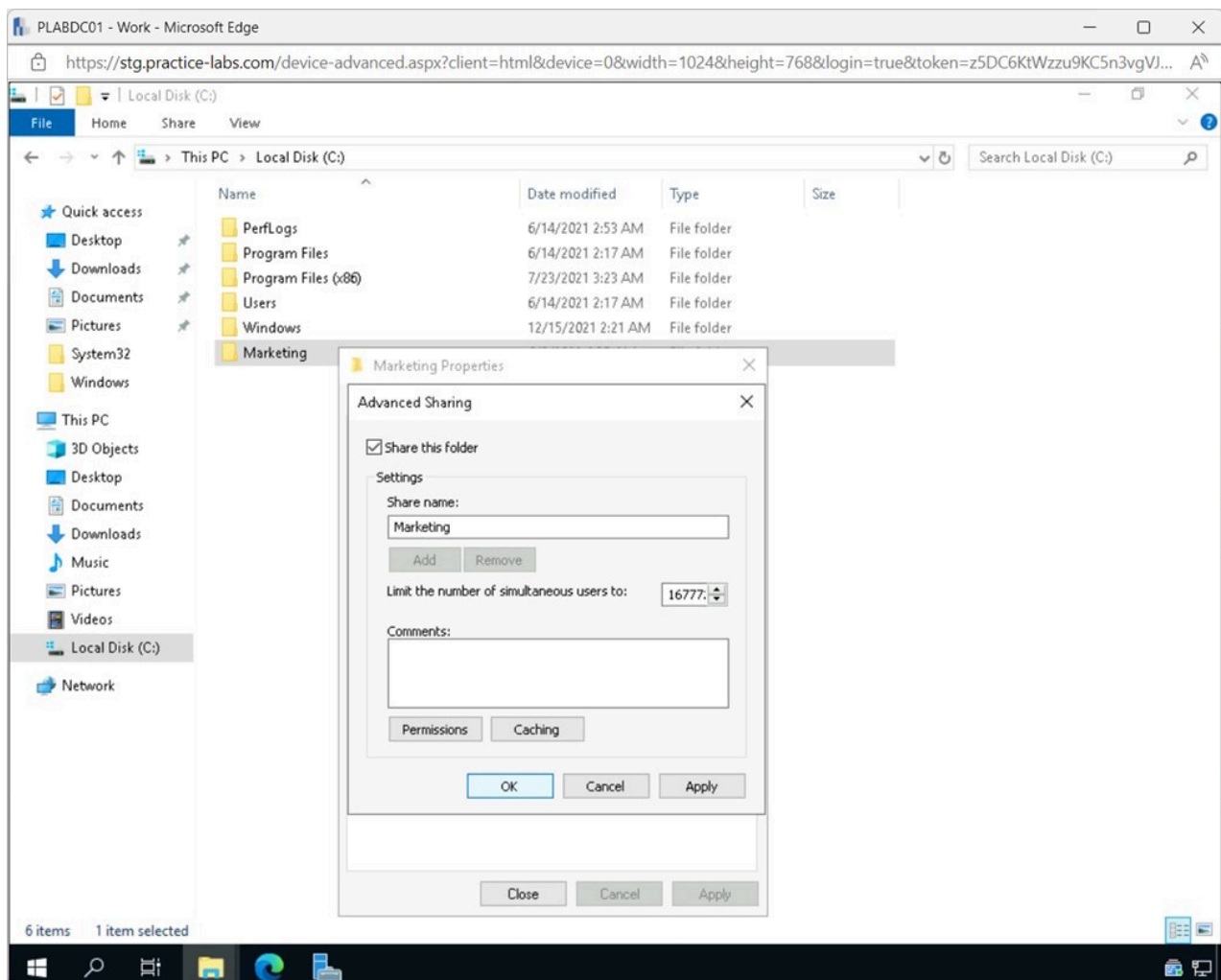


Figure 1.23 Screenshot of PLABDC01: Displaying ticking the Share this folder checkbox and clicking OK in the Advanced Sharing window.

Step 9

Click **Close** on the **Marketing Properties** window.

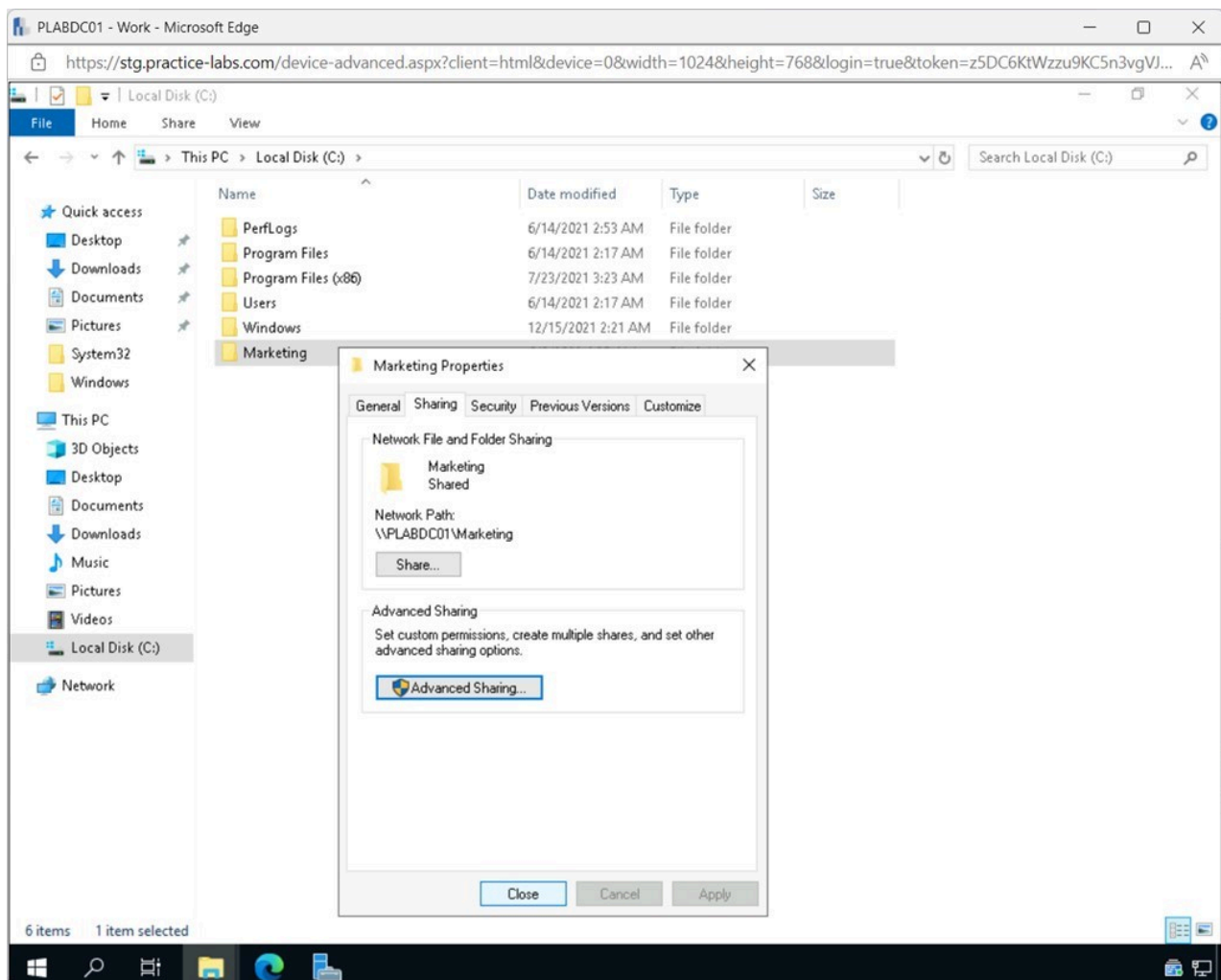


Figure 1.24 Screenshot of PLABDC01: Displaying selecting Close in the Marketing Properties window.

Step 10

Connect to **PLABWIN10**.

Right-click the **Start** charm and select **Windows Powershell (Admin)**.

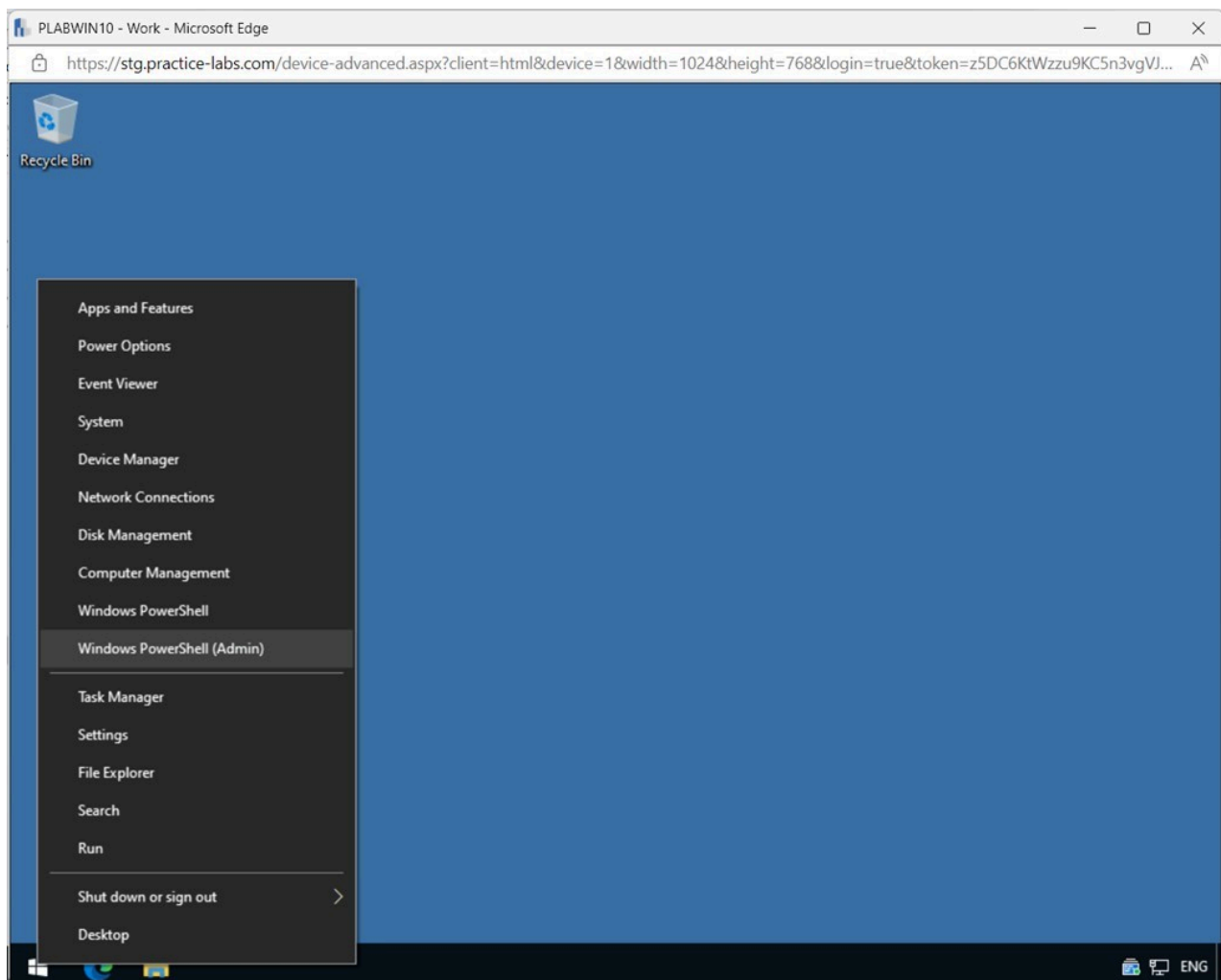


Figure 1.25 Screenshot of PLABWIN10: Displaying right-clicking Start and selecting Windows Powershell (Admin).

Step 11

In the **Administrator: Windows PowerShell** window, type the following and press **Enter**:

```
notepad.exe mapdrive.ps1
```

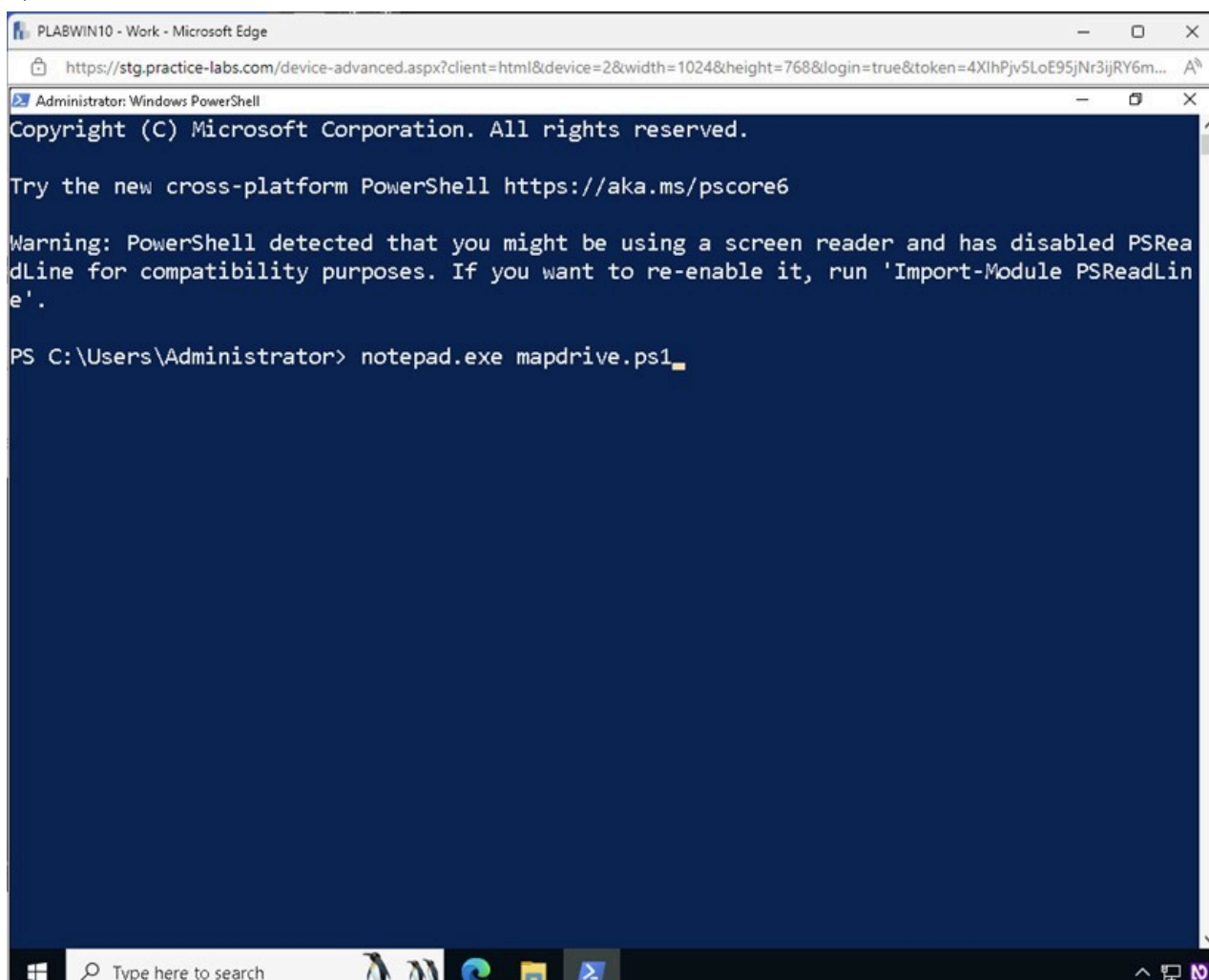


Figure 1.26 Screenshot of PLABWIN10: Displaying opening Notepad from Powershell.

Step 12

On the **Notepad** pop-up window, select **Yes**.

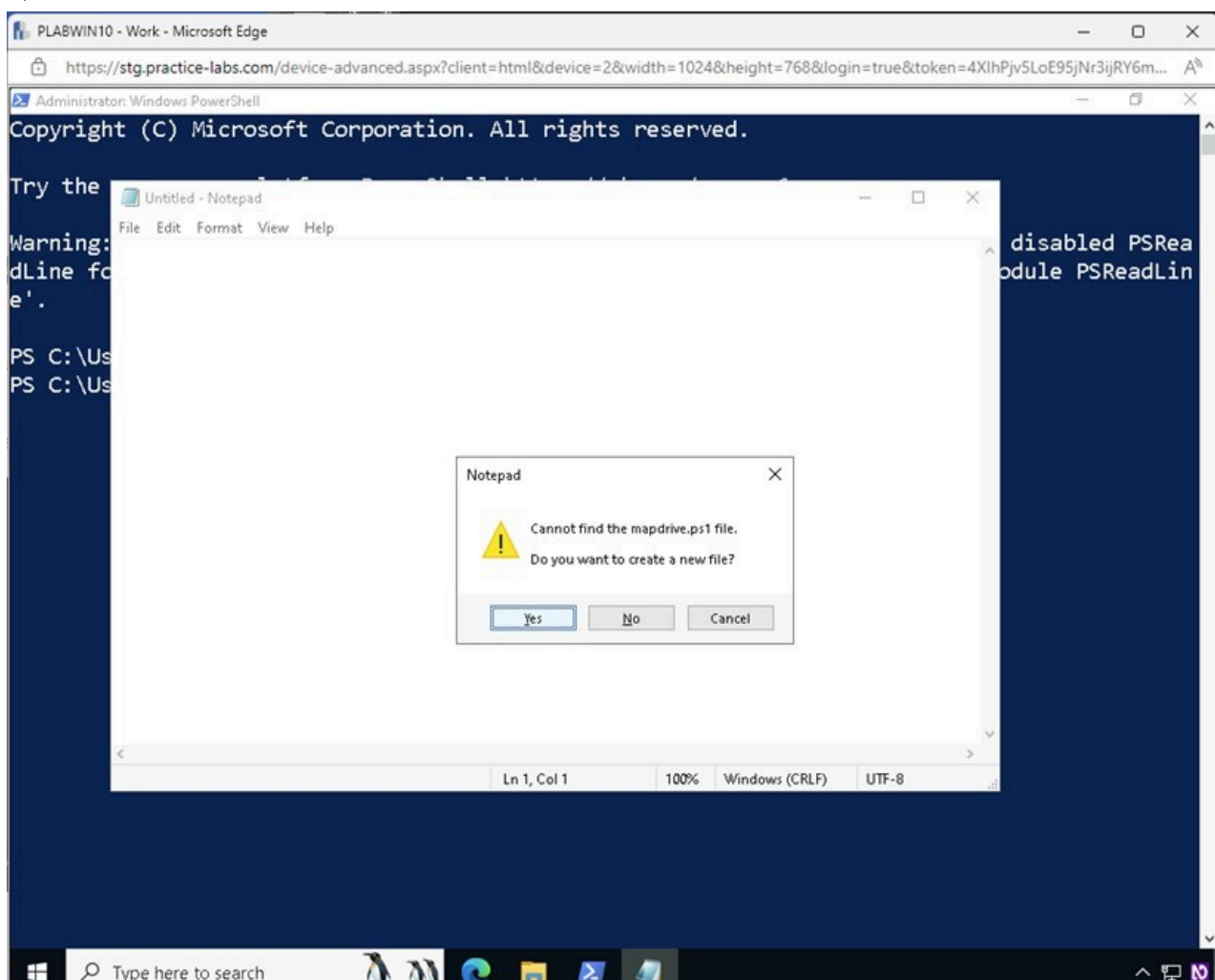


Figure 1.27 Screenshot of PLABWIN10: Displaying selecting Yes on the Notepad pop-up window.

Step 13

In the **mapdrive - Notepad** window, type the following:

```
New-PSDrive -Name K -PSProvider FileSystem -Root  
\\plabdc01\Marketing -Persist -Scope Global
```

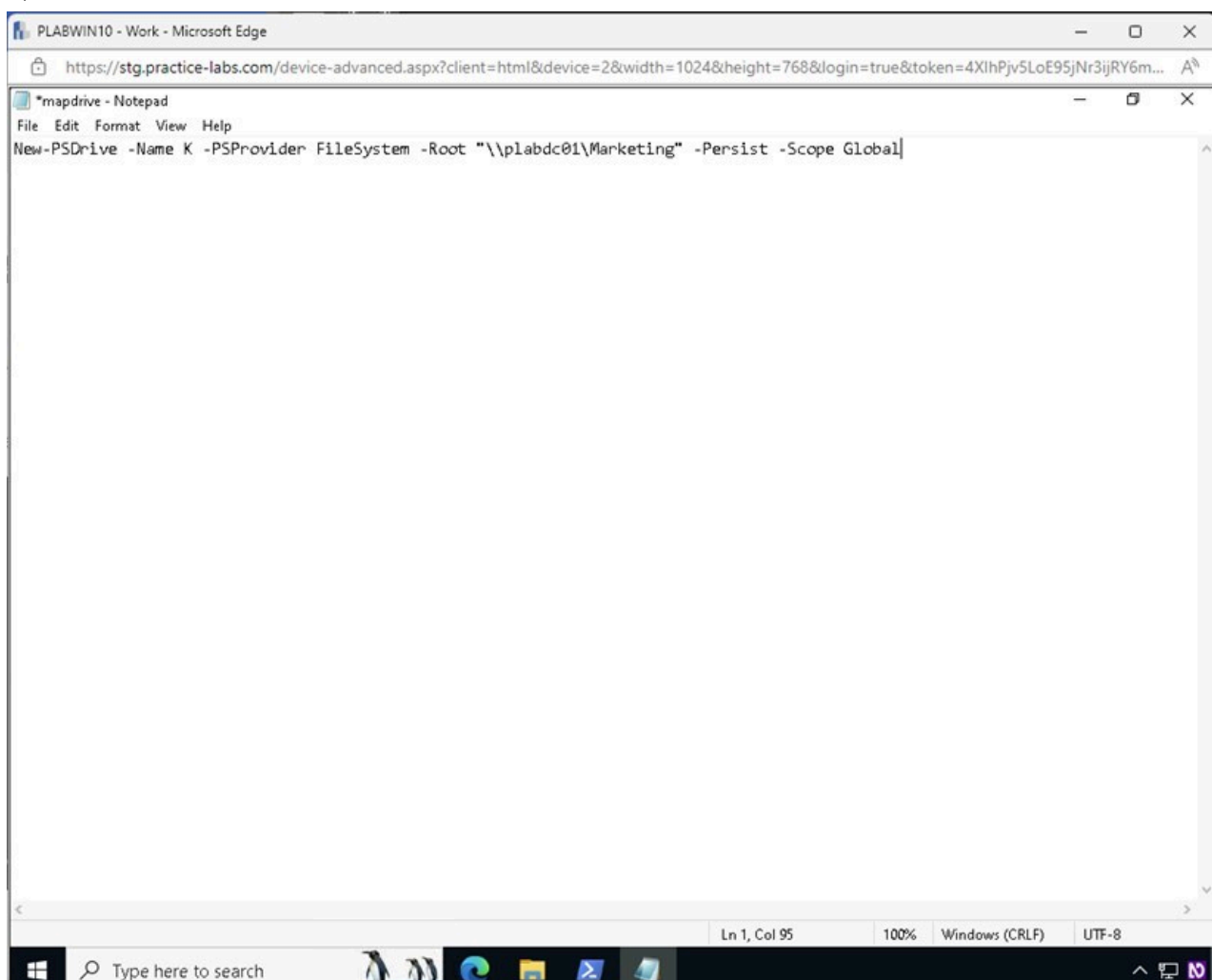


Figure 1.28 Screenshot of PLABWIN10: Displaying creating a Powershell script file on the Notepad window.

Step 14

In **Notepad**, click the **File** menu and select **Save**.

Close the **Notepad** window.

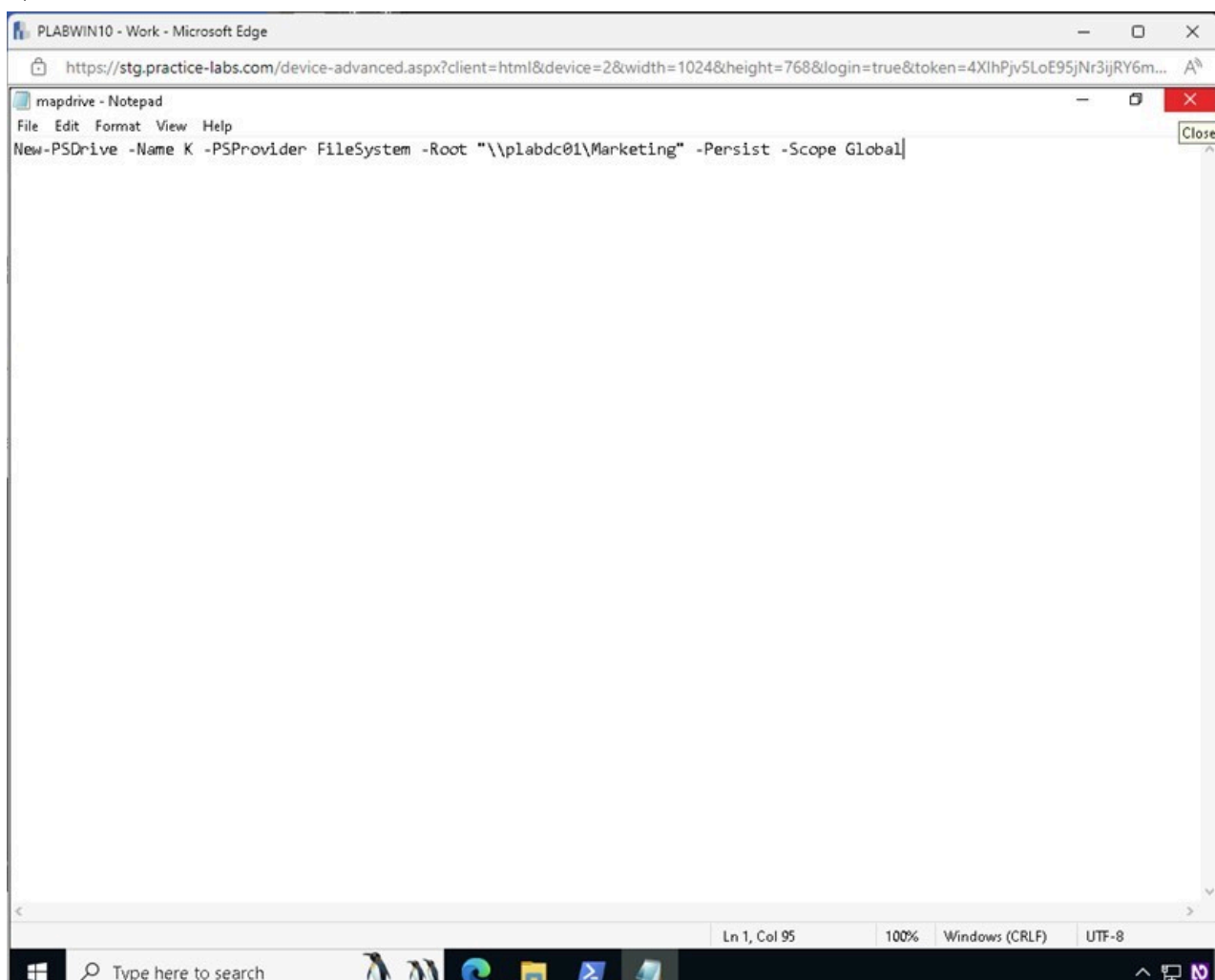


Figure 1.29 Screenshot of PLABWIN10: Displaying saving the Powershell script file and closing Notepad.

Step 15

In the **Administrator: Windows Powershell** window, type the following and press **Enter**:

```
Set-ExecutionPolicy Unrestricted
```

Type **a** and press **Enter**.

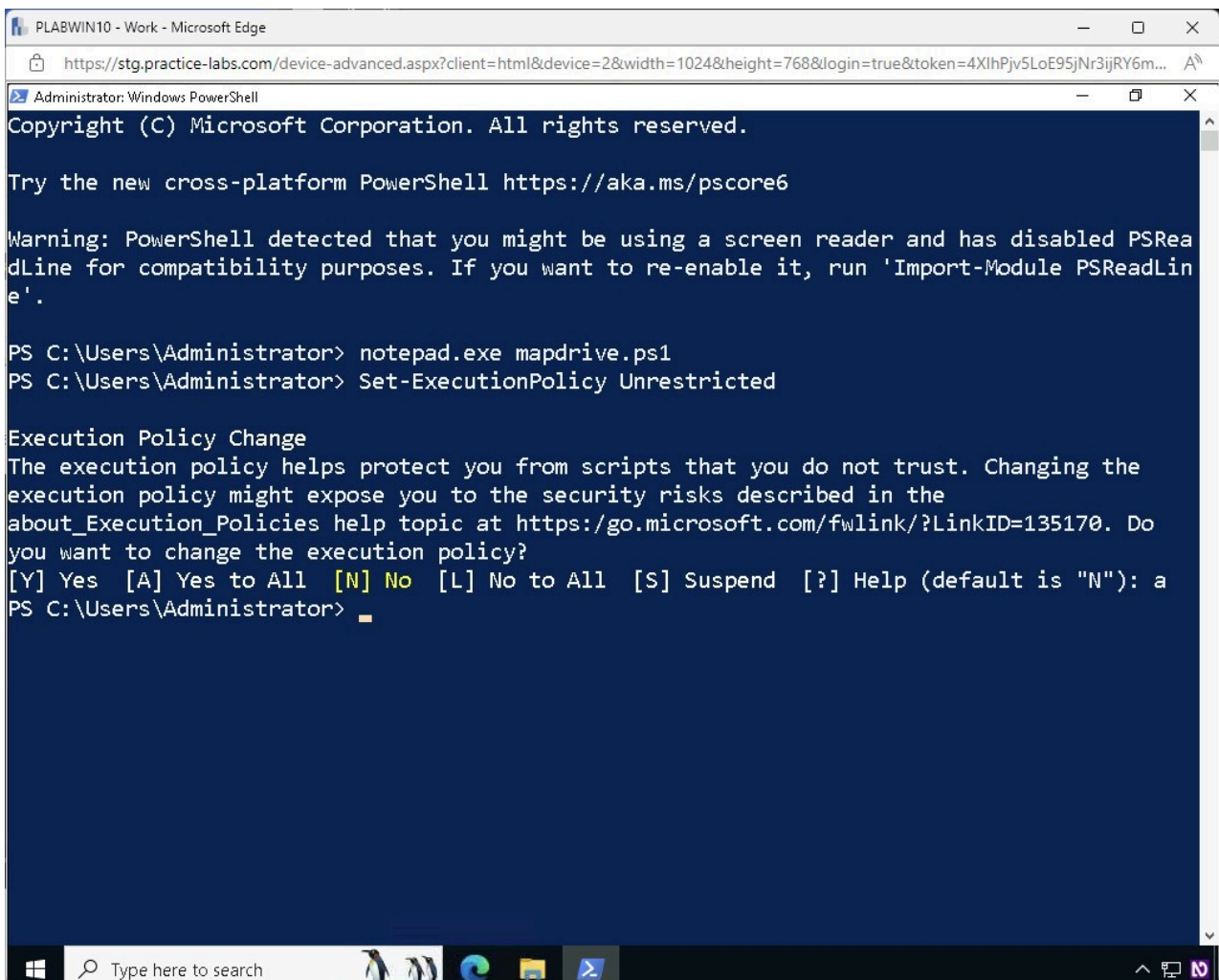


Figure 1.30 Screenshot of PLABWIN10: Displaying executing the required command in Powershell.

Step 16

Type the following in the **Administrator: Windows Powershell** window and press **Enter**:

```
.\mapdrive.ps1
```

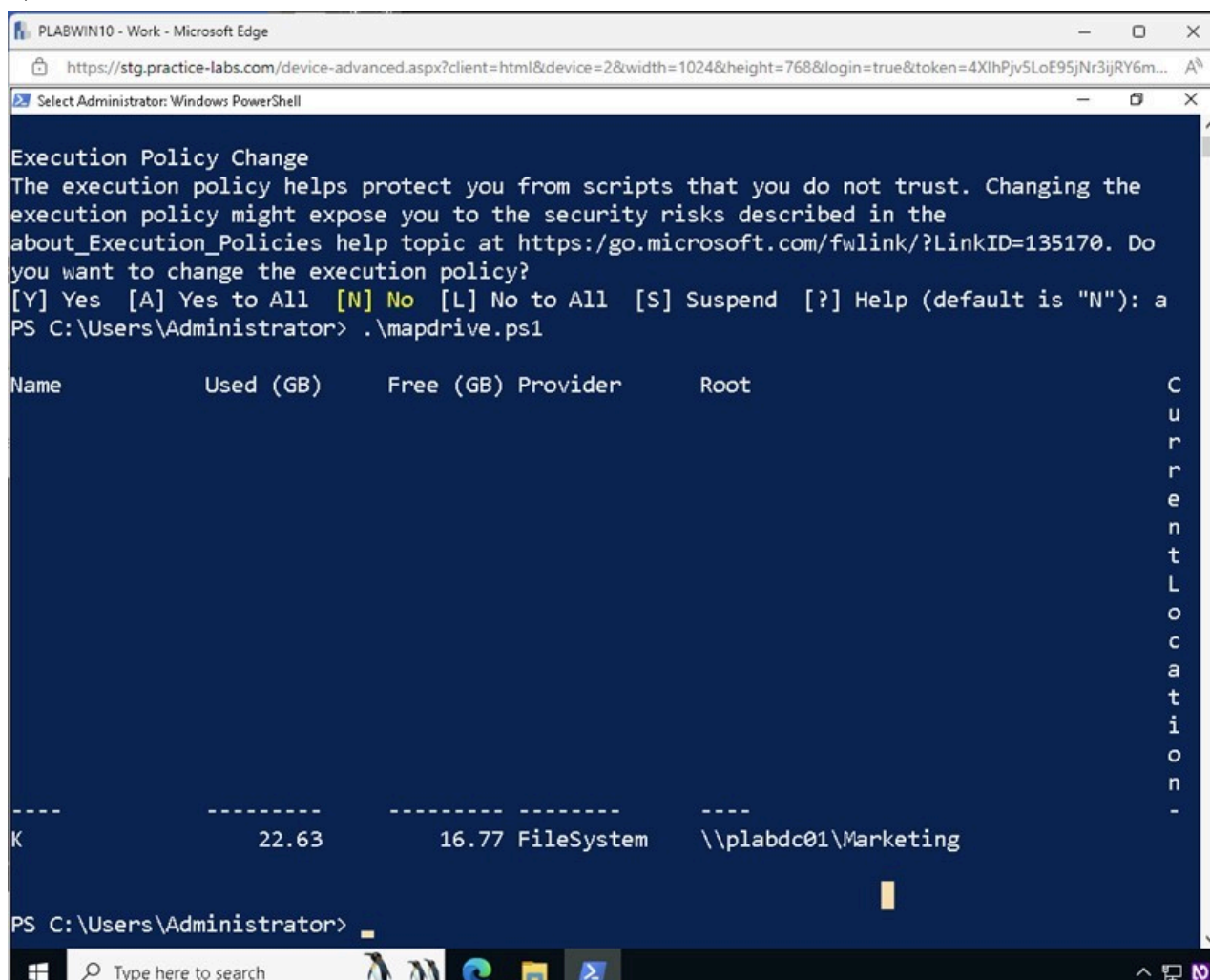


Figure 1.31 Screenshot of PLABWIN10: Displaying mapping a network drive using a Powershell script.

Step 17

Open **File Explorer** from the **Taskbar**.

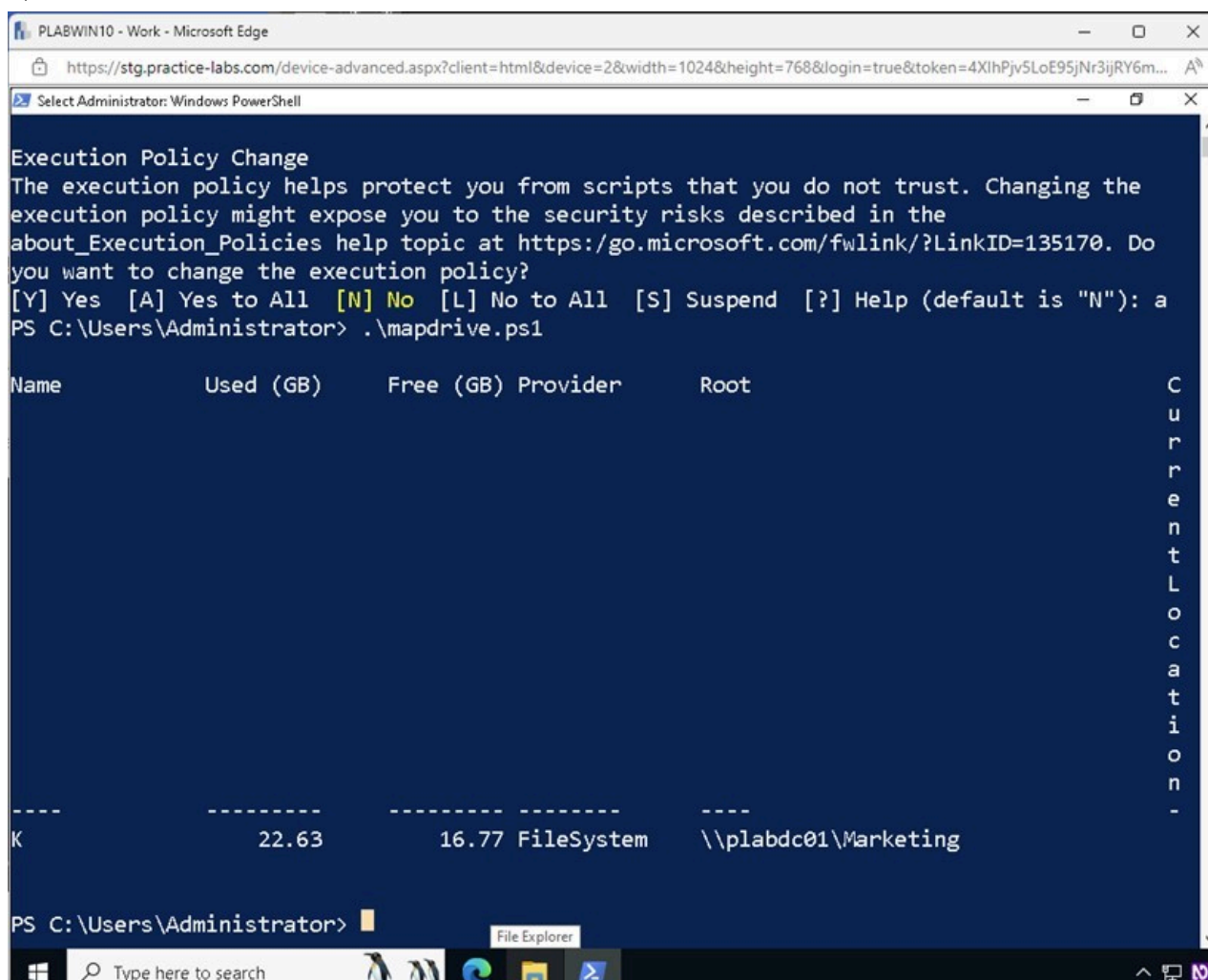


Figure 1.32 Screenshot of PLABWIN10: Displaying opening File Explorer from the Taskbar.

Step 18

The recently mapped network drive is displayed in **File Explorer**. The Powershell script can be used to map network drives on users' login.

Close **File Explorer** and **PowerShell** windows.

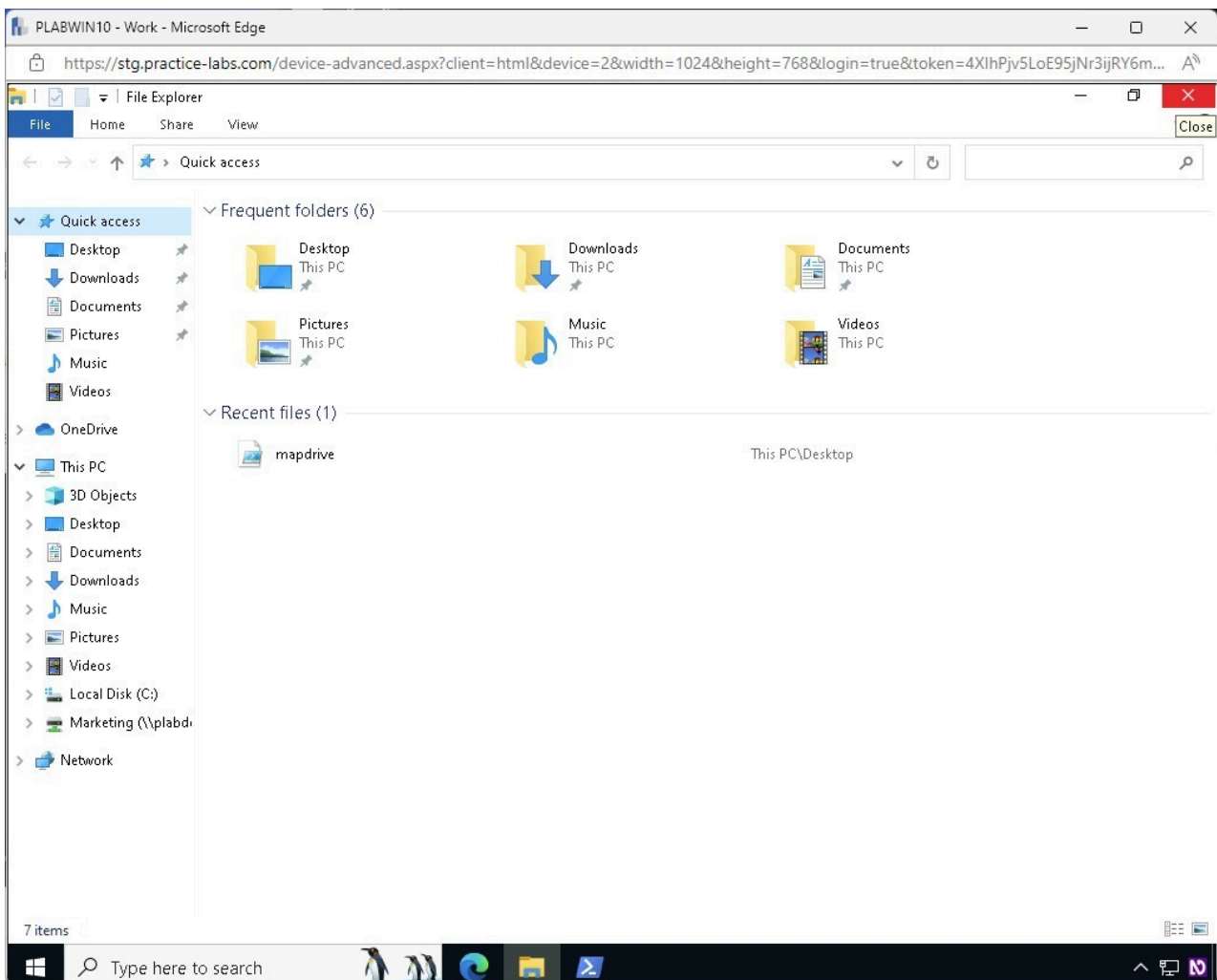


Figure 1.33 Screenshot of PLABWIN10: Displaying closing File Explorer.

Task 3 - Create a Basic Batch Script File

The **Windows Command Prompt** can be used to create a batch script file with a file extension of *.bat.

These script files can be used to automate tasks and reduce administrative effort.

In this task, a simple batch script file will be created through the **Command Prompt**.

Step 1

Connect to **PLABDM01**.

Click the **Start** charm and type:

```
cmd
```

Select **Command Prompt** from the **Best match** pop-up menu.

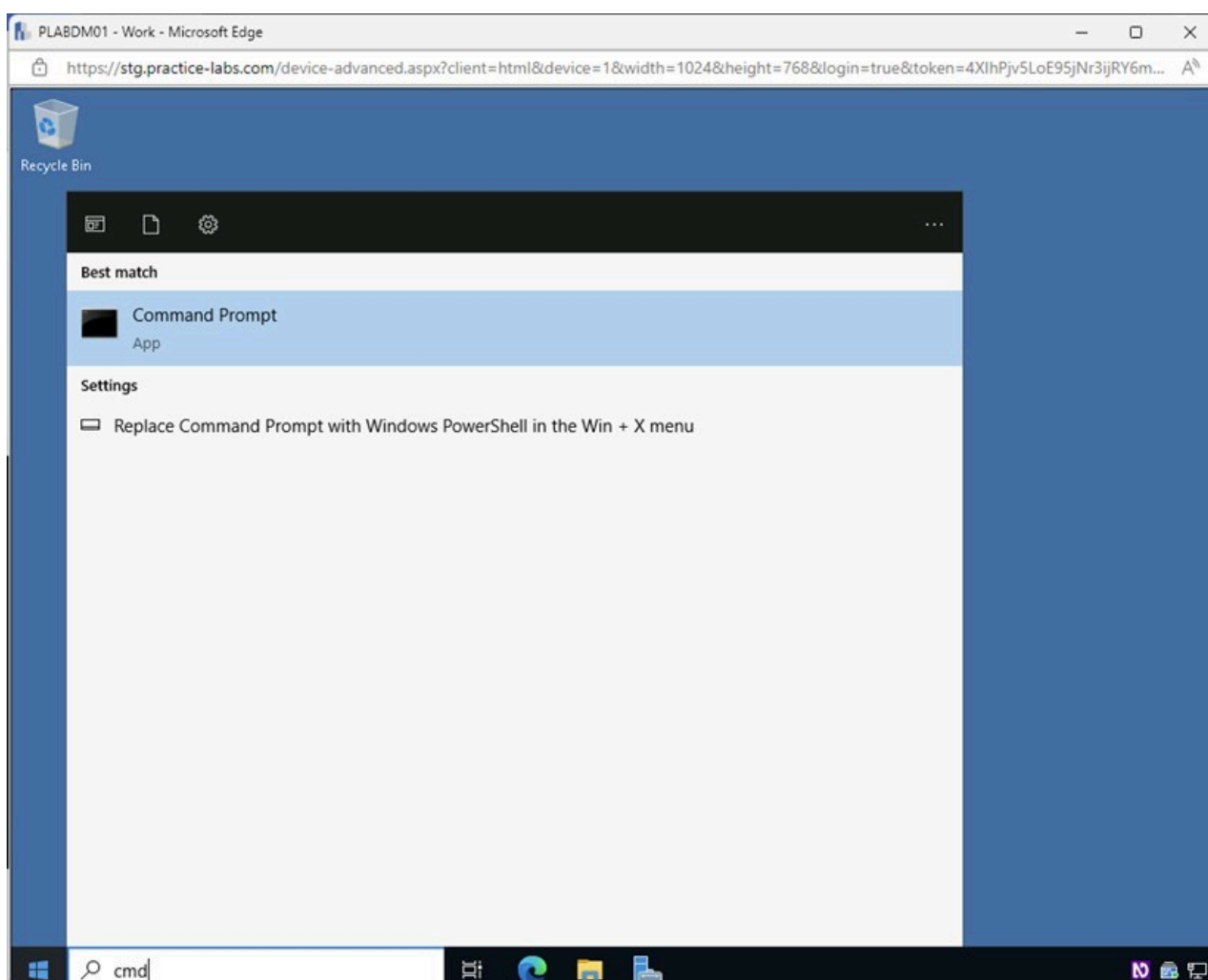


Figure 1.34 Screenshot of PLABDM01: Displaying selecting Command Prompt from the Best match pop-up menu.

Step 2

In the **Command Prompt** window, type the following and press **Enter**:

```
notepad.exe mapdrive.bat
```

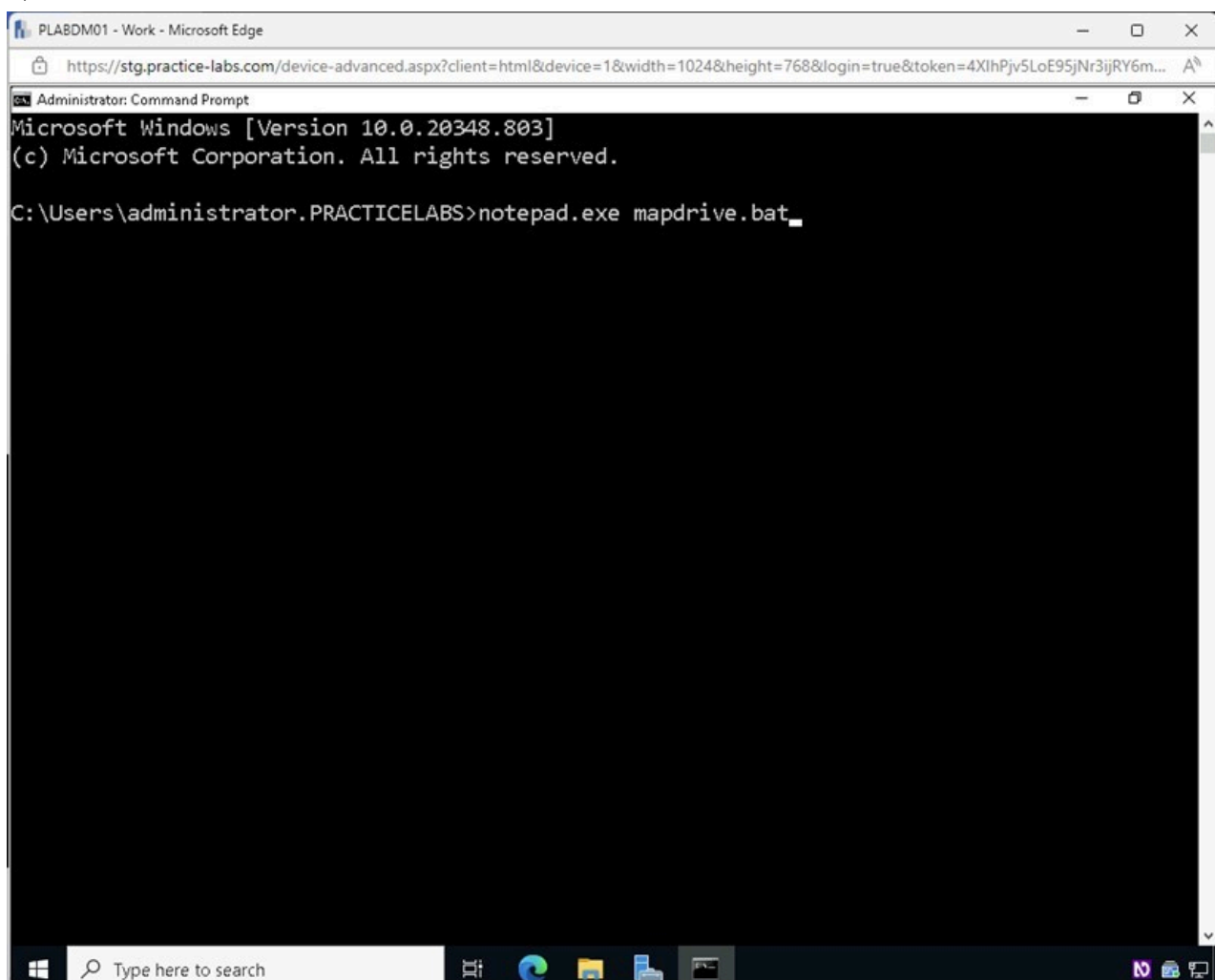


Figure 1.35 Screenshot of PLABDM01: Displaying executing the required command in the Command Prompt window.

Step 3

In the **Notepad** pop-up window, click **Yes**.

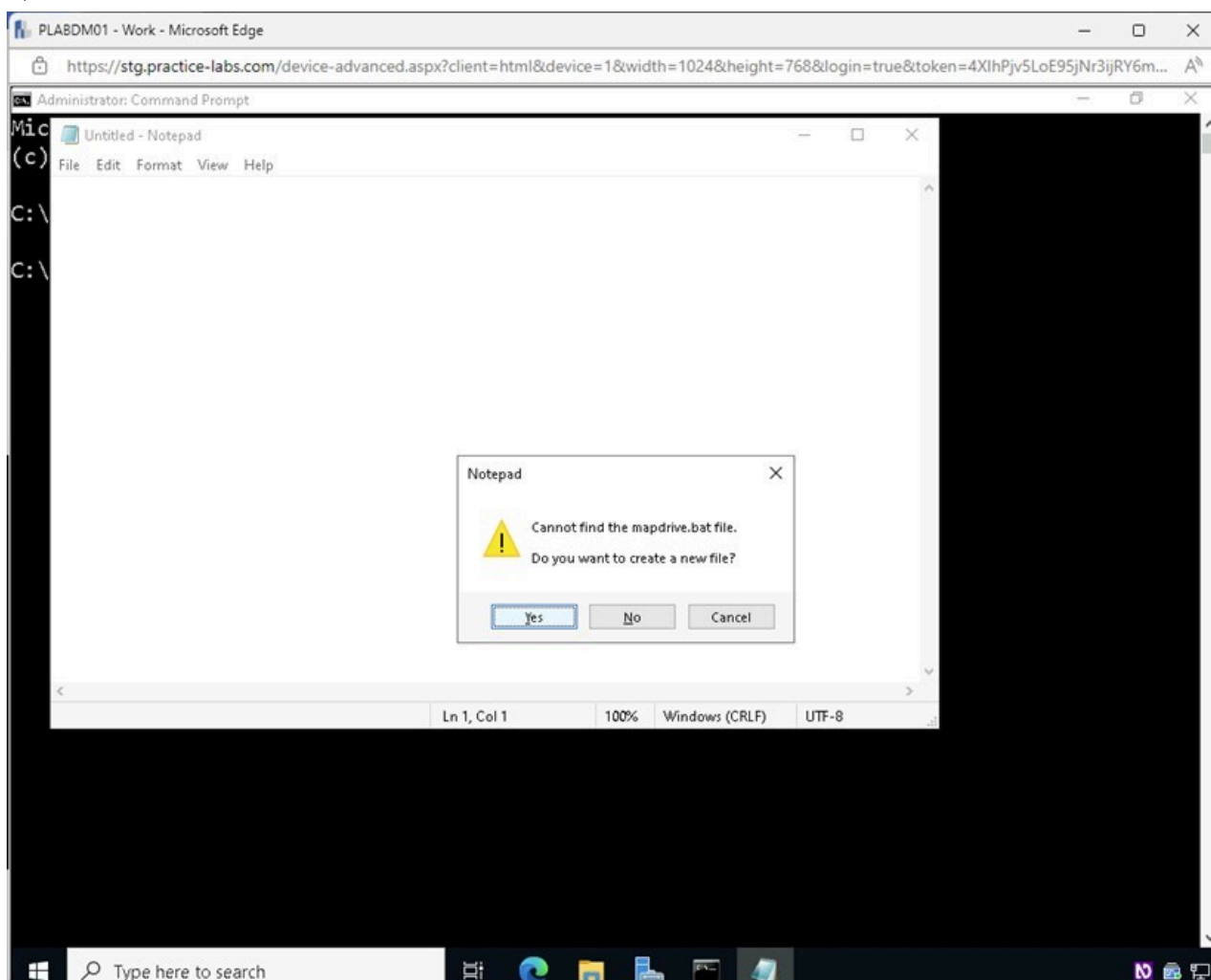


Figure 1.36 Screenshot of PLABDM01: Displaying clicking Yes on the Notepad pop-up window.

Step 4

Type the following in the **Notepad** file:

```
net use z: \\plabdc01\Marketing
```

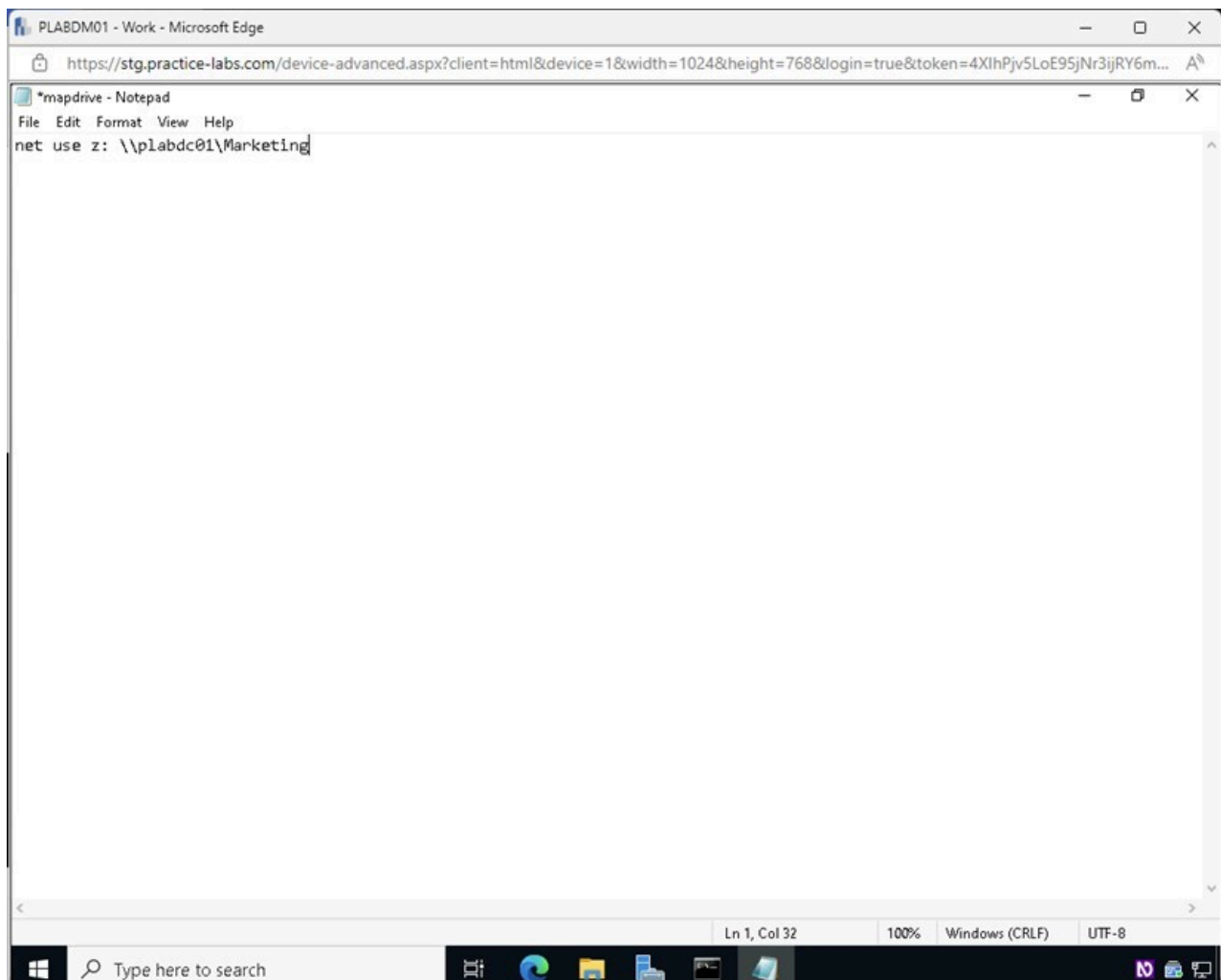



Figure 1.37 Screenshot of PLABDM01: Displaying typing the required command in the Notepad batch file.

Step 5

Click the **File** menu and select **Save**.

Close the **Notepad** batch file.

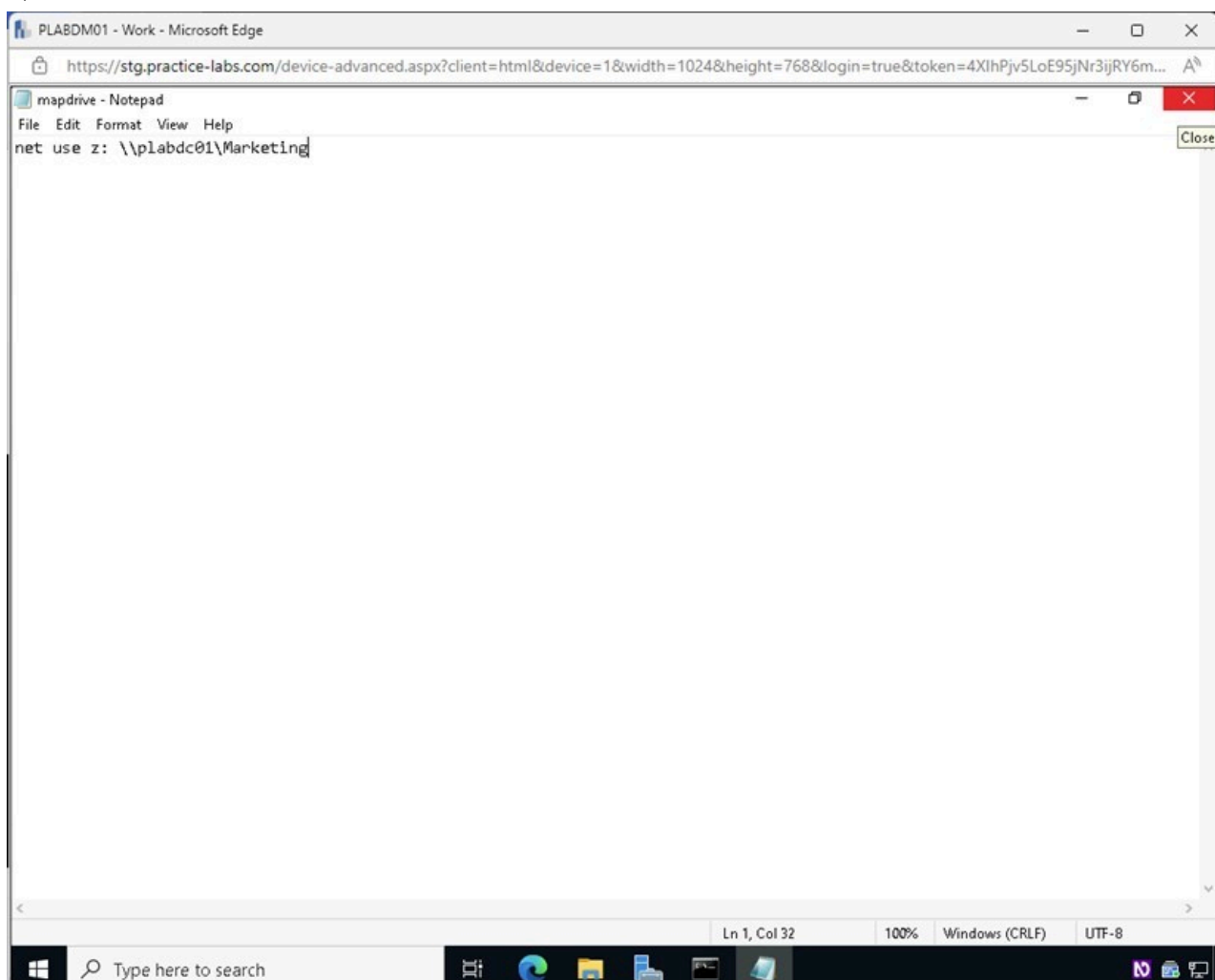


Figure 1.38 Screenshot of PLABDM01: Displaying saving and closing the Notepad batch file.

Step 6

In the **Command Prompt** window, type the following and press **Enter**:

```
.\mapdrive.bat
```

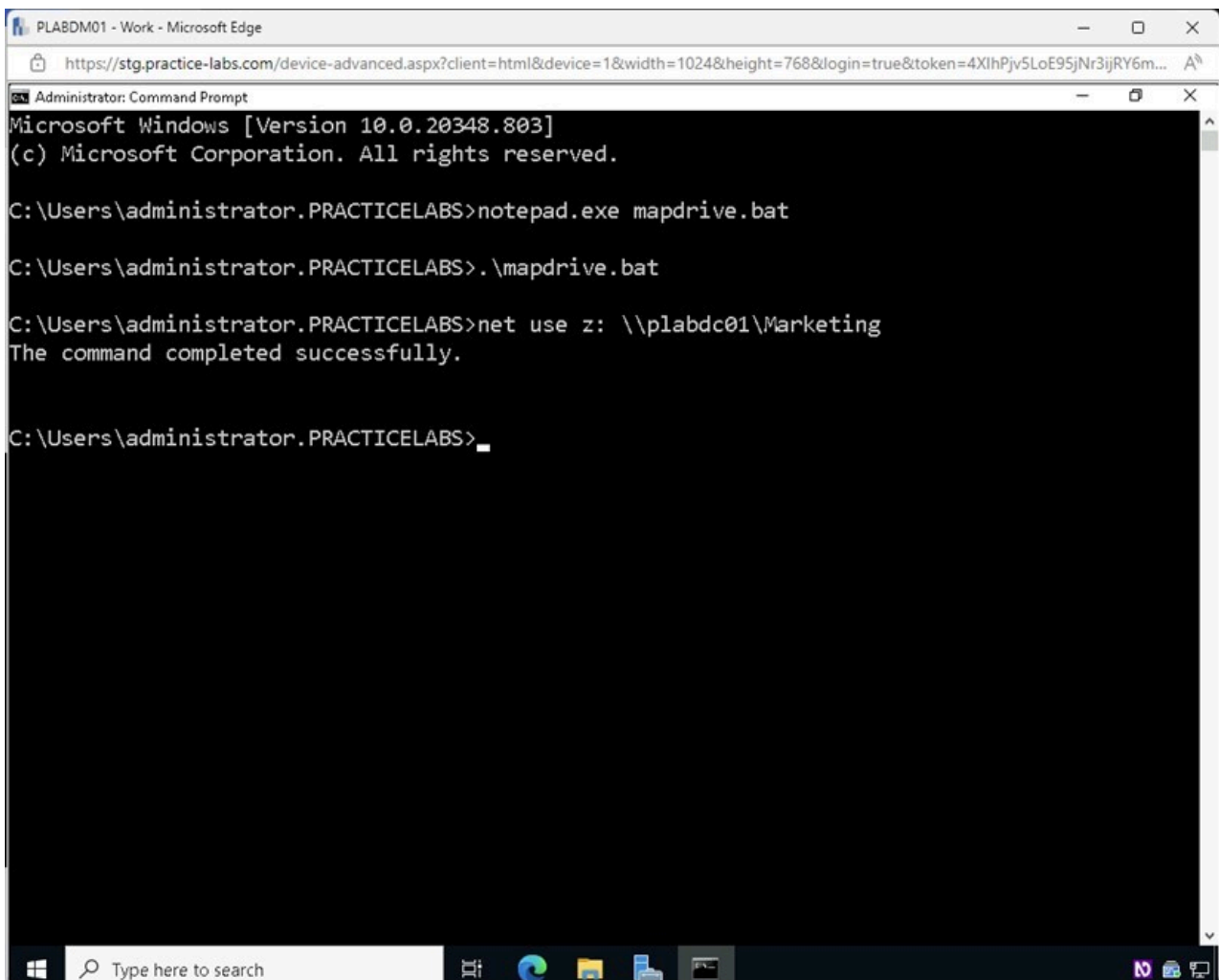


Figure 1.39 Screenshot of PLABDM01: Displaying entering and executing the required command in the Command Prompt window.

Note: The batch file was successfully executed, mapping the shared folder to the device as a drive. Like the Powershell script, a batch file can be associated with a user's profile to map a drive on login.

Step 7

Open **File Explorer** from the **Taskbar**.

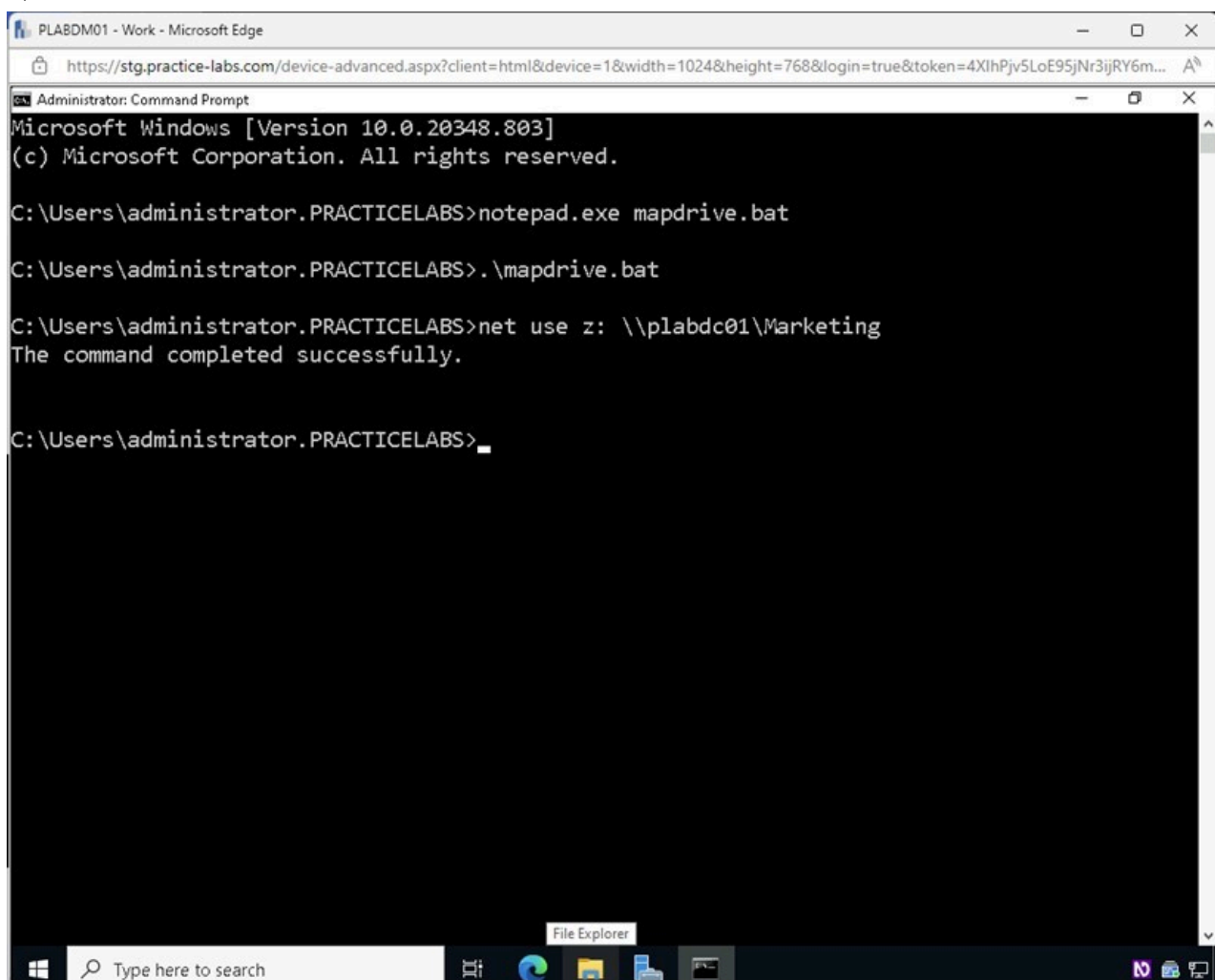


Figure 1.40 Screenshot of PLABDM01: Displaying opening File Explorer from the Taskbar.

Step 8

After executing the batch file, the mapped drive is visible in **File Explorer**.

Close the **File Explorer** and **Command Prompt** windows.

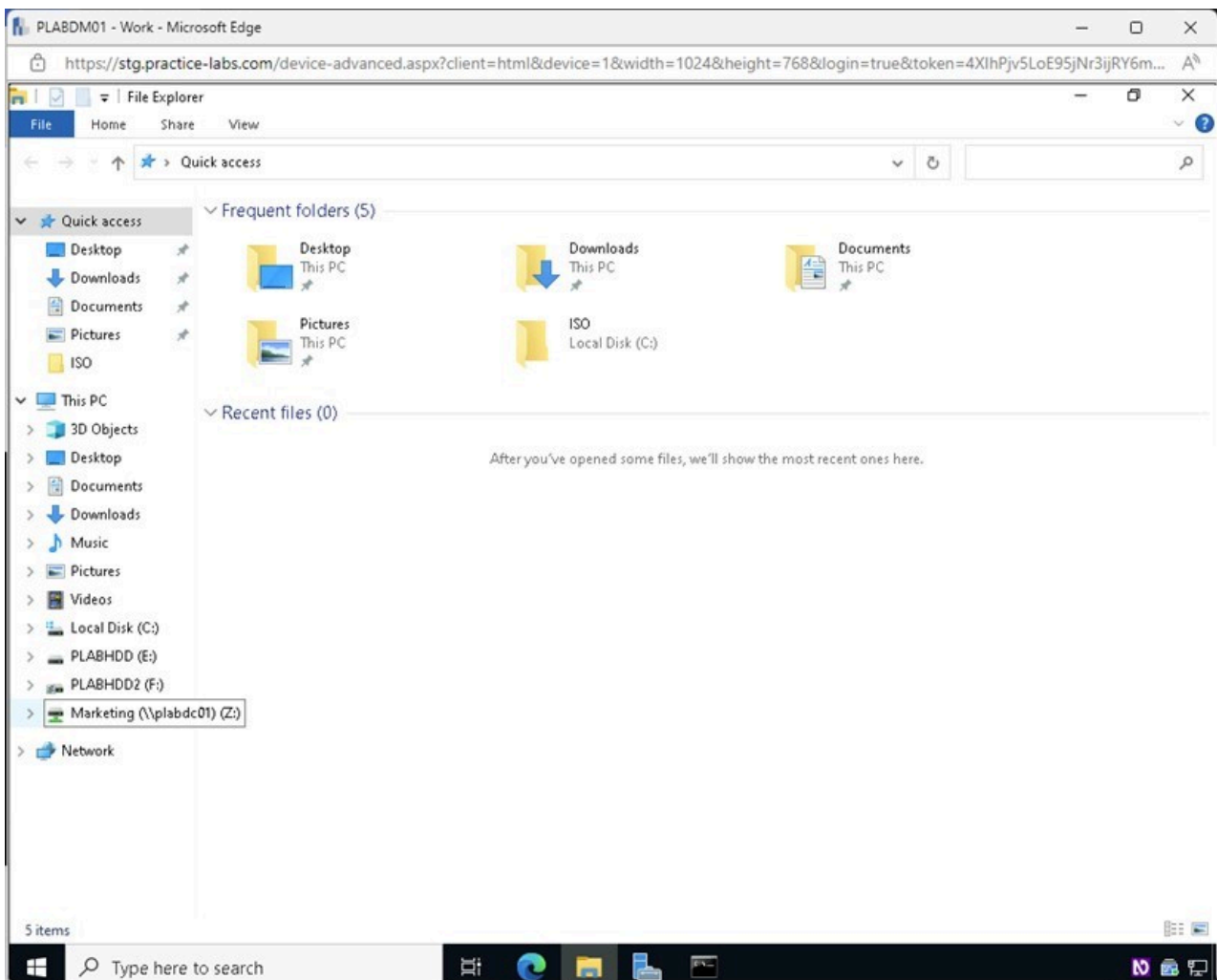


Figure 1.41 Screenshot of PLABDM01: Displaying the mapped drive in File Explorer.

Exercise 2 - Basic Script Constructs

A Linux script can contain conditions related to the current environment, for example, time or date variables. The script can then be used to display those specific variables if the conditions are met.

In this exercise, a script file will be created using specific conditions.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a Script File with Different Variables and Conditions

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABALMA** - (Alma Linux 8.6 - Stand-alone Linux Workstation)

Task 1 - Create a Variable Script File

In this task, a script file will be created that will match the specific conditions of the environment.

Step 1

Connect to **PLABALMA**.

Click **Activities** in the top left-hand corner.

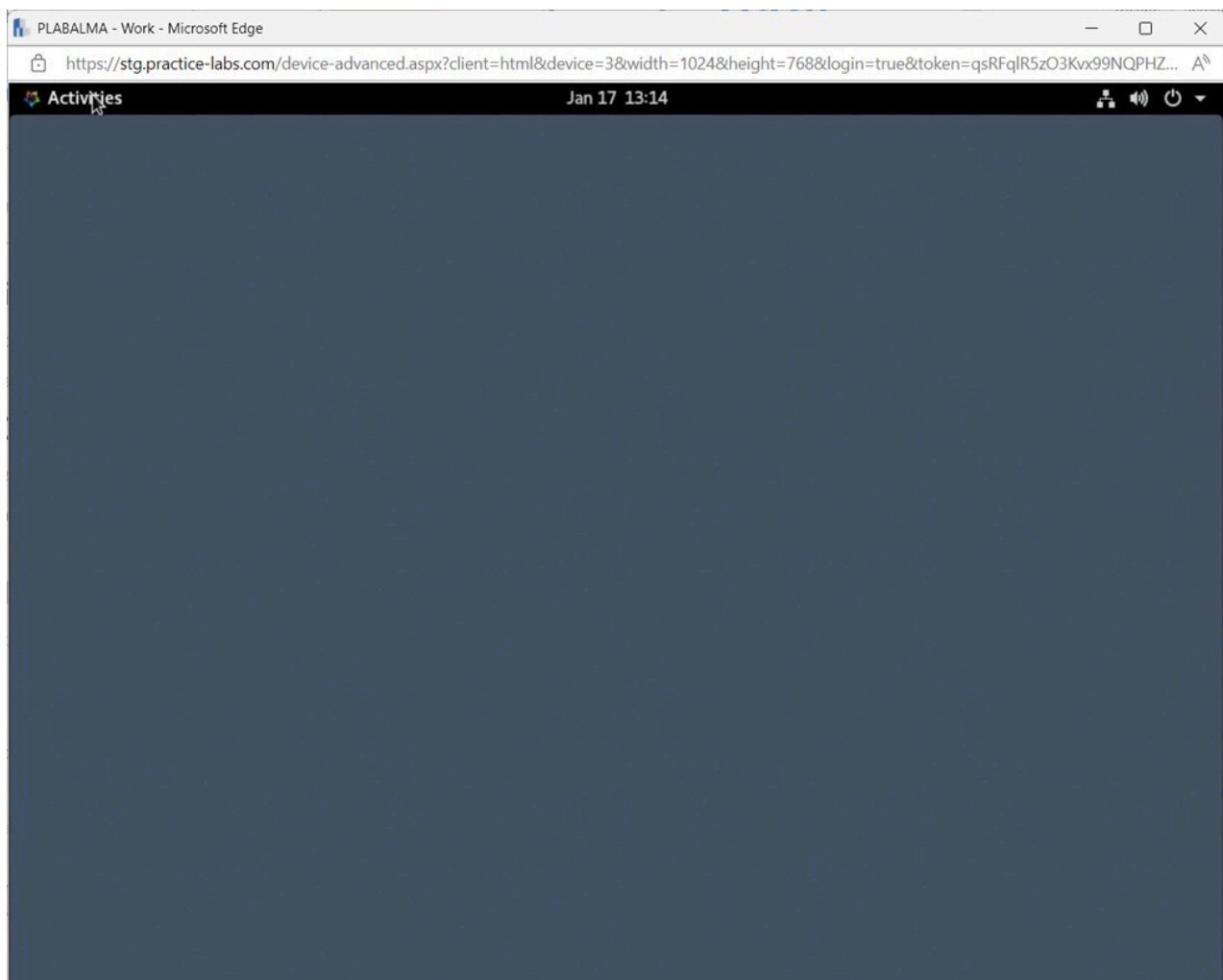


Figure 2.1 Screenshot of PLABALMA: Displaying clicking Activities in the top left-hand corner.

Step 2

In the **Activities** fly-out window, select **Terminal**.

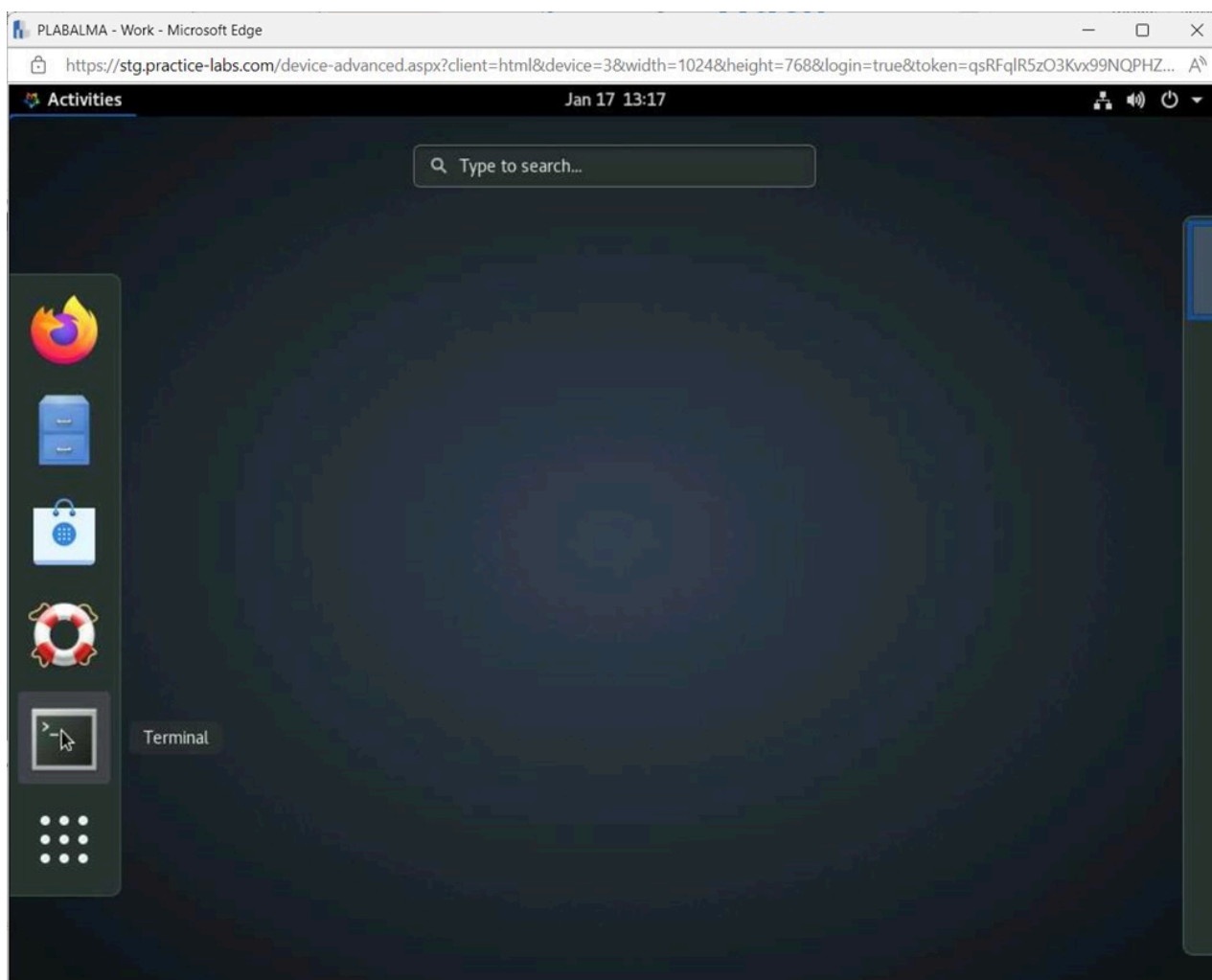


Figure 2.2 Screenshot of PLABALMA: Displaying selecting Terminal in the Activities fly-out window.

Step 3

In the **Terminal** window, type the following and press **Enter**:

```
vim daytime.sh
```

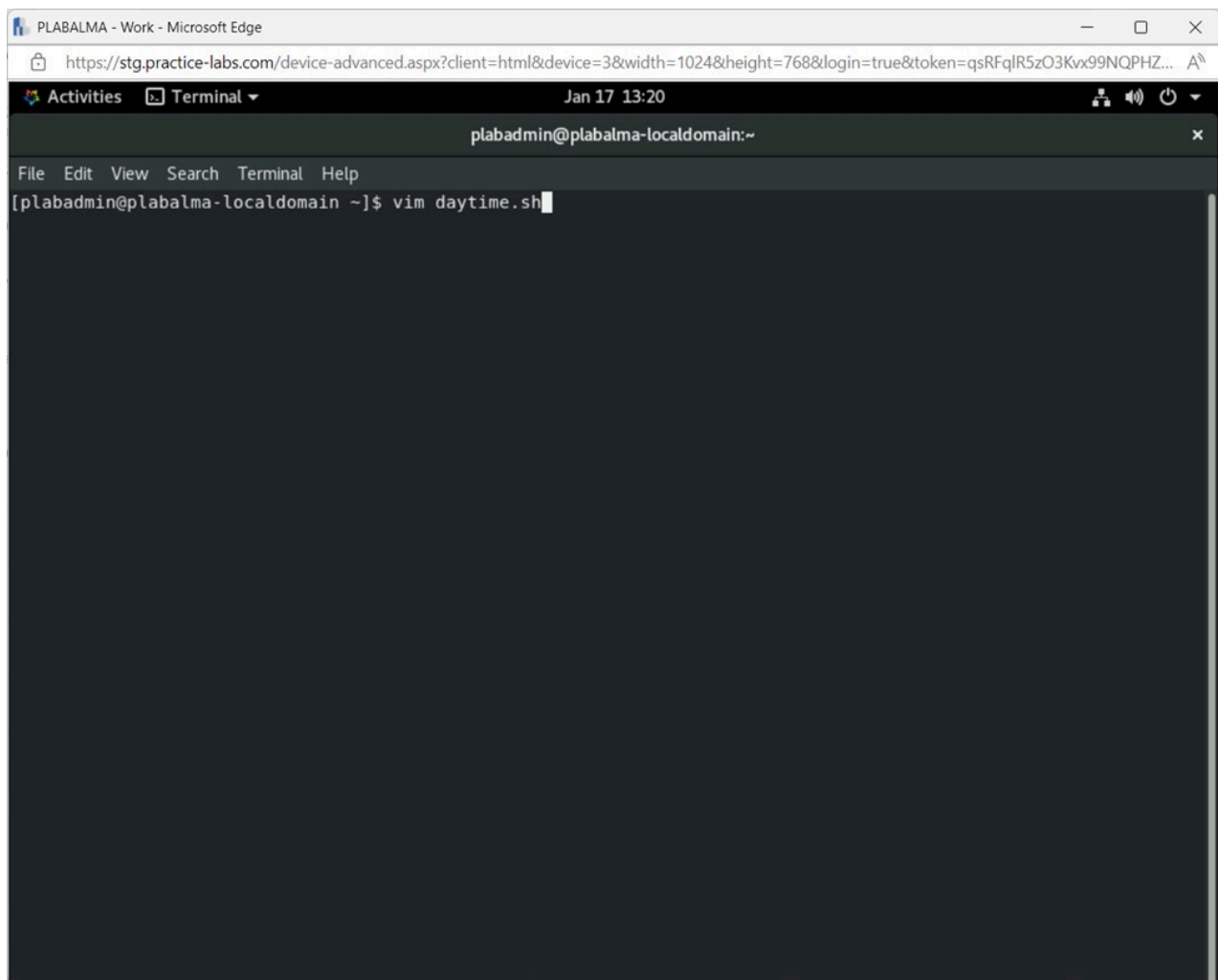


Figure 2.3 Screenshot of PLABALMA: Displaying typing the required command and pressing Enter.

Step 4

In the **Terminal** window, press 'i'.

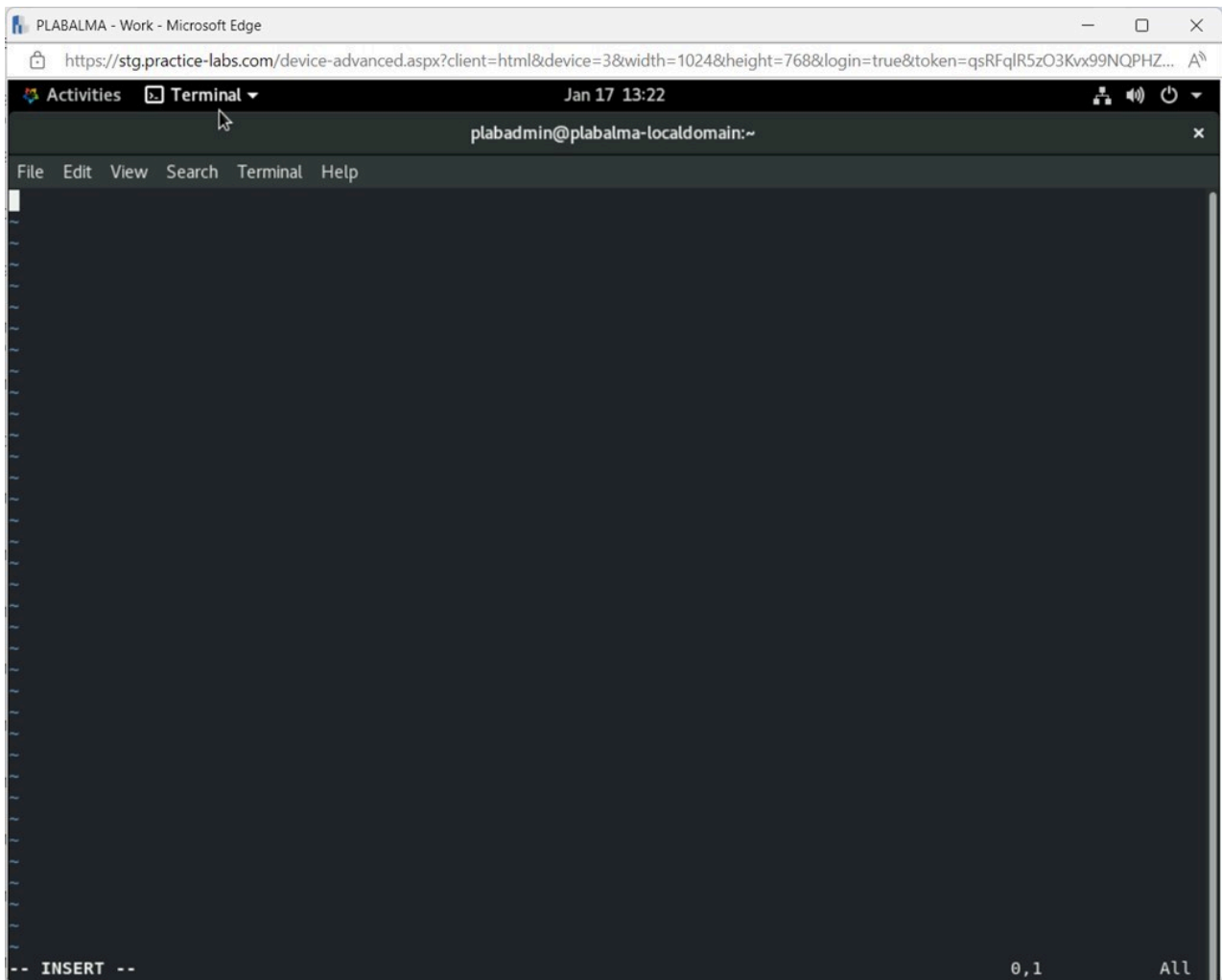


Figure 2.4 Screenshot of PLABALMA: Displaying pressing 'I' in the Terminal window.

Step 5

In the **Vim** text editor window, enter the following:

```
#!/bin/bash
# Determine if it is a working day or the weekend
and store it inside the $day variable
day=$(date +"%u")
# Determine if it is morning or afternoon and store
it in the $time variable
time=$(date +"%p")
# Check if the day of the week is between Monday and
Friday
if [ $day -le 5 ]; then
    #if it is a working day display the text below
```

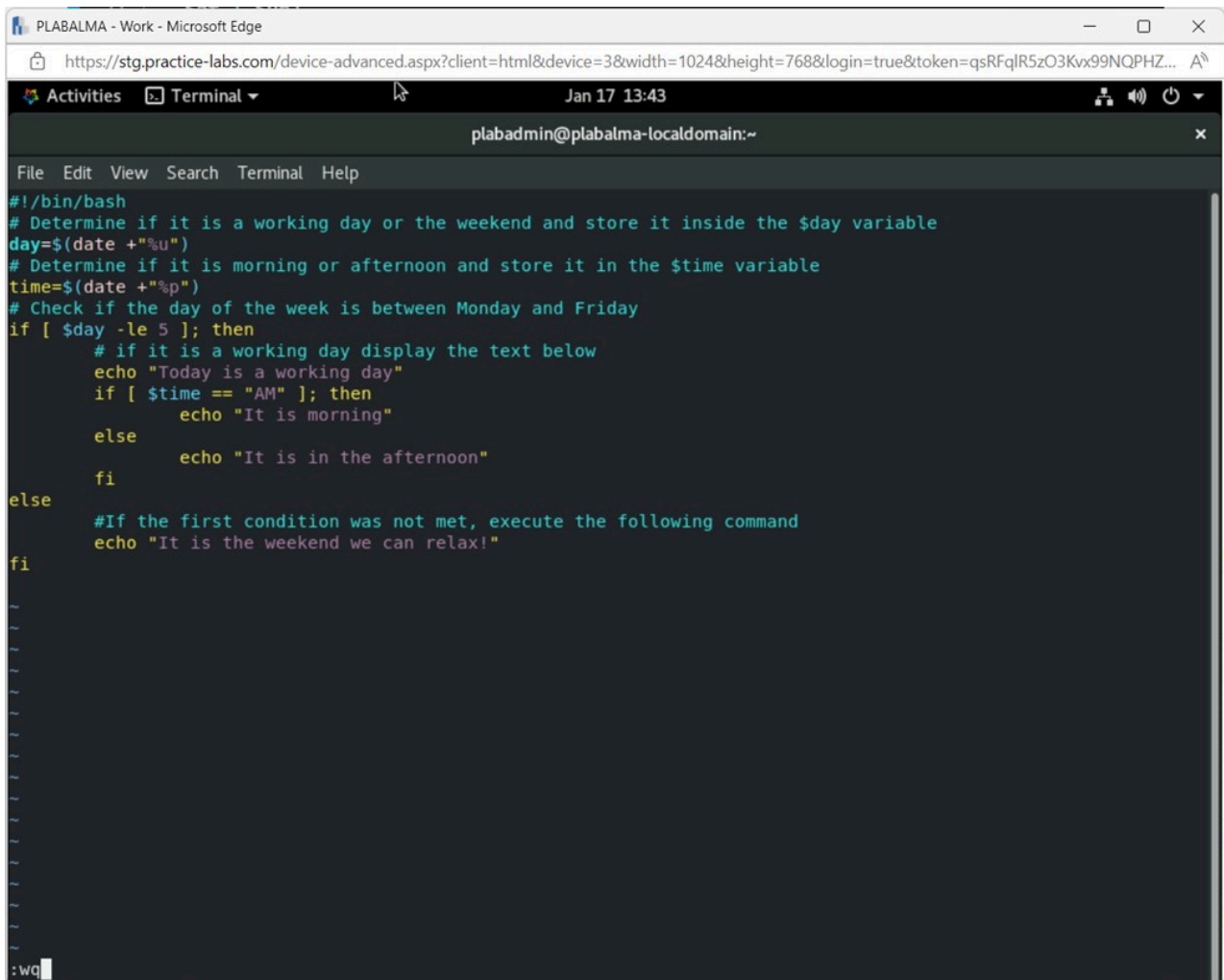
```
echo "Today is a working day"
if [ $time == "AM" ]; then
    echo "It is morning"
else
    echo "It is in the afternoon"
fi
else
    # If the first condition was not met, execute the
    following command
    echo "It is the weekend we can relax!"
fi
```

Step 6

Press **Esc** and type the following in the Vim text editor:

```
:wq
```

Press **Enter**.



The screenshot shows a Microsoft Edge browser window with the address bar displaying a URL from stg.practice-labs.com. The browser's top bar shows 'PLABALMA - Work - Microsoft Edge'. Below the address bar, there's a navigation bar with 'Activities' and 'Terminal' tabs, and a clock showing 'Jan 17 13:43'. The main content area is a terminal window titled 'plabadmin@plabalma-localdomain:~'. The terminal has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The terminal content shows a Bash script that determines if it's a working day or weekend based on the day of the week and time of day. The script uses variables for day and time, and conditional logic to print messages. The terminal prompt is ':wq'.

```
#!/bin/bash
# Determine if it is a working day or the weekend and store it inside the $day variable
day=$(date +%u)
# Determine if it is morning or afternoon and store it in the $time variable
time=$(date +%p)
# Check if the day of the week is between Monday and Friday
if [ $day -le 5 ]; then
    # if it is a working day display the text below
    echo "Today is a working day"
    if [ $time == "AM" ]; then
        echo "It is morning"
    else
        echo "It is in the afternoon"
    fi
else
    #If the first condition was not met, execute the following command
    echo "It is the weekend we can relax!"
fi
```

Figure 2.6 Screenshot of PLABALMA: Displaying entering the required command in the Vim text editor and pressing Enter.

Step 7

In the **Terminal** window, type the following and press **Enter**:

```
chmod +x daytime.sh
```

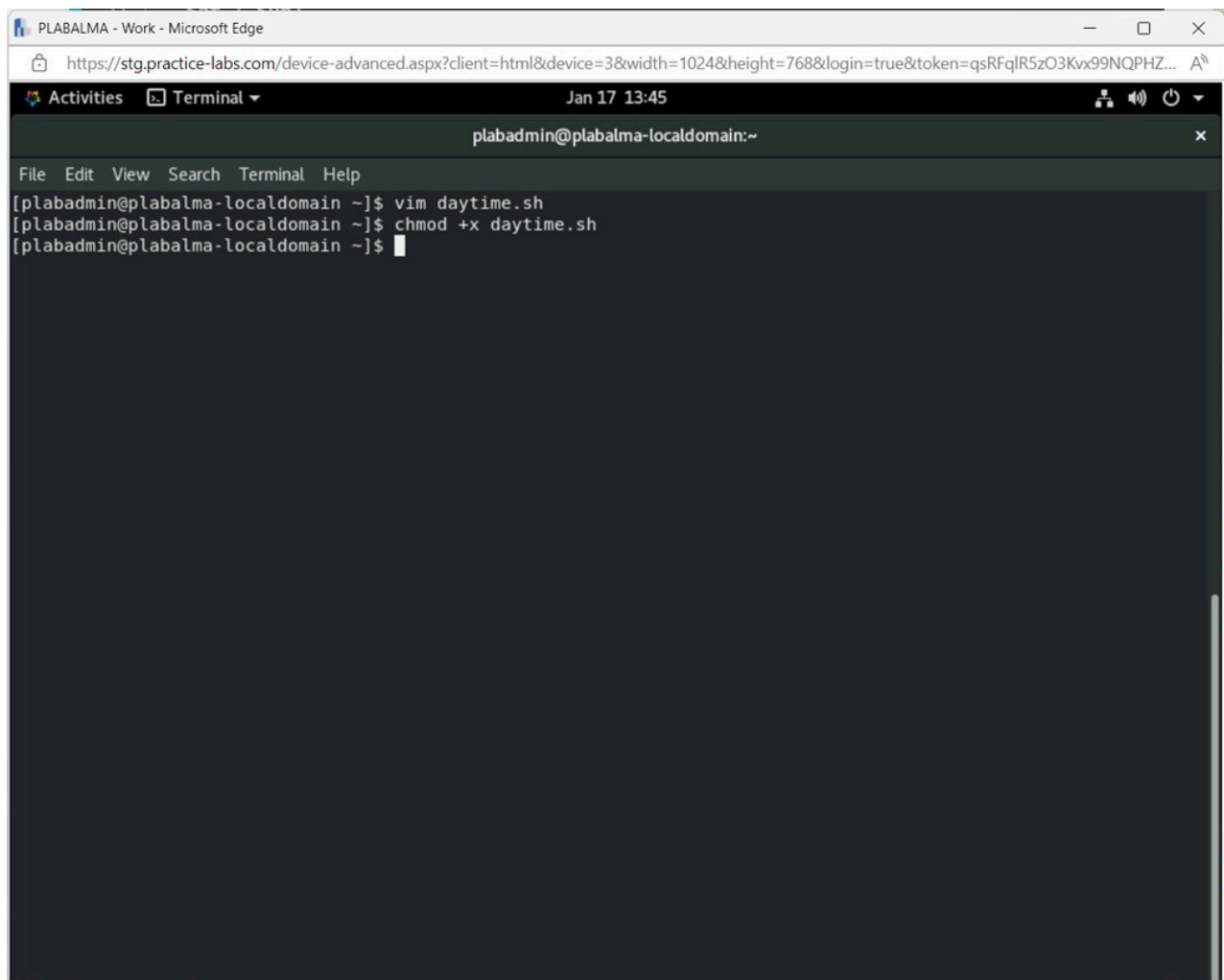


Figure 2.7 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window.

Note: By executing the *chmod* command with the *+x* parameter, the file permission was changed to make it executable.

Step 8

In the **Terminal** window, type the following and press **Enter**:

```
./daytime.sh
```

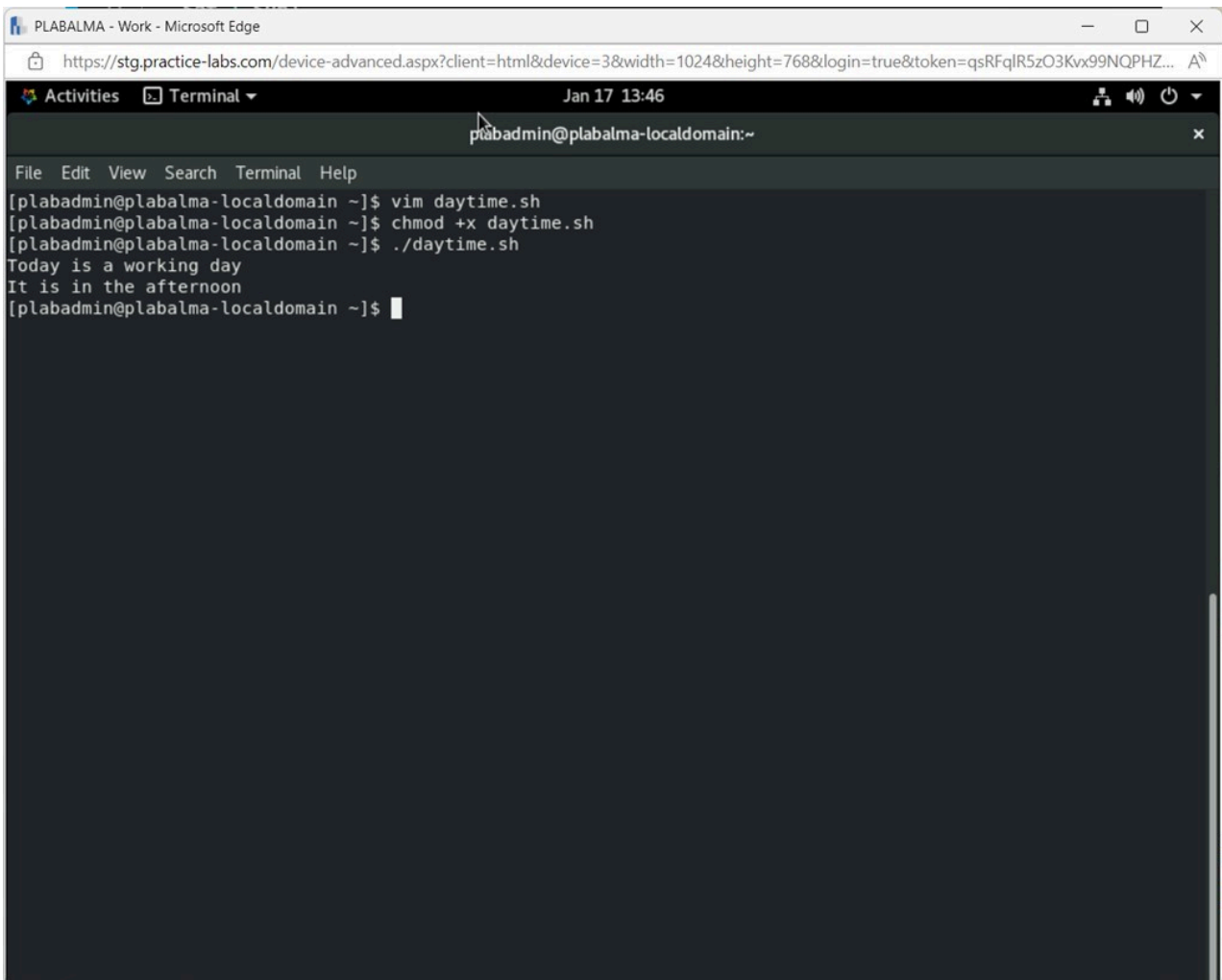
A screenshot of a web browser window displaying a terminal interface. The browser's address bar shows a URL from 'stg.practice-labs.com'. The terminal window has a title bar 'PLABALMA - Work - Microsoft Edge' and a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The terminal prompt is 'plabadmin@plabalma-localdomain:~'. The user has entered the following commands: 'vim daytime.sh', 'chmod +x daytime.sh', and './daytime.sh'. The output of the script is displayed: 'Today is a working day' and 'It is in the afternoon'. The terminal window also shows a system clock at 'Jan 17 13:46' and standard window controls.

Figure 2.8 Screenshot of PLABALMA: Displaying executing the required command in the Terminal window.

Note: A script file was created that uses the system date and time to display specific information about the system. It also includes the if and else statement, which determines the output of the script. These scripts can be used to minimize repetitive administrative duties.

Close all open windows.

Exercise 3 - User Creation Using Scripts

As a Server Administrator, one of the most repetitive tasks is creating user accounts, which can be very time consuming and repetitive and lead to user error. A script to provision user accounts will increase productivity and eliminate user error.

In this exercise, a script will be created that will automatically create users in Active Directory.

Learning Outcomes

After completing this exercise, you should be able to:

- Create a Script that Creates Users in Active Directory

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Creating Users using a Script

A Server Administrator can use Powershell to automate creating users using a script.

In this task, a Powershell script will create to provision a user account in Active Directory.

Step 1

Connect to **PLABDC01**.

The **Server Manager** window will open on startup.

In the **Server Manager** window, select **Tools**, and click **Active Directory Users and Computers**.

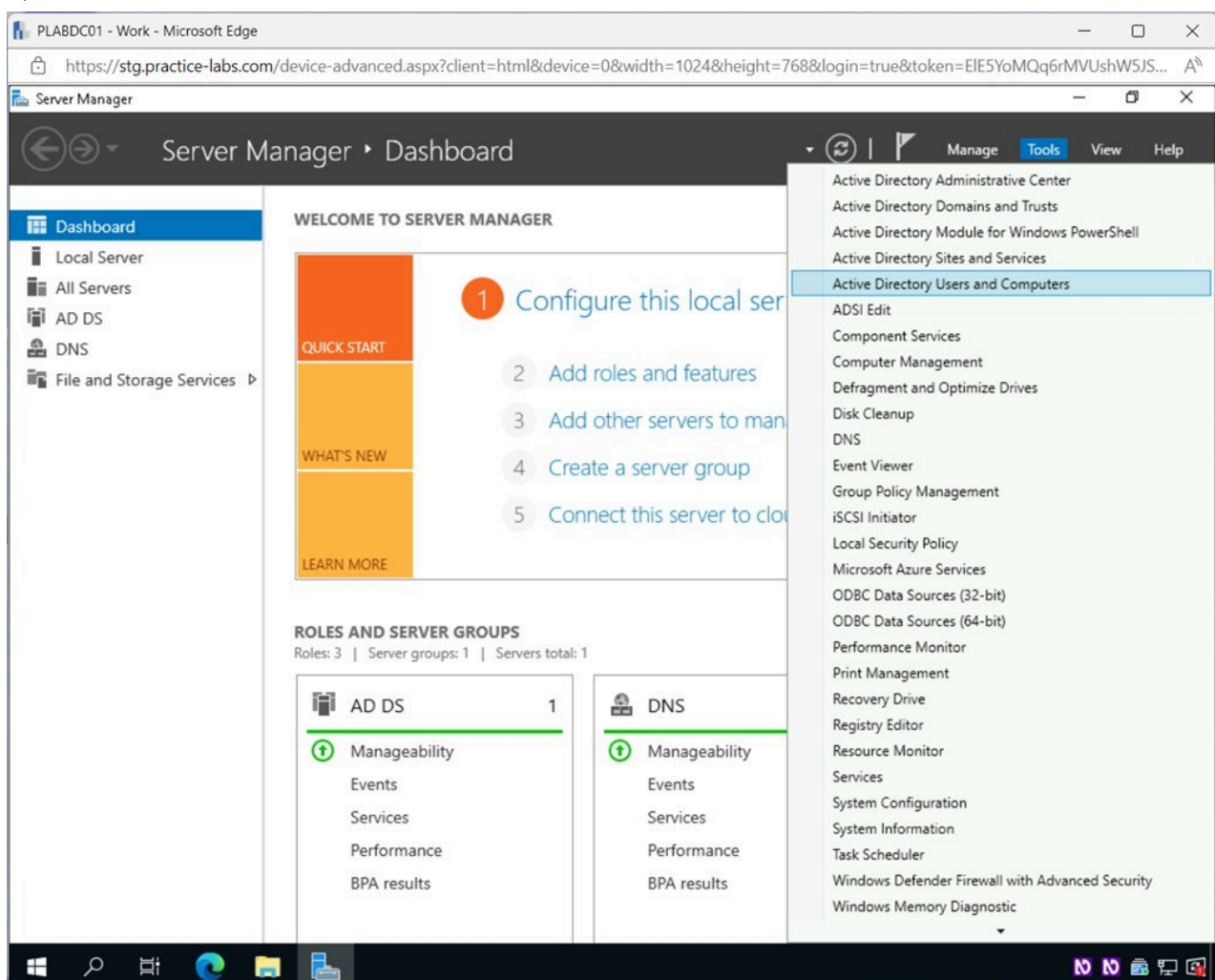


Figure 3.1 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Active Directory Users and Computers is selected.

Step 2

In the **Active Directory Users and Computers** window, right-click **practicelabs.com** and select **New > Organizational Unit**.

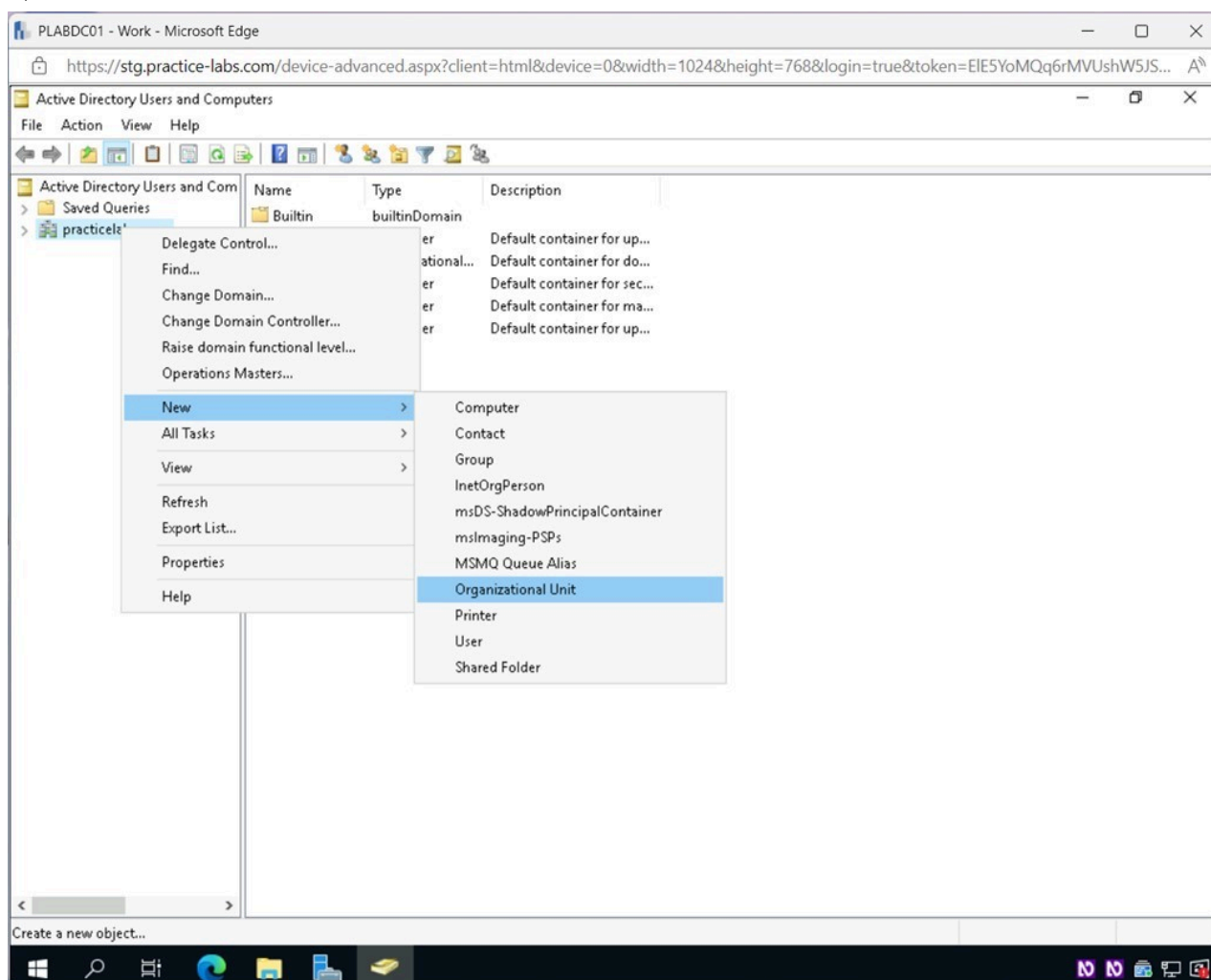


Figure 3.2 Screenshot of PLABDC01: Displaying right-clicking practicelabs.com and selecting New > Organizational Unit in the Active Directory Users and Computers window.

Step 3

In the **New Object - Organizational Unit** pop-up window, type the following for the **Name** field:

Marketing

Click **OK**.

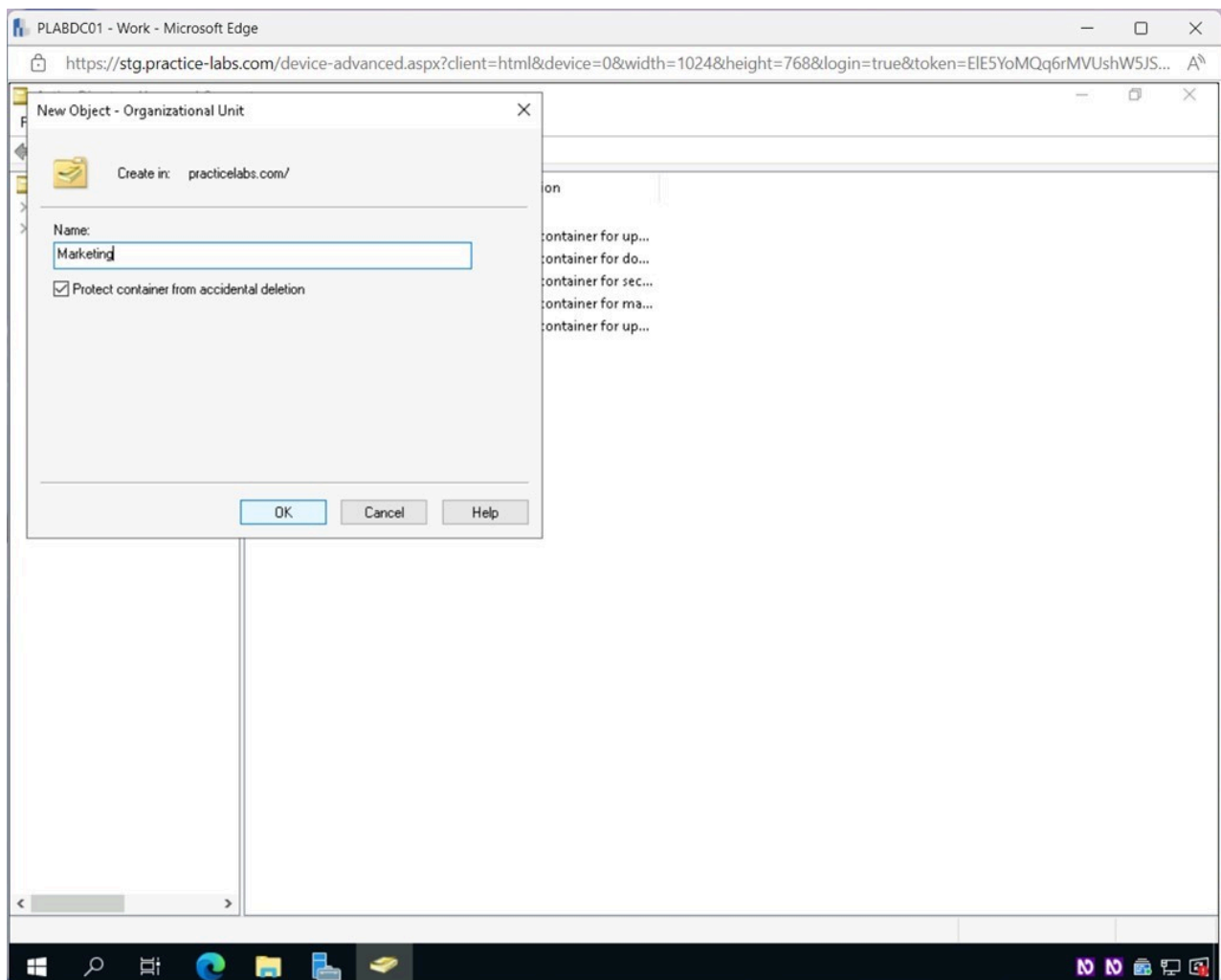


Figure 3.3 Screenshot of PLABDC01: Displaying the New Object - Organizational Unit pop-up window, showing Marketing typed in for the Name field and OK selected.

Step 4

The new **Marketing** Organizational Unit is now created.

Close the **Active Directory Users and Computers** window.

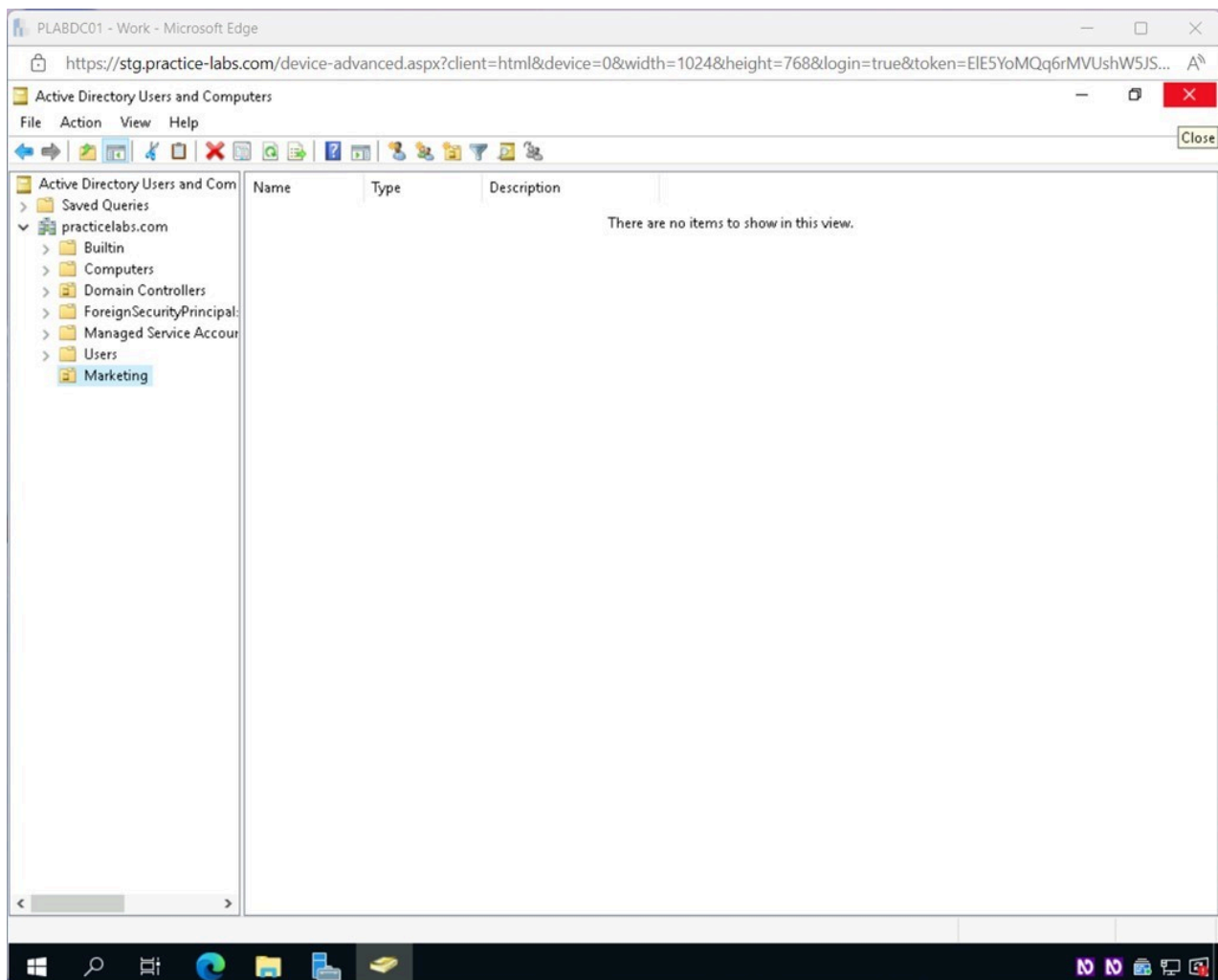


Figure 3.4 Screenshot of PLABDC01: Displaying closing the Active Directory Users and Computers window.

Step 5

Click the **Start** charm and type:

Powershell ISE

Select **Windows Powershell ISE** from the **Best match** pop-up menu.

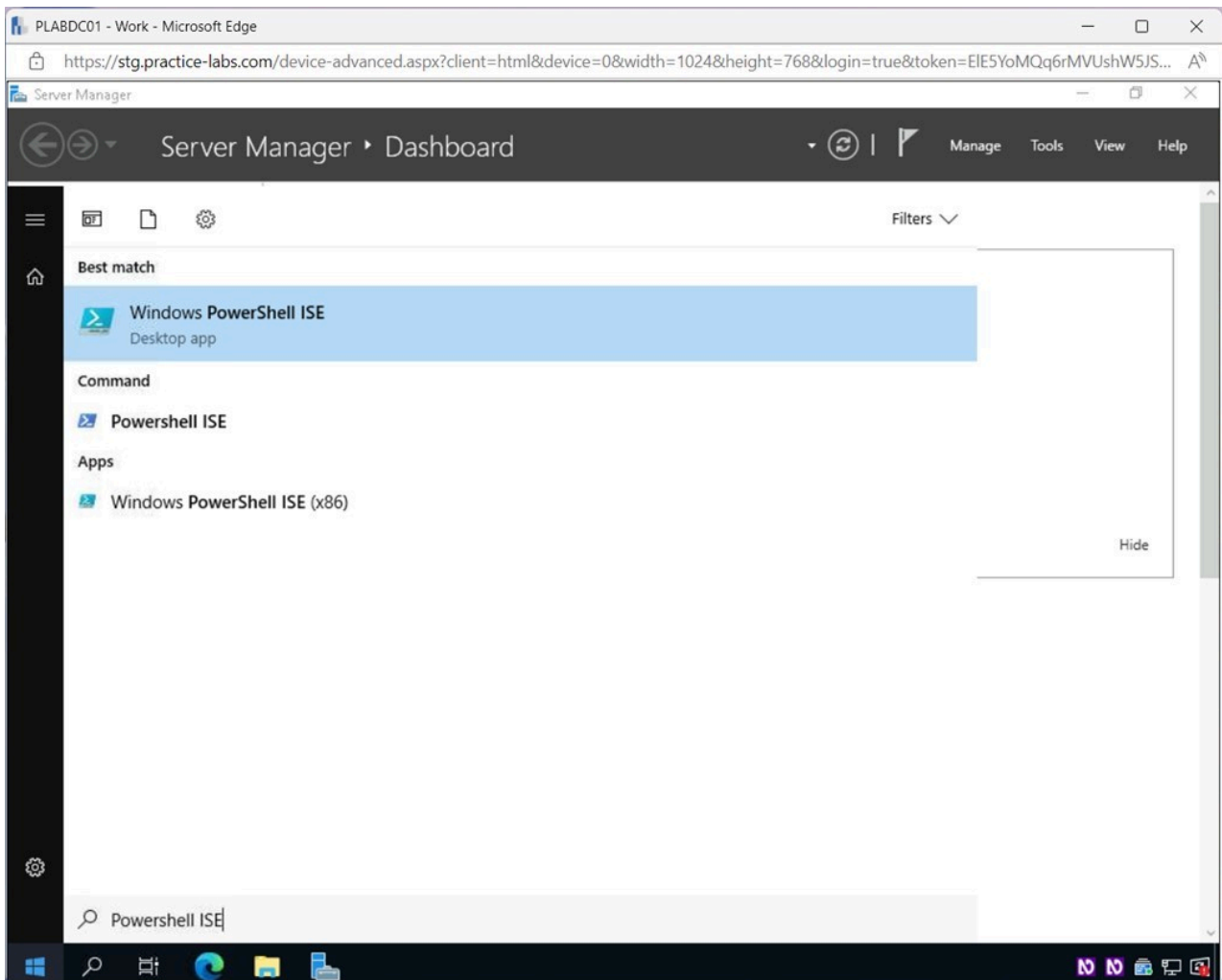


Figure 3.5 Screenshot of PLABDC01: Displaying selecting Windows Powershell ISE from the Best match pop-up menu.

Step 6

In the **Administrator: Windows Powershell ISE** window, type the following:

```
Import-Module activedirectory
New-ADUser -Name(Read-Host "Enter Name") -
GivenName(Read-Host "Enter Given Name ") -
Surname(Read-Host "Enter Surname") -
SamAccountName(Read-Host "Enter SamAccountName") -
UserPrincipalName(Read-Host "Enter UPN eg
@practicelabs.com") -Path
"Ou=Marketing,DC=practicelabs,DC=com" -
AccountPassword(Read-Host -AsSecureString "Enter
```

Secure ChPassword") -ChangePasswordAtLogon \$true -
Enabled \$true

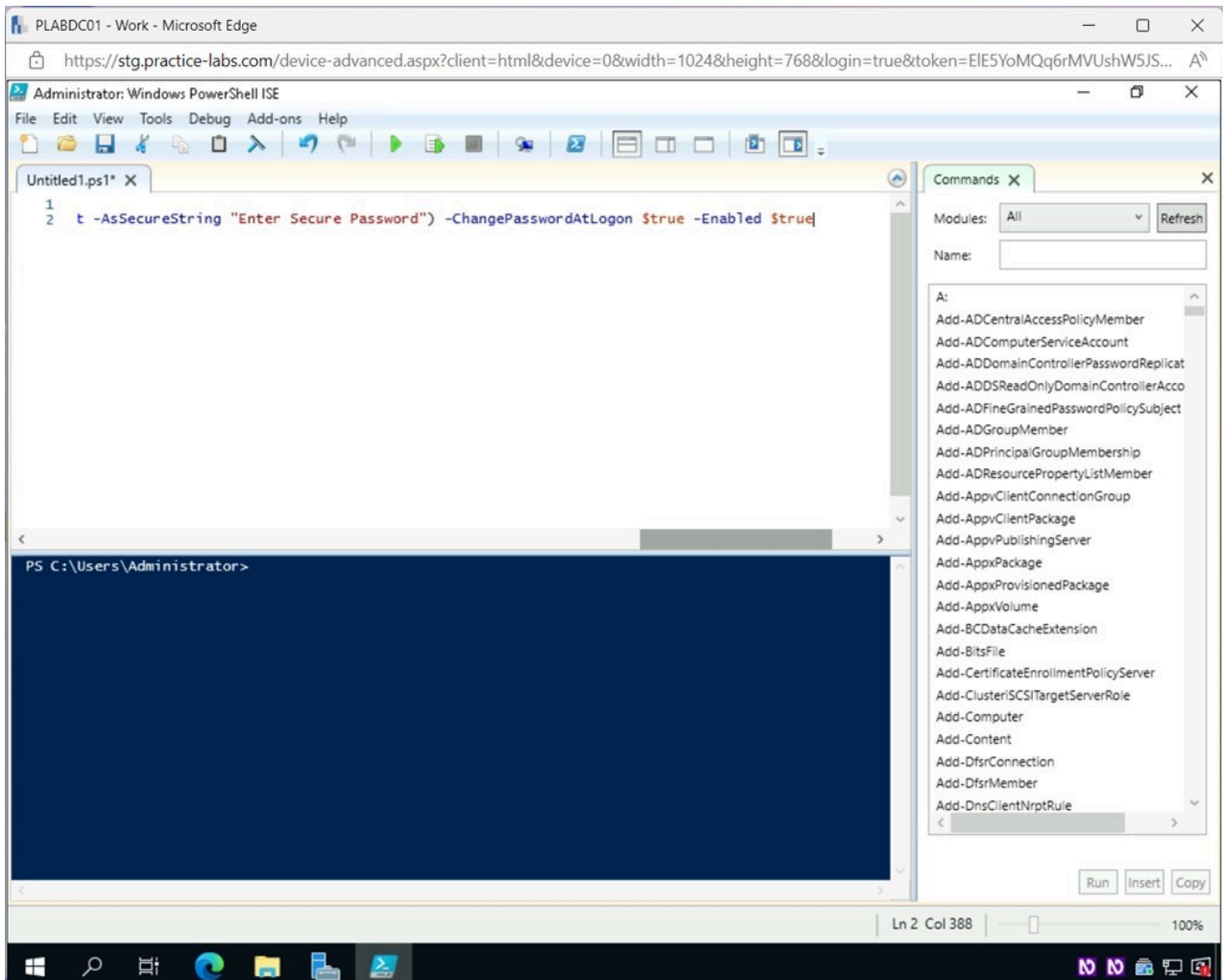


Figure 3.6 Screenshot of PLABDC01: Displaying entering the required command in the Powershell ISE window.

Step 7

In the **Administrator: Windows Powershell ISE** window, click **Save**.

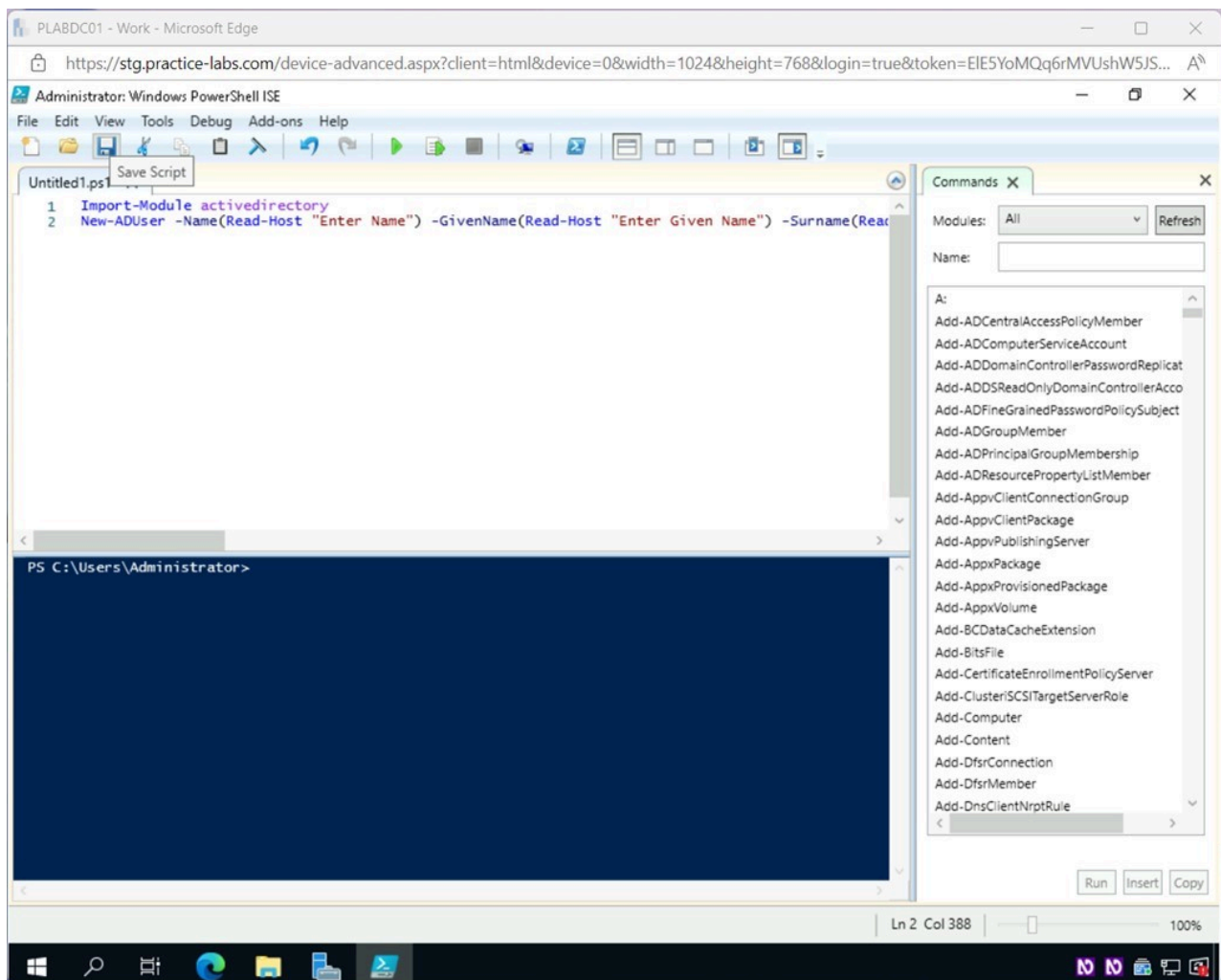


Figure 3.7 Screenshot of PLABDC01: Displaying clicking Save in the Powershell window.

Step 8

In the **Save as** dialog box, ensure the **Documents** folder is selected on the left pane.

Type the following in the **File Name** field and click **Save**:

ADUserCreate

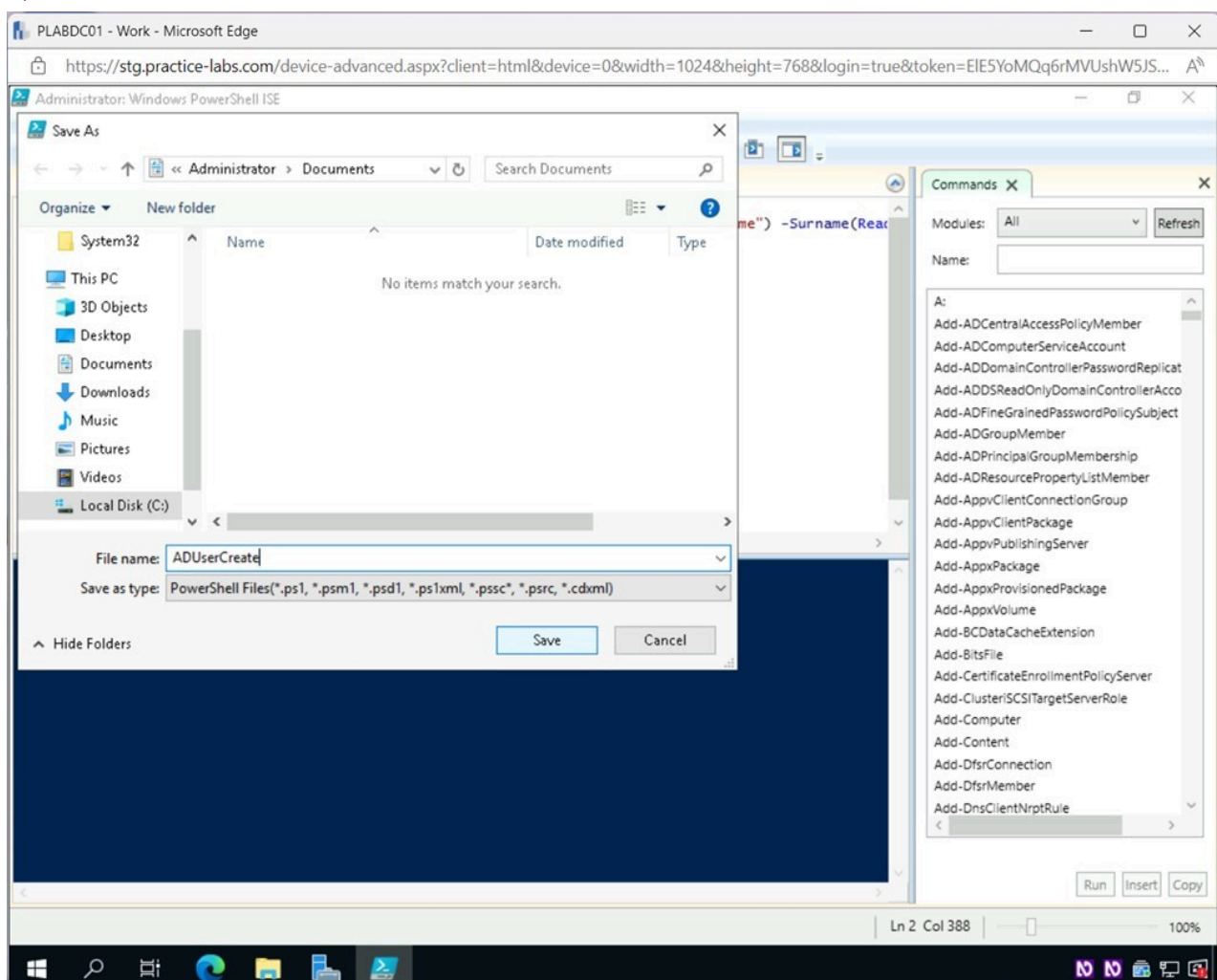


Figure 3.8 Screenshot of PLABDC01: Displaying the Save As dialog box, showing the required File name typed in and the Save button selected.

Step 9

Close the **Administrator: Windows Powershell ISE** window.

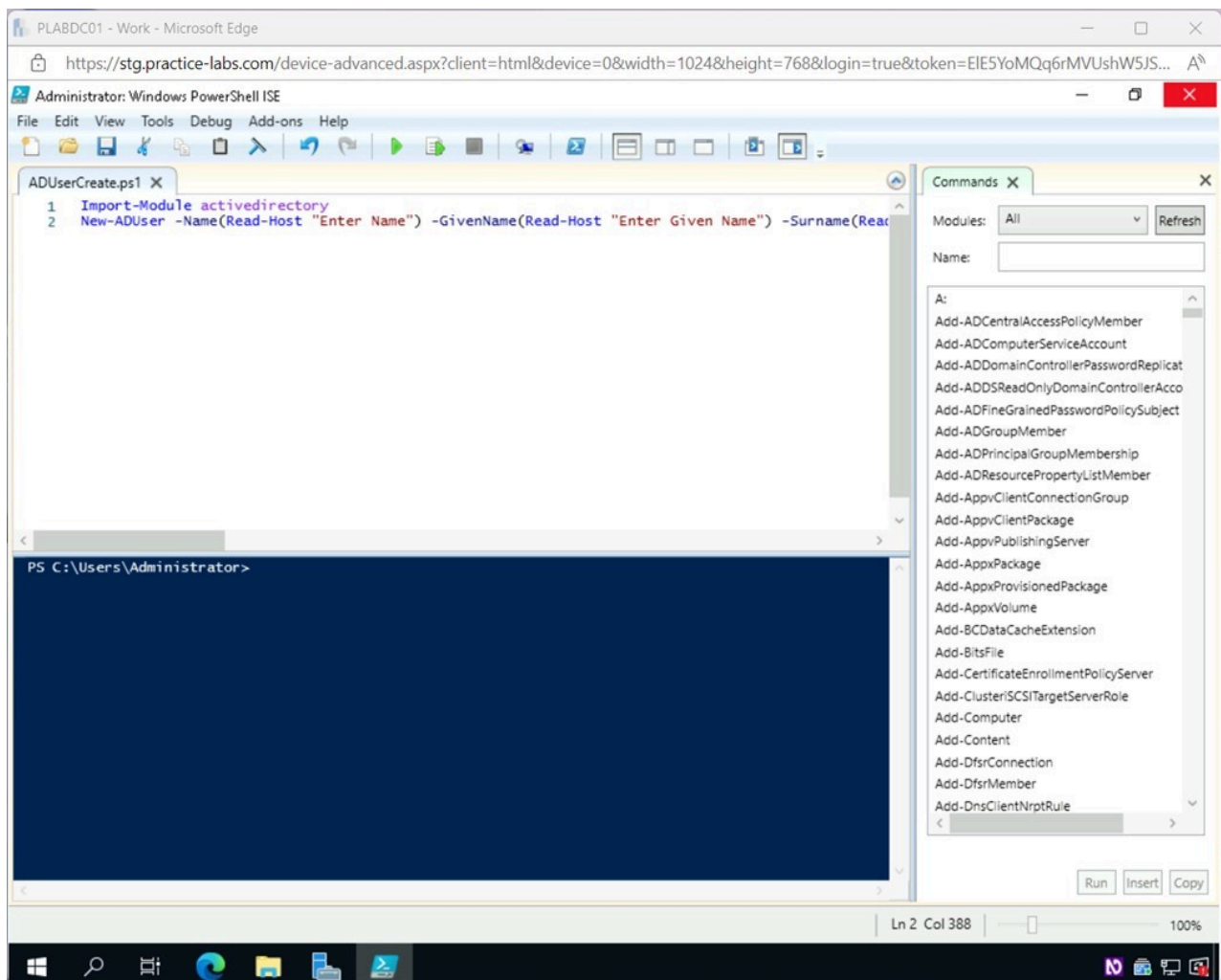


Figure 3.9 Screenshot of PLABDC01: Displaying closing the Windows Powershell ISE window.

Step 10

Right-click the **Start** charm and select **Windows Powershell (Admin)**.

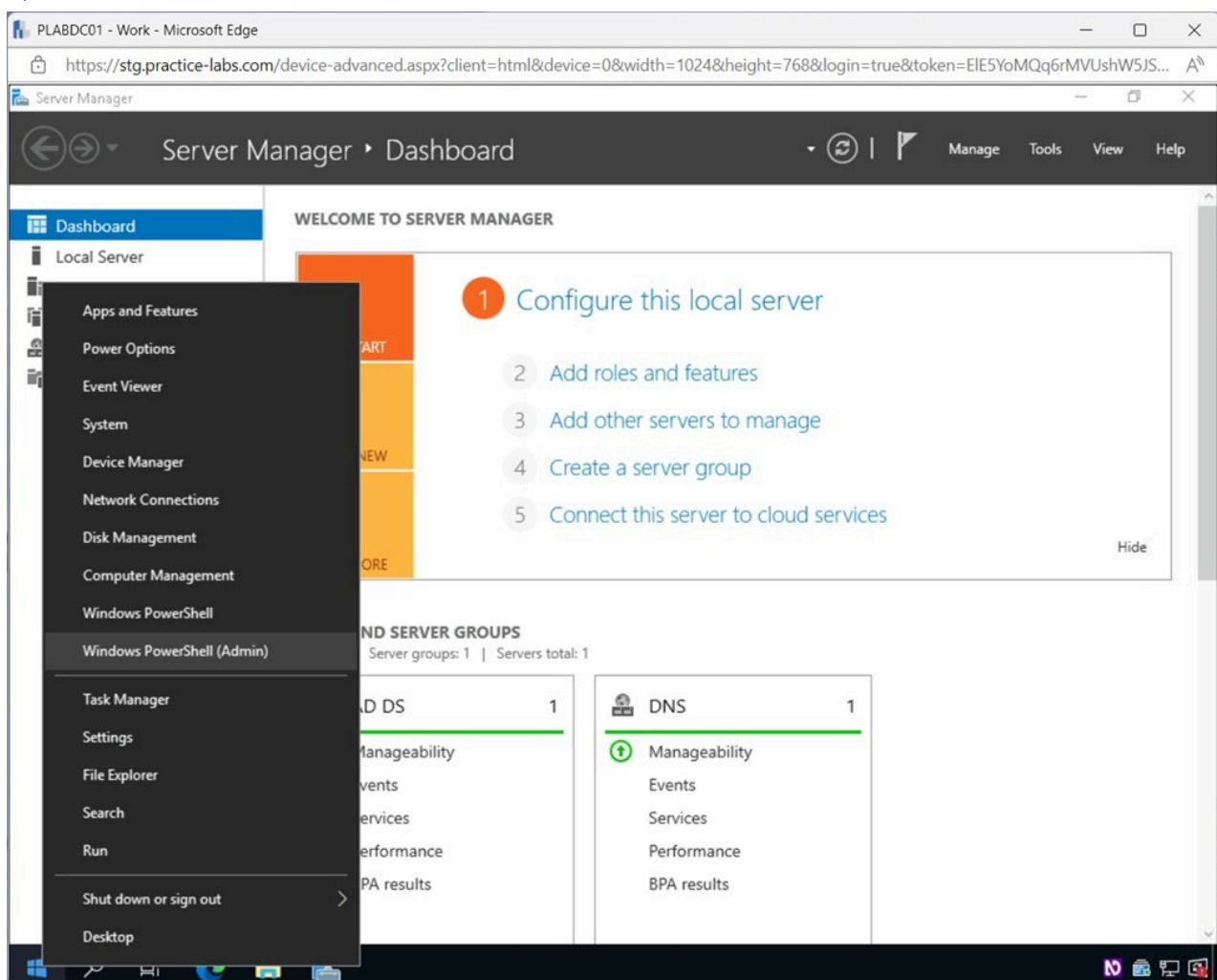


Figure 3.10 Screenshot of PLABDC01: Displaying right-clicking the Start charm and selecting Windows PowerShell (Admin).

Step 11

In the **Administrator: Windows Powershell** window, type the following and press **Enter**:

```
cd .\Documents\
```

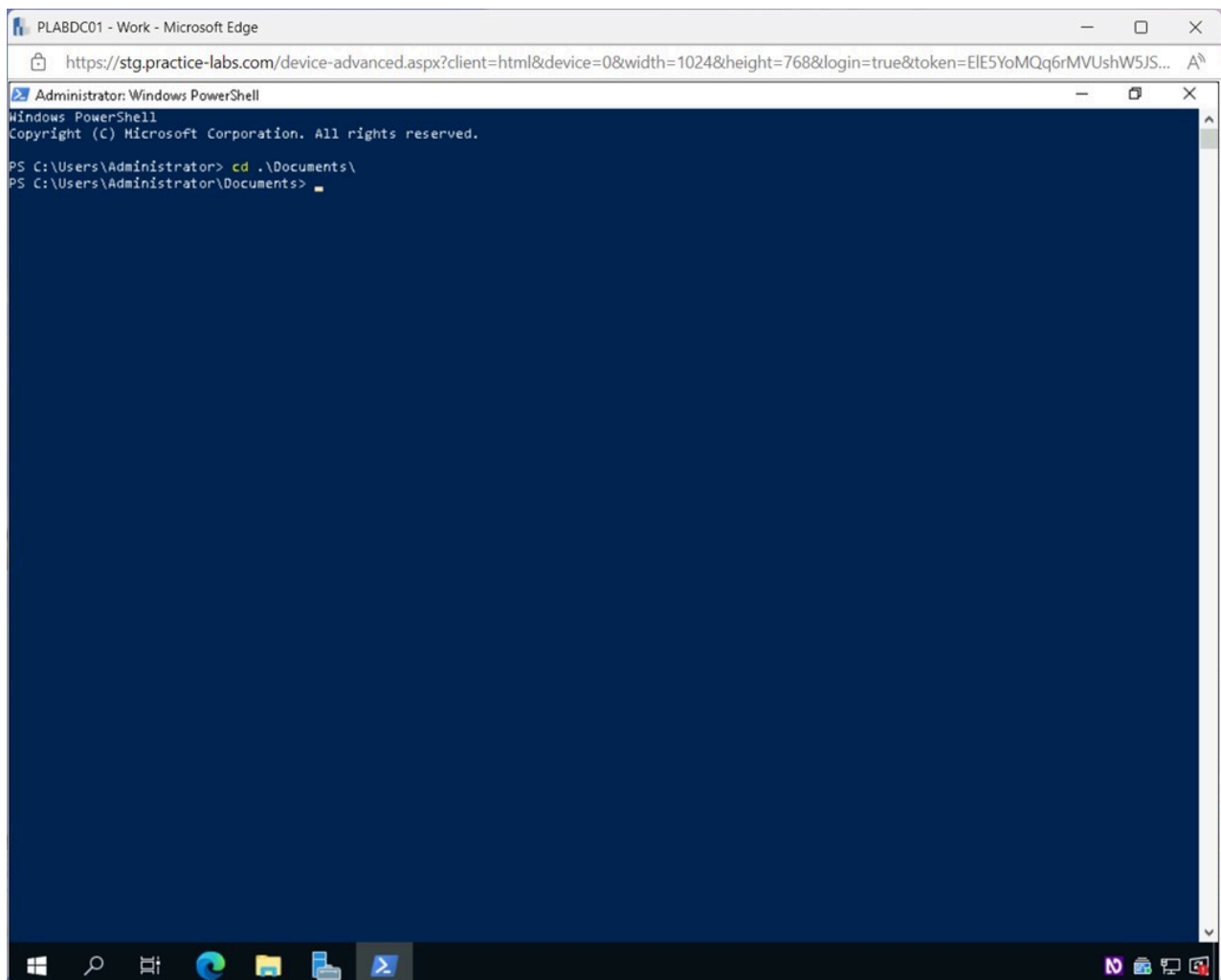



Figure 3.11 Screenshot of PLABDC01: Displaying executing the required command in Windows Powershell.

Step 12

In the **Administrator: Windows Powershell** window, type the following and press **Enter**:

```
.\ADUserCreate.ps1
```

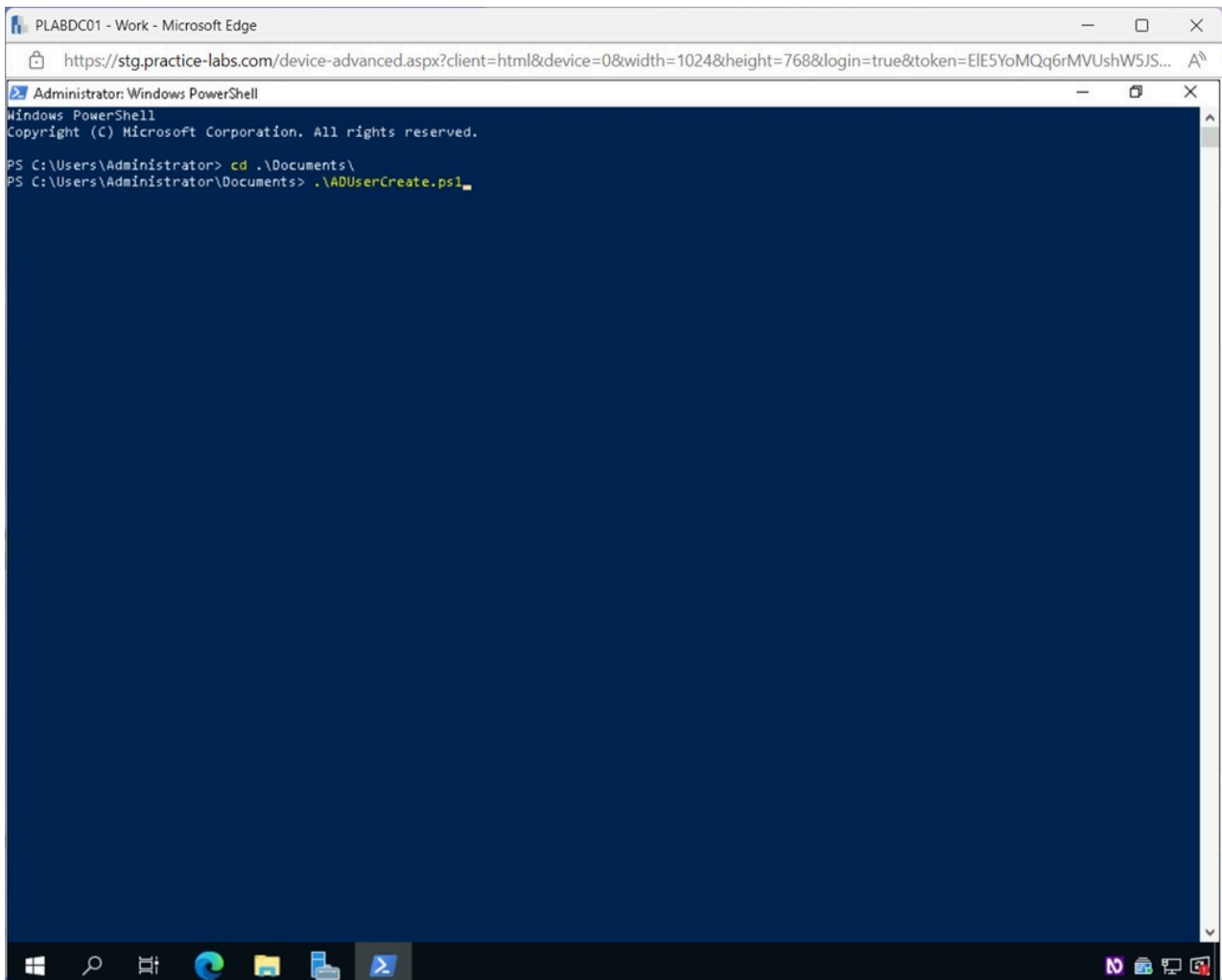


Figure 3.12 Screenshot of PLABDC01: Displaying executing the required command in Windows Powershell.

Step 13

Enter the following and press **Enter** after each field:

```
Enter Name: Subi
Enter Given Name: Subi
Enter Surname: ACI
Enter SamAccountName: Subi.a
Enter UPN eg @practicelabs.com:
subi.a@practicelabs.com
Enter Secure Password: Passw0rd
```

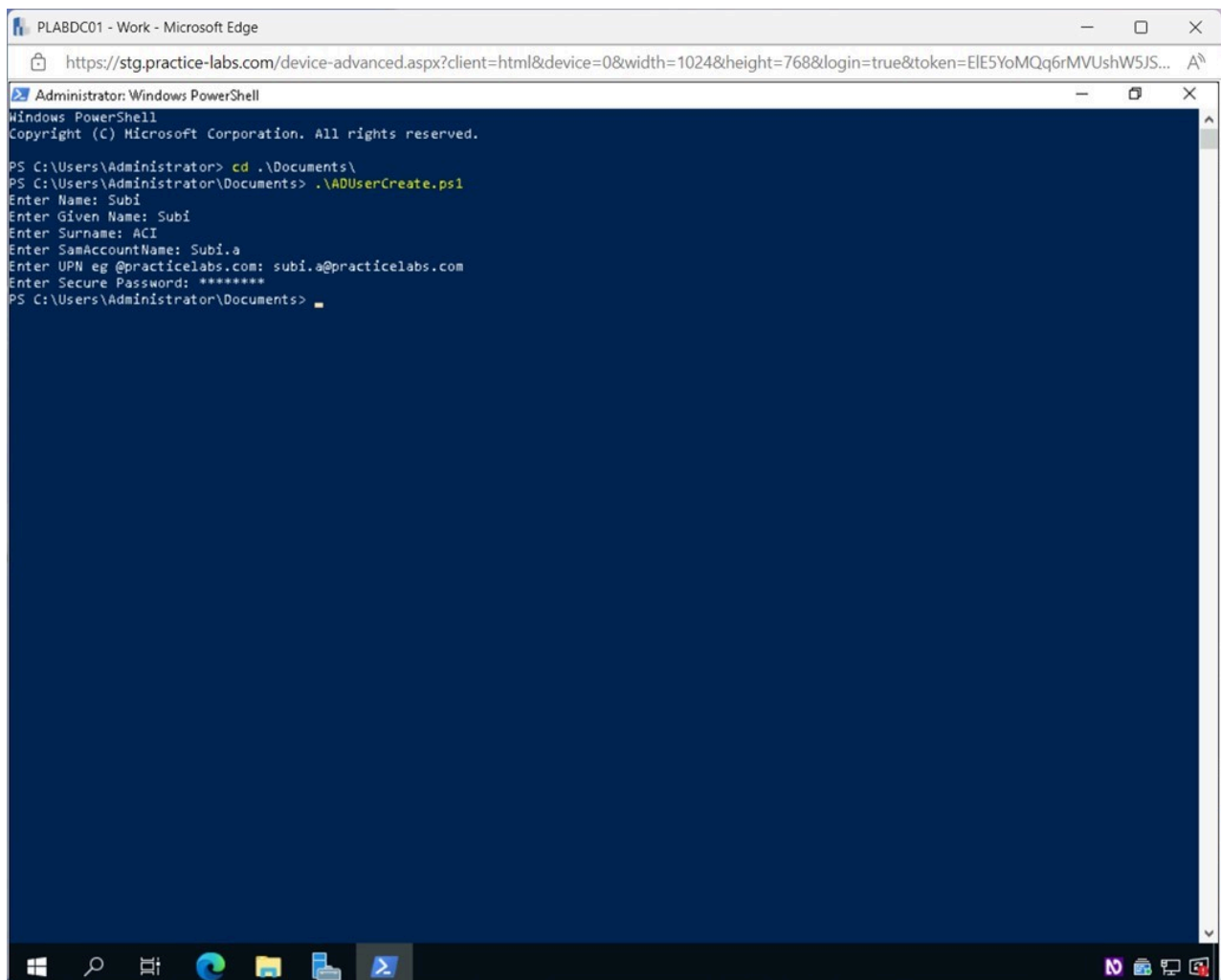


Figure 3.13 Screenshot of PLABDC01: Displaying entering the values into the different fields and pressing Enter.

Step 14

Restore **Server Manager** from the **Taskbar**.

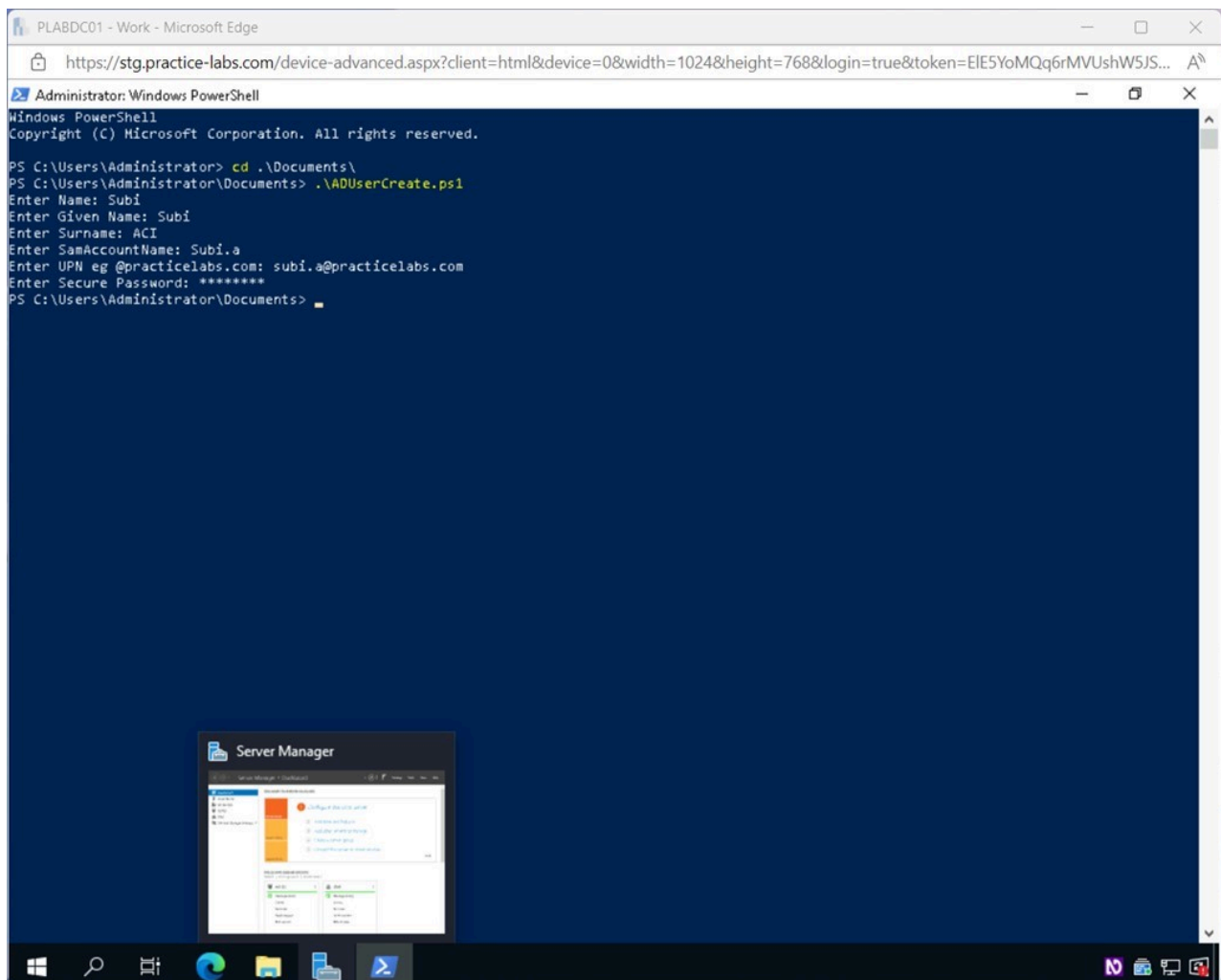


Figure 3.14 Screenshot of PLABDC01: Displaying opening Server Manager from the Taskbar.

Step 15

In the **Server Manager** window, click **Tools** and select **Active Directory Users and Computers**.

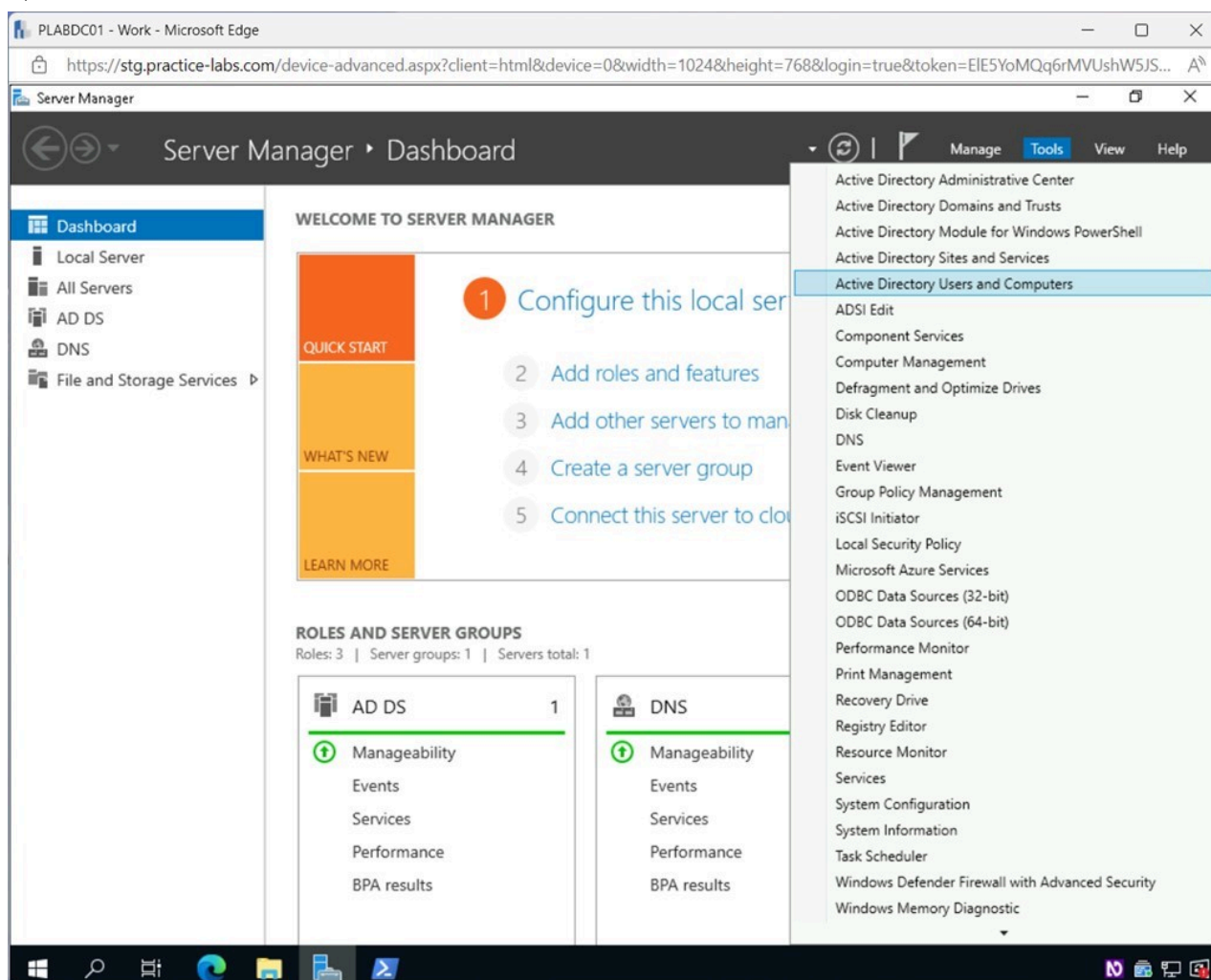


Figure 3.15 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Active Directory Users and Computers is selected.

Step 16

In the **Active Directory Users and Computers** window, ensure the **Marketing** Organizational Unit is selected.

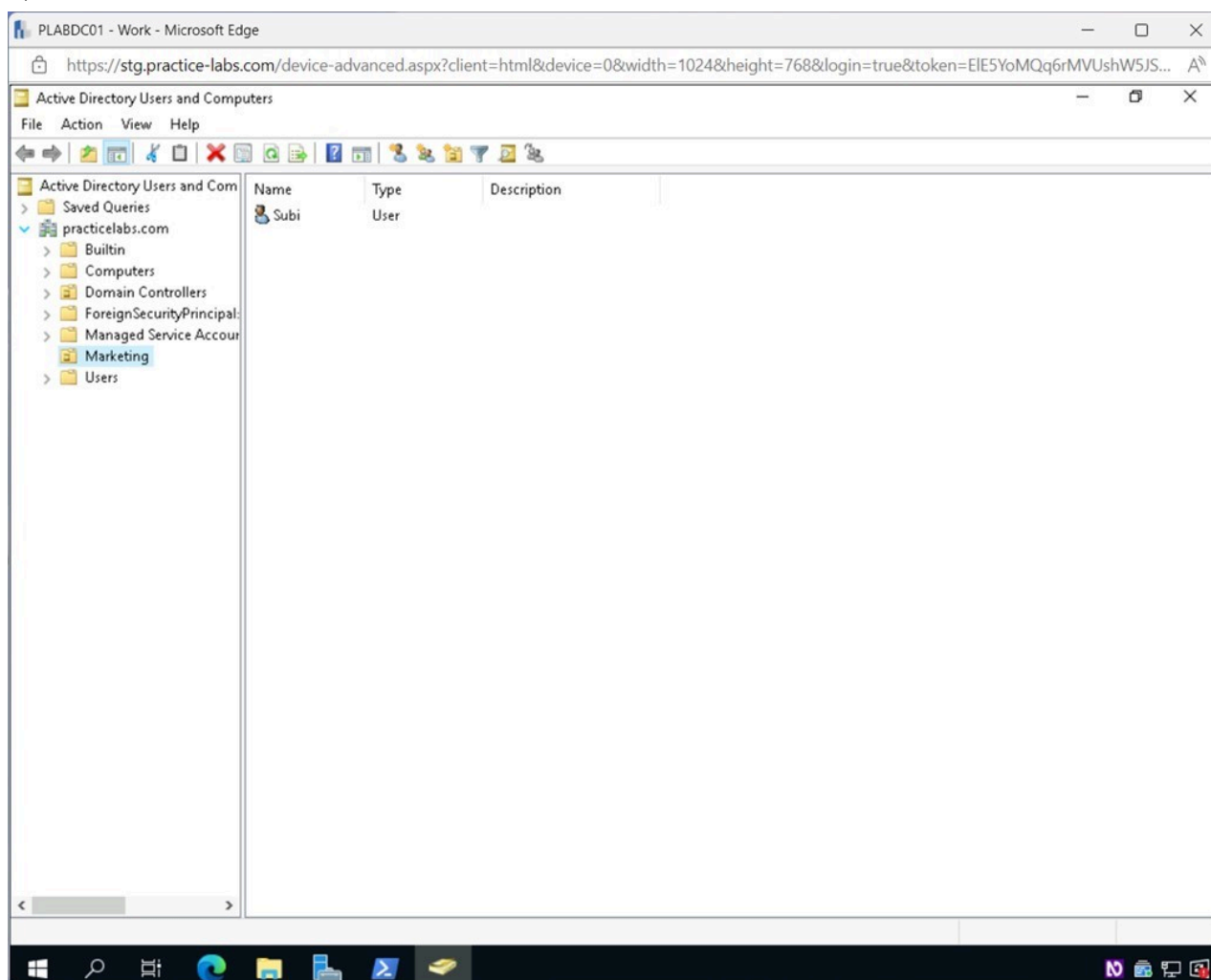


Figure 3.16 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window, showing the Marketing Organizational Unit selected.

Note: The user **Subi** was successfully created using the Powershell script.

Close all open windows.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Automation of Server Administration using Scripts** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Different Script Types
- Exercise 2 - Basic Script Constructs
- Exercise 3 - User Creation using Scripts

You should now be able to:

- Create a Linux Script
- Create a Basic Powershell Script
- Create a Basic Batch Script File
- Create a Script File with Different Variables and Conditions
- Create a Script that Creates Users in Active Directory

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.